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MILITARY SPECIFICATION

TERMINAL, COMMUNICATIONS
AN/UGC-74B(V) and AN/UGC-74C(V)

This specification is approved for use within the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Terminal, Communications AN/UGC-74B(V), which is composed of Teleprinter, TT-744A/UGC-74B(V), Keyboard, TT-745/UGC-74A(V), Case, CY-7735/UGC-74A(V), and Cover, MX-10038/UGC-74A(V) (see 6.8). The Appendix of this specification covers the Terminal, Communications AN/UGC-74C(V), which is composed of Teleprinter, TT-744B/UGC-74C(V), Keyboard, TT-745A/UGC-74C(V), Case CY-7735/UGC-74A(V), Cover MX-10038/UGC-74A(V), and Auxiliary Memory Module, MU-856/UGC-74(V) (see 60.8).

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

UU-P-547	Paper, Teletypewriter, Roll

MILITARY

MIL-S-901	Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-T-49261	Terminal, Communications AN/UGC-74A(V)3, TEMPEST Requirements for (U)
MIL-P-40023	Paper, Fanfold
MIL-C-46168	(ME) Coating, Aliphatic Polyurethane, Chemical Agent Resistant

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables Inspection by Attributes
MIL-STD-188-114	Electrical Characteristics of Digital Interface Circuits
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

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MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781	Reliability Test: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1280	Keyboard Arrangements
MIL-STD-1474	Noise Limits for Army Material
MIL-STD-1562	List of Standard Microcircuits

2.1.2 Other Government Documents, Drawings and Publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

A3041500	Teletypewriter AN/UGC-74B(V)
SM-D-972467	Mount Assembly
A3152457	Electronic Equipment Design Requirements for the Nuclear Radiation Environment

JOINT CHIEFS OF STAFF

JANAP-128	Automatic Digital Network (AUTODIN) Operating Procedures
ACP-127, US Supplement 1	Communication Instructions Tape Relay Procedures (Allied RESTRICTED)
ACP-127, NATO Supplement 3	Communication Instruction Tape Relay Procedures

(Copies of specifications, standards, documents, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets of MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing A3041500 (including the requirements for parts, materials and processes thereon).

3.3 Parts, Materials and Processes: General. In addition to the requirements of this specification, the requirements of MIL-P-11268 shall apply. Microelectronic devices shall be selected from MIL-STD-1562.

3.4 Finish. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings.

3.5 Marking.

3.5.1 General. Marking shall conform to Specification MIL-M-13231.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.7 Preconditioning. The equipment shall be capable of meeting the requirements herein, after subjection to the preconditioning of 4.6.1 and 4.6.2.

3.8 Functional Requirements for Terminal, Communications AN/UGC-74B(V).

3.8.1 Paper Usage. The equipment shall legibly print an original plus two carbon copies; accept roll paper IAW Specification UU-P-547; accept fanfold paper IAW Specification MIL-P-40023.

3.8.2 Visibility of Copy. No operator action shall be required to view each typed character after printing stops. The equipment shall accept and reliably feed both 15# and 20# weight roll and fanfold paper.

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3.8.3 Copy Illumination. With the POWER switch in the ON position, the copy lamps shall illuminate the paper. Illumination shall be adjustable by clockwise rotation of the ILLUM knob from the OFF position to the MAX position.

3.8.4 ASCII/BAUDOT Control. The setting of the control shall determine whether the teletypewriter shall transmit and receive either ASCII (USA Standard Code for Information Exchange) or BAUDOT Code (International Telegraph Alphabet American Variation (ITA-2)).

3.8.5 Printer Functions.

3.8.5.1 Keyboard/Print Codes. Tables I, II and III define the keyboard/print codes. Table I defines the Control Key Codes, Table II defines the Upper Case Key Codes and Table III defines the Lower Case Key Codes. These tables define the keyboard/print codes for 64/96/128 ASCII character font symbol sets; the selection of these is as specified in 3.8.5.7.3.2.

3.8.5.2 Local Printer Functions. Local machine functions shall be performed whether or not the machine is also responding to line signals.

3.8.5.3 Coded BAUDOT Printer Functions.

3.8.5.3.1 Carriage Return. This code shall return printing position to left margin and suppress printing and spacing during the operation.

3.8.5.3.2 Line Feed. This code shall advance paper one or two lines as determined by the parameter set in the LF Subcommand.

3.8.5.3.3 Letters. This code shall shift the printer to the printing letters condition.

3.8.5.3.4 Figures. This code shall shift the printer to the printing figures condition.

3.8.5.3.5 Parity Error. The parity function shall be disabled when receiving Baudot characters.

3.8.5.4 Coded ASCII Printer Functions.

3.8.5.4.1 HT (TAB). When this code is encountered, the print position shall be moved to the smallest set tab position which is greater than the current print position. The next character is printed in the position. If the present print position is greater than or equal to the largest tab position, the tab shall be replaced by a carriage return.

3.8.5.4.2 CR. This code shall return printing position to left margin and suppress printing and spacing.

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TABLE I. CONTROL CHARACTERS AND CODES

	LOCAL PRINT			TRANSMITTED CODE 1/	
	ASCII 3/				
CHARACTER	128 CHAR	96/64 CHAR	BAUDOT 2/	ASCII	BAUDOT
NUL	\	NONE	NONE	0000000	00000
SOH	⌐	SP	◇	0000001	NONE
STX	┬	SP	◇	0000010	NONE
ETX	└	SP	◇	0000011	NONE
EOT	┼	SP	◇	0000100	NONE
ENQ	┼	SP	◇	0000101	NONE
ACK	▯	SP	◇	0000110	NONE
BEL	▯	SP/ALARM	SP/ALARM/ APOSTROPHE	0000111	00101/ 01011
BS/FIGS	BACKSPACE	BACKSPACE	^ 4/	0001000	11011
HT (TAB)	TAB	TAB	TAB	0001001	00100
LF	LINE FEED	LINE FEED	LINE FEED	0001010	00010
VT	▽	SP	◇	0001011	NONE
FF	▽	SP	◇	0001100	NONE
CR	RETURN TO LEFT MARGIN	RETURN TO LEFT MARGIN	RETURN TO LEFT MARGIN	0001101	01000
SO	^	SP	◇	0001110	NONE
SI	v	SP	◇	0001111	NONE
DLE	○	SP	◇	0010000	NONE
DC1	○	SP	◇	0010001	NONE
DC2	○	SP	◇	0010010	NONE
DC3	○	SP	◇	0010011	NONE
DC4	○	SP	◇	0010100	NONE
NAK	○	SP	◇	0010101	NONE
SYN	○	SP	◇	0010110	NONE
ETB	○	SP	◇	0010111	NONE
CAN	○	SP	◇	0011000	NONE
EM	○	SP	◇	0011001	NONE
SUB	○	SP	◇	0011010	NONE
ESC	○	SP	◇	0011011	NONE
FS	○	SP	◇	0011100	NONE
GS	○	SP	◇	0011101	NONE
RS	○	SP	◇	0011110	NONE
US	○	SP	◇	0011111	NONE

1/ The least significant bit (LSB) of each code group is located in the far right position. In serial bit by bit transmission, the coded characters shall be transmitted low order (LSB) first.

2/ In BAUDOT local print, the ◇ symbol shall be printed by keyboard entry only, and shall not be put into memory, i.e., nothing shall be printed during transmission.

3/ If an ASCII character with a parity error is received, a ◇ symbol shall be printed locally.

4/ Symbol shall be printed upon keyboard entry only. When received from message memory, the Figures function shall be performed and symbol shall not be printed.

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TABLE II. UPPER CASE CHARACTERS AND CODES

	LOCAL PRINT			TRANSMITTED CODE 1/	
	ASCII 3/				
CHARACTER	128/96 CHAR	64 CHAR	BAUDOT 2/	ASCII	BAUDOT
A	A	A	A	1000001	00011
B	B	B	B	1000010	11001
C	C	C	C	1000011	01110
D	D	D	D	1000100	01001
E	E	E	E	1000101	00001
F	F	F	F	1000110	01101
G	G	G	G	1000111	11010
H	H	H	H	1001000	10100
I	I	I	I	1001001	00110
J	J	J	J	1001010	01011
K	K	K	K	1001011	01111
L	L	L	L	1001100	10010
M	M	M	M	1001101	11100
N	N	N	N	1001110	01100
O	O	O	O	1001111	11000
P	P	P	P	1010000	10110
Q	Q	Q	Q	1010001	10111
R	R	R	R	1010010	01010
S	S	S	S	1010011	00101
T	T	T	T	1010100	10000
U	U	U	U	1010101	00111
V	V	V	V	1010110	11110
W	W	W	W	1010111	10011
X	X	X	X	1011000	11101
Y	Y	Y	Y	1011001	10101
Z	Z	Z	Z	1011010	10001

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TABLE II. UPPER CASE CHARACTERS AND CODES - Continued.

CHARACTER	LOCAL PRINT		TRANSMITTED CODE 1/	
	ASCII 3/ 128/96 CHAR	BAUDOT 2/ 64 CHAR	ASCII	BAUDOT
SP	SP	SP	0100000	00100
!	!	!	0100001	01101
"	"	"	0100010	10001
#	#	#	0100011	10100
$	$	$	0100100	01001
%	%	%	0100101	NONE
&	&	&	0100110	11010
'	'	'	0100111	01011
(	(	(	0101000	01111
)	)	)	0101001	10010
=	=	=	0111101	NONE
{	{	[	1111011	NONE
}	}	]	1111101	NONE
`	`	@	1100000	NONE
¦	¦	\	1111100	NONE
+	+	+	0101011	NONE
*	*	*	0101010	NONE
¬	~	^	1111110	NONE
<	<	<	0111100	NONE
>	>	>	0111110	NONE
?	?	?	0111111	11001

1/ The least significant bit (LSB) of each code group is located in the far right position. In serial bit by bit transmission, the coded characters shall be transmitted low order (LSB) first.

2/ In BAUDOT local print, the Ø symbol shall be printed by keyboard entry only, and shall not be put into memory, i.e., nothing shall be printed diring transmission.

3/ If an ASCII character with a parity error is received, a Ø symbol shall be printed locally.

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TABLE III. LOWER CASE CHARACTERS AND CODES

	LOCAL PRINT			TRANSMITTED CODE 1/	
	ASCII 3/				
CHARACTER	96/128 CHAR	64 CHAR	BAUDOT 2/	ASCII	BAUDOT
a	a	A	A	1100001	00011
b	b	B	B	1100010	11001
c	c	C	C	1100011	01110
d	d	D	D	1100100	01001
e	e	E	E	1100101	00001
f	f	F	F	1100110	01101
g	g	G	G	1100111	11010
h	h	H	H	1101000	10100
i	i	I	I	1101001	00110
j	j	J	J	1101010	01011
k	k	K	K	1101011	01111
l	l	L	L	1101100	10010
m	m	M	M	1101101	11100
n	n	N	N	1101110	01100
o	o	O	O	1101111	11000
p	p	P	P	1110000	10110
q	q	Q	Q	1110001	10111
r	r	R	R	1110010	01010
s	s	S	S	1110011	00101
t	t	T	T	1110100	10000
u	u	U	U	1110101	00111
v	v	V	V	1110110	11110
w	w	W	W	1110111	10011
x	x	X	X	1111000	11101
y	y	Y	Y	1111001	10101
z	z	Z	Z	1111010	10001

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TABLE III. LOWER CASE CHARACTERS AND CODES (continued)

	LOCAL PRINT			TRANSMITTED CODE 1/	
	ASCII 3/				
CHARACTER	96/128 CHAR	64 CHAR	BAUDOT 2/	ASCII	BAUDOT
0	0	0	0	0110000	10110
1	1	1	1	0110001	10111
2	2	2	2	0110010	10011
3	3	3	3	0110011	00001
4	4	4	4	0110100	01010
5	5	5	5	0110101	10000
6	6	6	6	0110110	10101
7	7	7	7	0110111	00111
8	8	8	8	0111000	00110
9	9	9	9	0111001	11000
[	[	[	◇	1011111	NONE
]	]	]	◇	1011011	NONE
@	@	@	◇	1011101	NONE
\	\	\	◇	1000000	NONE
-	-	-	◇	1011100	NONE
DEL/LTRS	NONE	NONE	- 4/	0101101	00011
;	;	;	;	1111111	11111
:	:	:	:	0111011	11110
^	^	^	◇	0111010	01110
,	,	,	,	1011110	NONE
.	.	.	.	0101100	01100
/	/	/	/	0101110	11100
				0101111	11101

1/ The least significant bit (LSB) of each code group is located in the far right position. In serial bit by bit transmission, the coded characters shall be transmitted low order (LSB) first.

2/ In BAUDOT local print, the ◇ symbol shall be printed by keyboard entry only, and shall not be put into memory, i.e., nothing shall be printed during transmission.

3/ If an ASCII character with a parity error is received, a ◇ symbol shall be printed locally.

4/ Symbol shall be printed upon keyboard entry only. When received from message memory, the Figures function shall be performed and symbol shall not be printed.

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3.8.5.4.3 LF. This code shall advance paper one or two lines as determined by the parameter set in the LF Subcommand.

3.8.5.4.4 BS. This code shall move the print position one space to the left thus providing overprint capability.

3.8.5.4.5 Parity Error. The PARITY lamp shall light when an ASCII character with a parity error is received and parity is not inhibited. When the received message is printed, the parity error symbol (diamond) shall be printed in place of all characters that contained parity errors on reception.

3.8.5.5 Automatic Carriage Return. After the printer has reached the end of line position, and upon entry of a character which would normally cause the printer to move to the right (end of line +1), the printer shall automatically insert a carriage return and line feed.

3.8.5.6 Automatic Line Feed. In the Intelligent Communication State (ICT) mode, the printer shall perform an automatic line feed with each carriage return. In the Keyboard Send Receive (KSR) 'Character Mode', there shall be no automatic line feeds associated with a received carriage return code. In the KSR 'Line Mode', automatic line feed activation shall be an operator selectable option (see 3.8.8.6).

In the Receive Only (RO) mode, the printer shall perform a single automatic line feed for each string of received carriage return codes (1 to n) not immediately preceded or followed by a line feed code.

3.8.5.7 Printer Characteristics.

3.8.5.7.1 Printing Method. Printing shall be implemented via an impact type, 9 pin, print head.

3.8.5.7.1.1 Print Pattern. Print pattern shall be 9XN. Characters shall be defined by the character font Read Only Memory (ROM). The print pattern shall have a minimum resolution such that vertical and horizontal dot spacing is nominally a dot diameter.

3.8.5.7.1.2 Print Pin Pressure Selection. The capability shall be provided to permit operator selection of increased print-pin pressure for enhanced printing on multiple-copy paper. The default value for this function shall be the 'lightest' pressure value, i.e., for single copy paper.

3.8.5.7.1.3 Print Head Position Adjustment. The capability shall be provided to permit the operator to adjust the print head to paper distance for enhanced printing on single and on multiple-copy paper. No tools shall be required to perform the adjustment. As a minimum, two positive print head position stops shall be provided. One shall provide the optimum print head position for single copy paper, the other for original plus two copy carbon paper (i.e. 5-ply paper).

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3.8.5.7.2 Line Length. Provide line length from 40 up to and including 132 characters. Line length shall be selectable via keyboard entry.

3.8.5.7.2.1 Compressed Text Mode. The compressed text mode shall be provided for line lengths from 81 up to and including 132 characters.

3.8.5.7.3 Multiple Character Fonts. Multiple character font capability shall be provided. All character font software program coding shall be resident in a single memory microcircuit.

3.8.5.7.3.1 Character Font Memory Capacity. The character font memory shall have, as a minimum, the capacity to store 768 unique character symbols for retrieval and printout. Storage for a minimum of 6 character font sets, each set consisting of up to 128 characters/symbols shall be provided.

3.8.5.7.3.2 Character Font Selection. The capability shall be provided to select character font sets via keyboard entry.

3.8.5.7.3.3 Standard Character Font Symbol Sets. The following standard character font symbol sets shall be provided: (1) font 1-64 character ASCII, (2) font 2-96 character ASCII, and (3) font 3-128 character ASCII as specified in 3.8.5.1.

3.8.5.7.4 Paper/Out. When the supply of roll or fanfold paper is depleted, printing shall stop. In addition, when set for RO operation, a continuous spacing line (break) shall be transmitted.

3.8.6 Alarm and Indicators.

3.8.6.1 Audible Alarm.

3.8.6.1.1 Audible Alarm Adjustment. The audible alarm level shall be adjustable from maximum to no alarm.

3.8.6.1.2 Audible Alarm Actuation. The audible alarm shall be actuated under the following conditions:

a. By Figure S/J Codes being received (BAUDOT mode) as follows:

SWITCH POSITION	RECEIVED CODE	ACTION
S	Figure S	Alarm for 0.3 +/- 0.1 second
J	Figure J	Alarm for 0.3 +/- 0.1 second

b. A steady-state alarm shall be actuated when an ASCII character with a parity error is received (PARITY lamp lights). The PARITY RESET switch shall turn the parity alarm condition off and extinguish the lamp.

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c. The alarm shall be actuated for 0.3 +/- 0.1 second when a BEL code is received in the ASCII mode.

d. By LINE lamp conditions as follows:

INTERFACE	SIGNAL	ALARM
High Level Digital	Open-Line Condition	Steady-State Alarm
Low Level Digital	Continuous Space	Steady-State Alarm
Conditioned Diphase	Continuous Space	Steady-State Alarm

e. A steady-state alarm shall be actuated when a message, with precedence of IMMEDIATE or higher, has been received. The alarm condition shall be turned off when the message is printed.

3.8.6.2 Indicators.

3.8.6.2.1 Power Indicator. The power on indicator shall be the copy lamps, and they shall light when the POWER switch is activated if the ILLUM control is on and primary power is available.

3.8.6.2.2 Battery Back-Up Indicator. The BAT. LAMP shall be activated when the system is in the Battery Back-Up state and the POWER switch is "ON". All other indicators shall automatically be turned off.

3.8.6.2.3 Paper Low Indicator. The PAPER LOW lamp shall light when the roll paper supply is reduced to a diameter of 1-7/16" +/- 1/16". The Paper Low indicator is not applicable when fanfold paper is in use.

3.8.6.2.4 Line Indicator. The LINE lamp shall be on while mark and space signals are being received, and shall be off while the line is idle. For the high level digital interface only, an open line condition shall cause the LINE lamp to flash on and off. For the low level digital interface only, a continuous space shall cause the lamp to flash on and off. For the conditioned diphase interface only, a continuous space shall cause the lamp to flash on and off. The audible alarm shall be on while the LINE lamp is flashing.

3.8.6.2.5 Received Message Indicator. The MSG RCVD lamp shall be activated when a message has been received. If message memory has a message with precedence of IMMEDIATE or higher, the lamp shall flash until all the IMMEDIATE or higher precedent messages have been printed. For lower precedent messages, the lamp shall light in a steady state until all messages have been printed.

3.8.6.2.6 Transmit Indicator. The XMIT lamp shall light during message transmission.

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3.8.6.2.7 Message Memory Full Warning Indicator. The MEM FULL lamp shall be activated when 15% or less of message memory is available for allocation.

3.8.6.2.8 Parity Indicator. The PARITY lamp shall light when an ASCII character with a parity error is received and parity is not inhibited.

3.8.6.2.9 End of line signal. The END OF LINE warning lamp shall be on when the print position is within 6 character positions of the end of the line or during message composition and editing when the print position is within 6 character positions of the line length specified by the LINE subcommand.

3.8.7 Switches.

3.8.7.1 Internal Switches. The internal switches shall be located on the interface assembly.

3.8.7.1.1 PARITY Switch. The parity function shall be active when the ASCII/BAUDOT switch is in the ASCII position, and shall be inactive in the BAUDOT position. When active, the EVEN setting shall cause ASCII characters with even parity to be transmitted, and shall cause received characters to be checked for even parity. Setting the switch to the ODD position shall cause ASCII characters with odd parity to be transmitted and shall cause received characters to be checked for odd parity. Setting the switch to the INHIBIT position shall cause ASCII characters with odd parity to be transmitted. No parity check shall be made on received characters when the switch is set to the INHIBIT position.

3.8.7.1.2 STOP BITS Switch. The STOP BITS switch shall provide for transmission of one, or two stop bits.

3.8.7.1.3 STATE Switch. The AN/UGC-74B(V) shall operate in any of three switch selectable states. The first shall be a Receive Only (RO) state which shall operate as a buffered printer. In the Keyboard Send/Receive (KSR) state, a line shall be transmitted as soon as it is composed unless the character-by-character mode is selected. The character-by-character mode shall transmit a character as it is typed. When characters are received in the KSR state, they shall be automatically buffered and printed as they are received. In the Intelligent Communications Terminal (ICT) state, the AN/UGC-74B(V) shall have full or half duplex capability and shall have the capability to be used to compose, edit, transmit, receive, print, and store messages.

3.8.7.1.4 SELF TEST Switch. The SELF TEST switch shall activate the Self Test state.

3.8.7.1.5 MODE Switch. The MODE switch shall determine whether the AN/UGC-74B(V) will transmit and receive ASCII or BAUDOT code.

3.8.7.1.6 FIGURES Switch. The FIGURES switch shall determine which BAUDOT character code is used as a bell (FIGS S or FIGS J).

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3.8.7.1.7 CLOCK (+/-) Switch. The CLOCK (+/-) switch shall provide for selection of the leading or falling clock waveform as the point in time used to sample the receive and transmit data.

3.8.7.1.8 BAUD RATE Switch. The BAUD RATE switch shall be used to determine the rate for data reception and transmission.

3.8.7.1.9 REC MODE Switch. The REC MODE switch shall provide the means to invert (or not invert) the signal sense of received digital low level data. The REC MODE switch shall also provide for selection of 20 MA or 60 MA high level current loop and 48 volt (NIK) operation.

3.8.7.1.10 XMIT MODE Switch. The XMIT MODE switch shall provide the means to invert (or not invert) the signal sense of transmitted digital low level data. The XMIT Mode switch shall also provide for selection of 20 MA or 60 MA high level current loop and 70 microamp (NIK) operation.

3.8.7.1.11 SIGNAL Switch. The SIGNAL switch shall provide for selection of NRZ or Conditioned Diphase data signals for Interface (MIL-STD-188-114).

3.8.7.1.12 CLOCK (INT/KG-30/EXT) Switch. The CLOCK (INT/KG-30/EXT) switch shall provide for selection of the internal clock, KG-30 clock or external clock for determination of bit rate.

3.8.7.2 External Switches. The external switches shall be located on the front panel.

3.8.7.2.1 ABORT Switch. Activation of the momentary ABORT switch shall abort all current processing and return control to the operator.

3.8.7.2.2 Audio Alarm Reset Switch. The AUDIO ALM RESET switch shall reset all current steady-state audible alarm conditions.

3.8.7.2.3 External POWER Switch. The external POWER switch located on the front panel shall apply or disconnect primary power to the terminal.

3.8.7.2.4 Illumination Control. With the POWER switch in the ON position the brightness of the copy lamps and all other indicators shall be controllable. Illumination shall be adjustable by clockwise rotation of the Illum knob from the OFF position to the BRT position.

3.8.7.2.5 TRANSFER (PTT) Switch. This switch shall be wired between pin R of the J1 communication connector and signal ground.

3.8.7.2.6 PARITY RESET Switch. This switch shall allow the operator to reset the parity lamp; turn off the parity alarm, and when held on shall perform lamp tests. This switch shall be used to step through the terminal self test.

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3.8.7.2.7 LINE FEED Switch. Depressing the LINE FEED switch shall advance the paper continuously until the switch is released.

3.8.7.2.8 Audio Control. This control shall be adjustable by clockwise rotation of the AUDIO knob from the OFF position to MAX position to control audible alarm volume.

3.8.8 System States.

3.8.8.1 Off State. This nonoperational state shall be entered whenever the POWER switch is off and the system is powered down. This state shall also be entered whenever prime power is lost and no battery backup is available. In this state, all other states and processes shall be inoperative. Prior to starting up the system, manual switch settings shall be made in this state.

3.8.8.2 Cold Start State. When power is applied, this nonoperational state shall be entered. The state of the system and memory shall be initialized to no messages present in message memory and the following message shall be printed.

'SYSTEM INITIALIZED'

The transfer to the next state (see 3.8.8.3) shall be automatic when initialization is complete.

3.8.8.3 Operation Validation/State Determination State. Upon entry into this nonoperational state, the operation validation test shall be performed without operator intervention. This test shall determine if the system shall be operational without destroying the state of the machine. The system shall sample the internal switches and configurations. Based on this information, the state of the system shall be initialized and stored. The operational state of the system shall be determined by the following rules:

State Switch Position	Keyboard Present	Operational State
ICT	Yes	ICT
ICT	No	RO
KSR	Yes	KSR
KSR	No	RO
RO	Not Applicable	RO

The software system configuration shall be 'UGC-74B.' Any other software system configuration (e.g., UGC-74A-1, UGC-74A-2, or UGC-74C) shall be indicative of a system fault.

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The system shall print out the following information - ICT - ASCII:

SOFTWARE SYSTEM CONFIGURATION = UGC-74B
SWITCH STATE = ICT
OPERATIONAL STATE = ICT
MODE = ASCII
BAUD RATE = 75, 150, 300, 600, or 1200
STOP BITS = 1 or 2
PARITY OPTION = ODD, EVEN, or INHIBIT
DUPLEX MODE = FULL
END OF LINE OPTION = CR CR LF
RECEIVE ENVELOPE OPTION = VZCZC:NNNN
TRANSMIT ENVELOPE OPTION =
VZCZC:NNNN DE DE DE DE DE DE DE DE DE DE DE DE
SPACE OPTION = OFF
CAPITAL LETTER OPTION = ON
LINE LENGTH = 80
LINE FEEDS = 1
END OF MESSAGE LINE FEEDS = 20
FONT = 1 ENG 64 CHAR
LINES PER PAGE = 66
NUMBER OF PRINT COPIES = 1

The system shall print the following for Initialization - KSR, RO - ASCII:

SOFTWARE SYSTEM CONFIGURATION = UGC-74B
SWITCH STATE = ICT, KSR or RO
OPERATIONAL STATE = KSR or RO
MODE = ASCII
BAUD RATE = 75, 150, 300, 600, or 1200
STOP BITS = 1 or 2
PARITY OPTION = ODD, EVEN, or INHIBIT
DUPLEX MODE = FULL
SPACE OPTION = OFF
CAPITAL LETTER OPTION = ON
LINE LENGTH = 80
LINE FEEDS = 1
END OF MESSAGE LINE FEEDS = 20 (RO ONLY)
FONT = 1 ENG 64 CHAR
LINES PER PAGE = 66
NUMBER OF PRINT COPIES = 1

The system shall print the following for Initialization - ICT - Baudot:

SOFTWARE SYSTEM CONFIGURATION. UGC-74B
SWITCH STATE. ICT
OPERATIONAL STATE = ICT
MODE = BAUDOT
BAUD RATE = 45.5, 50, 75, 150, 300, 600 or 1200

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STOP BITS = 1 or 2
BELL OPTION = FIGURES S or FIGURES J
DUPLEX MODE = FULL
END OF LINE OPTION = CR CR LF
RECEIVE Envelope OPTION = VZCZC:NNNN
TRANSMIT ENVELOPE OPTION =
VZCZC:NNNN DE DE DE DE DE DE DE DE DE DE DE DE
SPACE OPTION = ON
LINE LENGTH = 80
LINE FEEDS = 1
END OF MESSAGE LINE FEEDS = 20
FONT = 1 ENG 64 CHAR
LINES PER PAGE = 66
NUMBER OF PRINT COPIES = 1

The system shall print the following for Initialization - KSR, RO - Baudot:

SOFTWARE SYSTEM CONFIGURATION = UGC-74B
SWITCH STATE = ICT, KSR or RO
OPERATIONAL STATE = KSR or RO
MODE = BAUDOT
BAUD RATE = 45.5, 50, 75, 150, 300, 600 or 1200
STOP BITS = 1 or 2
BELL OPTION = FIGURES S or FIGURES J
DUPLEX MODE = FULL
SPACE OPTION = ON
LINE LENGTH = 80
LINE FEEDS = 1
END OF MESSAGE LINE FEEDS = 20 (RO ONLY)
FONT = 1 ENG 64 CHAR
LINES PER PAGE = 66
NUMBER OF PRINT COPIES = 1

If any combination of switches produces an illegal configuration, an error message shall be printed. The system shall stop until power is turned off, the switches are properly set and power is turned on again.

Invalid options shall be as follows:

(1) ASCII at less than 75 baud.
(2) Conditioned diphase at less than 300 Baud or in Baudot mode.
(3) Conditioned diphase with external clock.

When the processing in this state is complete, the control shall be transferred to the appropriate operational state based on the STATE switch position. The receive process shall be initialized and started.

3.8.8.4 ICT State. Upon entry into ICT state the following shall be printed:

HELP COMMAND AVAILABLE

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In this operational state, the operator shall be able to control system operation via keyboard command entry. The system commands shall specify the functions available on the system. The Edit mode shall allow the operator to create and change messages. The Parameter mode shall allow system parameters and options to be specified. If power loss is detected, the system shall go to the Battery Back-Up state. If the SELF TEST switch is activated, the system shall go to Self Test state. If the RESTART command is entered, the system shall go to the Warm Restart state (see paragraph 3.8.8.10).

3.8.8.5 RO State. In this operational state, the machine shall operate as a buffered printer. Data shall be printed as it is received. In the event of receiving data faster than the machine can print, this data shall be buffered in message memory (up to the capacity of the memory). When 15% or less of memory is available for allocation, the MEM Full lamp shall light and a spacing line (break) shall be transmitted. The spacing line (break) shall be removed when memory is less than 75% full. The machine shall continue printing until all received data has been printed. Printing shall be stopped when the paper supply is depleted and a spacing line (break) transmitted. Once the paper supply is replenished, the ABORT switch shall be used to remove the spacing line (break). The RO state shall not interpret communications envelopes, end of line sequences, or redundant carriage returns.

If the keyboard is available, nondefault printer options shall be available in RO by setting them in ICT, then performing RESTART (see 3.8.8.10).

3.8.8.6 KSR State. In this operational state two modes of transmission shall be available. As each line is composed and a CR is entered, it shall be transmitted (Line mode) or as each character is entered, it shall be transmitted (Character mode). The mode of transmission shall be operator selectable. The basic editing functions (DLC, DLL and REV keys) shall be available in the Line Mode only. During message transmission operator input shall not be accepted. Received characters shall be buffered through message memory and then printed as they are received. In the event of receiving data faster than the machine can print, no data shall be lost unless message memory is filled. The KSR state shall not transmit or interpret communication envelopes, end-of-line sequences or redundant carriage returns.

Upon power-up initialization following the system initialization printout, when KSR is selected, the following questions shall be asked:

1) Character or line transmission? (C/L)
2) Local entry print? (Y/N) - character mode only
3) Automatic Line Feed with carriage return? (Y/N) - Line Mode only

All received characters shall be printed. All typed characters shall be printed unless the response to the 'LOCAL ENTRY PRINT' query was N. If local entry print is off, only received characters shall be printed. If the response to 'AUTOMATIC LINE FEED WITH CARRIAGE RETURN? (Y/N)' was N, then there shall be no automatic line feeds associated with a received carriage

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return code. If the response to 'AUTOMATIC LINE FEED WITH CARRIAGE RETURN? Y/N)' was Y, then the printer shall perform a single automatic line feed for each string of received carriage return codes (1 to n) not immediately preceded or following by a line feed code (see 3.8.5.6).

Non-default printer options shall be available in KSR by setting them in ICT then performing RESTART (see 3.8.8.10).

3.8.8.7 Battery Back-up State. When this state is entered, all data in message memory shall be retained and all current processing shall be terminated. Any composed data being used for current processing may be lost. If no back-up battery power is provided, the system shall revert to the OFF state. If back-up battery power is provided, the system shall turn on the BAT lamp and go into a nonoperational state of minimum power consumption. When the system is in Battery Back-Up state, the BAT. lamp shall be the only indicator active. All other indicators shall be off. When prime power is returned, the system shall return to the System Command level.

3.8.8.8 Self Test State. The Self Test State shall be entered unconditionally via depression of the SELF TEST switch, except in Battery Back-Up state. In the Self Test state, a series of tests shall be performed that indicate the system status as operational or nonoperational. The operator must participate in the testing. Upon successful completion of the tests, the word 'READY' shall be printed, followed by a carriage return with a line feed. Upon failure, faults shall be isolated to the optimum replaceable subassembly. All data in message memory shall be lost upon completion of the self test.

3.8.8.9 Factory Self Test State. The Factory Self Test state shall provide the capability to exercise the Self Test functions available in the UGC-74 without the requirement for operator intervention. This software function shall reside in ROM and shall be evoked only when the Factory Self Test jumper is present and power is applied. The particular Self Test sequence exercised shall be dependent on the baud rate switch setting.

The normal operator interventions required for Self Test shall be bypassed and the Self Test (See 3.8.8.8) routines shall be exercised automatically when power is applied to the unit under test.

The Factory Self Test shall be functionally equivalent to the operational Self Test with the exception of the keyboard test and printouts. The keyboard test shall not be exercised during Factory Self Test operation.

3.8.8.10 Warm Restart State. This state shall be entered from the ICT state when the RESTART command is entered. In this state, the operator shall be able to change switch settings and restart the system without turning the power off. When restarted, system parameters set by the PARAMETER subcommands while in the ICT mode shall be retained in the operational state existing following the restart as follows:

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OPERATIONAL STATE

PARAMETER SUBCOMMAND	ICT	KSR	RO
LINE	x	x	x
SETAB	x	x	x
CAPIT	x	x	
REMIND	x		
EOL	x		
LF	x	x	x
SPACE	x 1/	x 1/	x 1/
RENVELOPE	x		
TENVELOPE	x		
EOMLF	x		x (EMER PRT)
FONT	x	x	x
COPIES	x	x	x
DUPLEX	x		
PAGE	x	x	x

1/ If the ASCII/BAUDOT mode switch was in the ASCII position prior to and after restart.

If the state switch is left in "ICT," any messages stored in message memory shall be retained. All message processing (transmission/reception) shall be terminated while in this state.

3.8.9 Message Processing. A line edit input processing capability shall be provided. This capability shall permit three basic editing functions. The functions shall be:

a. Delete last character entered (DLC key)
b. Delete current input line (DLL key)
c. Reprint current input line (REV Key)

These basic edit keys shall be active at all times for all keyboard inputs during the current line (KSR Character mode excepted). The current line shall be defined as any key-strokes up to but not including the carriage return or the number of characters specified in the LINE parameter subcommand, whichever comes first.

3.8.9.1 Message Storage. Messages shall be stored up to the maximum of memory capacity, as specified in 3.8.10.2.

3.8.9.1.1 Memory Full.

3.8.9.1.1.1 Operator Input. When message memory available becomes 1024 characters or less due to an operator input, the line entered by the operator shall not enter message storage, and an information message shall be printed. To alleviate the message memory full condition, the operator shall be able to clear memory by deleting messages via a processing command.

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3.8.9.1.1.2 Incoming Messages.

3.8.9.1.1.2.1 ICT State Incoming Messages. When message memory available becomes 1024 characters or less due to an incoming message, the operator shall be pre-empted, and the printer taken over by the system (keyboard shall be locked out). The current line (data or command) shall be lost and an information message shall be printed, followed by five (5) line feeds. The system shall respond in one of the two following ways depending upon the presence or absence of a received message in memory. When there are received messages in memory, they shall be printed out and subsequently deleted from memory in order of oldest-highest precedence first. These messages shall be separated by five (5) line feeds. As memory blocks of data have been printed, the incoming message shall use these memory blocks for storage. When there are no received messages in message storage, the message being received shall be printed from the beginning. As memory blocks of data have been printed, the incoming message shall use these memory blocks for message storage. This process shall continue until all received messages are printed and deleted from message storage. Then control shall be returned to the operator. No part of the incoming message shall be lost.

3.8.9.1.1.2.2 RO State Incoming Messages. Received characters shall be buffered through message memory as they are received. In the event of receiving data faster than the machine can print, no data shall be lost unless the spacing line (break) memory full warning is ignored by the sender and message memory is filled (see 3.8.8.5).

3.8.9.1.1.2.3 KSR State Incoming Messages. Received characters shall be buffered through message memory as they are received. In the event of receiving data faster than the machine can print, no data shall be lost unless message memory is filled.

3.8.9.2 Message Receipt. The AN/UGC-74B(V) shall be able to receive messages at all times that prime power is on unless in Self Test mode. This shall be an automatic nonoperator selectable function. The equipment shall recognize message envelopes only in the ICT operational state. Received messages shall be assigned a unique local message number between 1 and 9999. The MSG RCVD lamp shall be activated when the message is marked as received. When in half-duplex mode of operation, the receive process shall be inactive during a transmit operation. When in half duplex, all received data shall be discarded up to 1 second after completion of the transmit operation.

3.8.9.2.1 Receive Message Process. The Receive process shall consist of the following sequence of events:

a. If the front of receive envelope is not null, then the incoming characters shall be compared and discarded until the envelope sequence is recognized. The next character shall be considered the first of the message. If the front of receive envelope sequence is null, the first character received shall be considered the first character in the message.

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b. If the system is in ASCII mode, the characters shall be saved in memory. If a parity error is detected and the PARITY switch is in the ODD or EVEN position, the PARITY lamp shall be turned on and a diamond code stored. If the system is in BAUDOT mode, the characters shall be saved in memory. During message receipt, the end of line sequence shall be automatically replaced with a single CR.

c. If the close of receive envelope sequence is not null and no characters are received for 10 seconds, the message shall be assumed lost. The memory used so far shall be released and the receive process shall go back to looking for another message.

If the close of receive envelope character sequence is detected, the message shall be considered as complete and the information shall be provided to the system so it may become a received message. The Receive process shall return to looking for another message.

d. If the close of receive envelope sequence is null and no characters are received for 10 seconds, the message shall be considered as complete. If the front of receive envelope sequence is not null and this sequence detected again, the message shall be considered as complete. The information shall be provided to the system so it may become a received message and the Receive process shall return to looking for another message.

e. If the front of receive envelope is null or defined and found, the message shall be considered open and active until terminated as described below:

FRONT ENVELOPE		CLOSE ENVELOPE		TIME-OUT 10-SEC.	NEXT FRONT	ABORT SWITCH	CURRENT MESSAGE STATUS
NULL	DEFINED AND FOUND	DEFINED	FOUND			ENVELOPE FOUND	
X				X			SAVE
X		X		X			DUMP
X		X	X				SAVE
X						X	DUMP
	X			X			SAVE
	X				X		SAVE
	X	X		X			DUMP
	X	X			X		DUMP
	X	X	X				SAVE
	X					X	DUMP

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f. The Receive process shall be terminated by the system if the ABORT switch is activated. In this case, the message currently being received (if any) shall be discarded, the memory used so far shall be released, and Receive process shall return to looking for another message.

3.8.9.3 Transmit_Message_Process. The Transmit Message process shall be operator initiated through a system command. The Transmit Message process shall consist of first sending out the front of transmit envelope sequence. The message shall be accessed from memory and the characters transmitted sequentially in BAUDOT or ASCII code. In serial bit by bit transmission, the coded character shall be transmitted low order, least significant bit (LSB) (b1) first. When a CR is encountered in the message, it shall be replaced by the end of line sequence. The close of transmit envelope shall be appended to the message. The system shall now recognize the message as having been transmitted and the process terminated. The transmit process can be terminated before completion by the system if the ABORT switch is activated. The transmitted message shall be automatically printed, unless the HLT key is activated. The HLT key shall terminate printing, but not transmission and shall return control to the operator. When in half-duplex mode of operation, the transmit operation shall not be activated if the receive message process has recognized and is saving a message.

3.8.9.4 System_Commands. All commands shall be leading and trailing space insensitive and truncated forms of commands shall be allowed. Either capital or lower case characters shall be accepted. The system shall prompt with a System Command prompt sequence (@ symbol). An error message shall be printed if the command is not recognized.

3.8.9.4.1 PARAMETER_Command. The system shall enter the PARAMETER mode and prompt with the PARAMETER prompt sequence (- symbol). PARAMETER subcommands shall be the only valid commands or subcommands until the EXIT subcommand is executed and local control shall be returned to the operator.

3.8.9.4.1.1 LINE_Subcommand. This subcommand shall allow the operator to specify the length of the text line for message composition and editing. If no value is specified, then 80 shall be used. The value specified shall be between 40 and 132 inclusive or an error message shall be printed and the subcommand completed. On power-up initialization, the line length shall be set at 80.

3.8.9.4.1.2 SETAB_Subcommand. This subcommand shall allow the operator to specify print positions at which tabs shall be set. This subcommand shall take a list of one or more tab values. If any of the tab values are greater than 132 or the current line length, an error message shall be printed and only the tab values that precede the invalid entry in the subcommand shall be changed. If a specified value is already a Set Tab, then no change shall be made. If the combination of previously set and currently being set tabs results in more than 12 tabs, the entries preceding the entry which causes

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the setting of the thirteenth tab shall be accepted, then the subcommand shall be considered completed and an error message shall be printed. The tabs shall be displayed after the subcommand.

3.8.9.4.1.3 CLRTAB Subcommand. This subcommand shall allow the operator to clear tabs in the system. If '*' is specified, all tabs shall be removed. On power-up initialization, all tabs shall be cleared. This subcommand shall take a list of one or more Clear Tab values. If a specified tab value is not set, no action shall be taken. The remaining tabs set shall be displayed after this command. If none are set, an information message shall be printed. Only the CLRTAB entries preceding an invalid entry shall be accepted.

3.8.9.4.1.4 DISPLAY Subcommand. All tab settings of the system shall be displayed in ascending order. If none are set, an information message shall be printed.

3.8.9.4.1.5 CAPIT Subcommand. This subcommand (ASCII mode only) shall allow the operator to set or reset the capitalize option. This option shall be always set ON in the BAUDOT mode. If this option is set, all lower case letters (a through z) input from the keyboard
shall be converted to capitals and stored as such in memory. If this option is not set, no change shall be made. On power-up initialization, this option shall be set (ON).

3.8.9.4.1.6 REMIND Subcommand. This subcommand shall allow the operator to set or reset the Remind option. If the option is set, the following reminder shall be printed whenever a TRANSMIT or TTY (in ICT mode) command is given:

KEY TRANSMITTER AND ENTER CR

On power-up initialization, this option shall be reset (OFF).

3.8.9.4.1.7 EOL Subcommand. This subcommand shall allow the operator to input as ASCII characters, the end of the line (EOL) sequence for transmission. During transmission of a message, when a CR is found in the buffer, the EOL character sequence shall be sent in place of the CR. The EOL character sequence shall consist of zero to eight characters. The ASCII carriage return character shall be input as 'CR' (no space between characters). The ASCII line feed character shall be input as 'LF' no space between characters). If more than eight characters are specified or if the EOL sequence is improperly specified, an error message shall be printed. On power-up initialization, the EOL sequence shall be:

Characters: CR CR LF

If transmission is in process when the EOL Subcommand is recognized by the system, an error message shall be printed, no change shall be made and the subcommand shall be complete.

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3.8.9.4.1.8 LF Subcommand. This subcommand shall allow the operator to specify either one or two line feeds following a CR when printing in the ICT state. The operator must specify either 1 or 2 or an error message shall be printed and the value shall not be changed. On power-up initialization, the LF parameter shall be set at 1.

3.8.9.4.1.9 SPACE Subcommand. This subcommand (ASCII mode only) shall allow the operator to set the spaces option. This option shall be always set ON in the BAUDOT mode. When the "SPACE" option is either OFF or ON, and there are no more tabs between the present print position and the EOL position, a CR (carriage return) shall be inserted into memory whenever a tab character is recognized as input from the keyboard. When the "SPACE" option is ON, and there is a tab between the present print position and the EOL position, space codes shall be inserted into memory (up to but not including the tab position) whenever a tab character is recognized as input from the keyboard. When the "SPACE" option is OFF, and there is a tab betweenthe present print position and the EOL position, the tab code shall be inserted into memory whenever a tab character is recognized as input from the keyboard. If "ON" or "OFF" is not inserted in the parameter field, an error message shall be printed and no change made. On power-up initialization, the space option shall be OFF when in the ASCII mode.

3.8.9.4.1.10 PAPER Subcommand. This subcommand shall turn off the printer so that the paper and/or ribbon may be changed without loss of memory integrity. No further inputs shall be printed until the printer is restarted. Any keys may be activated but they shall be ignored until the specific sequence GO (CR) is entered. The LINE FEED switch shall be active to assist in loading the paper. After GO (CR) is entered, the system shall respond by turning the printer on and returning to the previous command or subcommand level. Receive data shall be accepted and stored until memory is full. Messages shall be lost if the memory becomes full while the printer is inactive. If the paper runs out, the system shall automatically initiate this subcommand and stop the printer.

3.8.9.4.1.11 RENVELOPE Subcommand. This subcommand shall allow the operator to input the envelope which shall be removed from a message on reception by the system. This sequence shall never appear in message storage and shall be used for communications purposes only. The envelope shall consist of zero to eight characters of start sequence and zero to sixteen characters of stop sequence. The carriage return character shall be input as "CR" (no spaces between characters). The line feed character shall be input as LF no spaces between characters). The Delete character (DEL) shall be input as "DE" (no spaces between characters). If more than the required characters are specified, or improperly specified, an error message shall be printed. If a message is currently being received when the RENVELOPE Subcommand is recognized by the system, an error message shall be printed, no change shall be made, and the subcommand shall be complete. On power-up initialization, the envelope shall be:

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FRONT: V Z C Z C

END: N N N N

3.8.9.4.1.12 TENVELOPE Subcommand. This subcommand shall allow the operator to input the envelope which shall be put around a message for transmission by the system. This sequence shall never appear in the message storage and shall be used for communications purposes only. The envelope shall consist of zero to eight characters of start sequence and zero to sixteen characters of stop sequence. The carriage return character shall be input as CR (no spaces between characters). The line feed character shall be input as LF (nospaces between characters). The Delete (DEL) character shall be input as "DE" (no spaces between characters). If more than the required characters are specified, or improperly specified, an error message shall be printed. If a message is currently being transmitted when the TENVELOPE Subcommand is recognized by the system, an error message shall be printed, no change shall be made, and the subcommand shall be complete. On power-up initialization, the envelope shall be:

FRONT: V Z C Z C

END: N N N N DEL (12X)

3.8.9.4.1.13 EOMLF Subcommand. The End of Message Line Feed (EOML) subcommand shall allow the operator to set the number of line feeds which shall be automatically inserted by the system following message printout. The number of line feeds shall be from one (1) to twenty (20). If more or less are specified, or improperly specified, an error message shall be printed. On power-up initialization, the EOMLF shall be set to twenty (20). During emergency printing (see 3.8.9.1.1.2.1), the EOMLF shall temporarily be five (5). When set for the KSR Operational State, the system shall not automatically insert any line feeds following message printout.

3.8.9.4.1.14 FONT Subcommand. This subcommand shall allow the operator to select the character font set used by the printer. If a selection number other than those included in the available font set is specified, an error message shall be printed. The selected font set name (up to 12 characters in length) shall be displayed by the QUERY Command. On power-up initialization, font set number 1 shall be active (see 3.8.5.7.3.3) and the font set name (up to 12 characters in length) shall be displayed in the initialization printout.

3.8.9.4.1.15 COPIES Subcommand. This subcommand shall allow the operator to select the number of parts (copies) in the multi-part paper being used. This value shall be used to adjust the print-pin pressure of the DMP printhead, applying more pressure for increased number of copies. The value specified shall be from one (1) to three (3). If more or less are specified or improperly specified, an error message shall be printed. On power-up initialization, the COPIES value shall be set to one (1).

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3.8.9.4.1.16 DUPLEX Subcommand. This subcommand shall allow the operator to select HALF or FULL duplex mode of operation. The inputs shall be "H" or "F". If the input is improperly specified, an error message shall be printed. On power-up initialization, the DUPLEX value shall be "FULL."

3.8.9.4.1.1 HEADERS Subcommand. This subcommand shall allow the operator to print the current header information for the JANAP-128, ACP-127 NATO Supplement 3 and ACP-127 USA Supplement 1 headers. If the headers have not been created, an information message shall be printed.

3.8.9.4.1.18 HELP Subcommand. This subcommand shall allow the operator to print a list of all the PARAMETER subcommands. The following list contains the subcommand format and a brief description of the subcommands functionality.

PARAMETER PROMPT CHARACTER IS: -

DLC KEY DELETES THE LAST CHARACTER INPUT
DLL KEY DELETES THE CURRENT INPUT LINE
REV KEY REVIEWS (REPRINTS) THE CURRENT INPUT LINE

PARAMETER SUBCOMMAND	COMMAND FORMAT(S)	DESCRIPTION
DUPLEX	DU F	SELECTS FULL DUPLEX OPERATION ON COMM PORT.
	DU H	SELECTS HALF DUPLEX OPERATION ON COMM PORT.
REMIND	REMI ON	SETS TRANSMISSION REMINDER MESSAGE.
	REMI OFF	RESETS TRANSMISSION REMINDER MESSAGE.
EOL	EOL AAA	SETS END OF LINE SEQUENCE FOR TRANSMISSION TO ASCII VALUES SPECIFIED.
EOMLF	EOM XX	SETS NUMBER OF END OF MESSAGE LINE FEEDS (XX=1 TO 20).
TENVELOPE	TENV AAAA:AAAA	SETS ENVELOPE FOR MESSAGE TRANSMISSION, UP TO 8 ASCII START ENVELOPE VALUES, THEN UP TO 16 ASCII STOP ENVELOPE VALUES.
RENVELOPE	RENV AAAA:AAAA	SETS ENVELOPE FOR MESSAGE RECEPTION, UP TO 8 ASCII START ENVELOPE VALUES, THEN UP TO 16 ASCII STOP ENVELOPE VALUES.

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SPACE	SP ON	SETS SPACE OPTION.
	SP OFF	RESETS SPACE OPTION.
CAPIT	CAP ON	SETS CAPITALIZATION OPTI ON.
	CAP OFF	RESETS CAPITALIZATION OPTION.
LINE	LIN	SETS LINE LENGTH TO 80 CHARACTERS.
	LIN XX	SETS LINE LENGTH (XXX = 40 TO 132).
LF	LF 1	SETS UP FOR 1 LINE FEED AFTER CARRIAGE RETURN.
	LF 2	SETS UP FOR 2 LINE FEEDS AFTER CARRIAGE RETURN.
FONT	FO X	SELECTS CHARACTER FONT SET (X = FONT NUMBER).
PAGE	PAG XX	SELECTS NUMBER OF LINES PER PAGE (XX=1 TO 66).
COPIES	CO X	SELECTS NUMBER OF PRINTED COPIES (X=1 TO 3).
PAPER	PAP	TURNS PRINTER OFF TO ALLOW PAPER/ RIBBON CHANGES.
GO	GO	TURNS PRINTER ON.
SETAB	SETA XX	SETS TAB MARK TO VALUE SPECIFIED.
	SETA XX XX XX, ECT.	SETS UP TO 12 TAB MARKS TO VALUES SPECIFIED (XX = 1 TO CURRENT LINE LENGTH).
CLRTAB	CLR XX	CLEARS TAB MARK SPECIFIED (XX = 1 TO CURRENT LINE LENGTH).
	CLR XX XX XX, ETC.	CLEARS TAB MARKS SPECIFIED (XX = 1 TO CURRENT LINE LENGTH).
	CLR *	CLEARS ALL TAB MARKS.
DISPLAY	DI	DISPLAYS ALL TAB MARK SETTINGS.
HEADERS	HEADERS	PRINTS CURRENT PRESTORED JANAP-128, ACP-127 US SUPP 1, ACP-127 NATO SUPP 3 HEADERS.
SETUP	SETU 128	PROMPTS OPERATOR THROUGH CREATION/ MODIFICATION OF JANAP-128 HEADER FOR STORAGE.

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SETU 1271	PROMPTS OPERATOR THROUGH CREATION/ MODIFICATION OF ACP-127 USA SUPP 1 HEADER FOR STORAGE.
SETU 1273	PROMPTS OPERATOR TROUGH CREATION/ MODIFICATION OF ACP-127 NATO SUPP 3 HEADER FOR STORAGE.

HALT KEY WILL ABORT ANY SETUP ACTIVITY. ENTER NULL (CR) TO RETAIN PREVIOUS INPUT.

HELP	HEL	PRINTS LIST OF PARAMETER SUBCOMMANDS.
EXIT	EX	RETURNS TO SYSTEM COMMAND LEVEL (@).

NOTE: In EOL and envelope commands, the ASCII carriage return character is input as 'CR' (no spaces between characters), the ASCII line feed character is input as 'LF' (with no spaces between characters), and the ASCII DELETE character is input as 'DE' (no spaces between characters).

3.8.9.4.1.19 PAGE Subcommand. This subcommand shall allow the operator to select the number of lines to be printed prior to the automatic insertion of enough line feeds to avoid the fanfold paper perforation. A fixed length (11") page shall be required. The number of the lines to be printed shall be from 1 to 66. On acceptance of the PAGE subcommand, the operator shall be prompted to align the paper. Selection of 66 lines shall mean no automatic insertion of line feeds. If the input is improperly specified, an error message shall be printed. On power up initialization, the PAGE value shall be 66.

3.8.9.4.1.20 SETUP Subcommand. This subcommand shall allow the operator to create JANAP 128, ACP-127 NATO Supp. 3, and ACP-127 USA Supp. 1 headers. The selection of which header to create shall be specified by the operator as an operand in the SETUP subcommand. If improperly specified, an error message shall be printed. Each header shall be created through a series of prompts for header information. The prompting shall consist of printing a series of questions followed by a carriage return and line feed and accepting a series of inputs from the operator. The number and type of characters to be input are given in parenthesis at the end of each question. If the input does not conform in number or type, an error message shall be printed and the question repeated. The system shall retain and display previous header information (if any). If no change to the previous answer is desired, a NULL line (CR) shall be entered. Whenever data is being input, leading and trailing spaces shall be allowed. The correction keys (DLC, DLL, REV) and the CAPIT subcommand shall be active. All numbers shall be entered with or without leading zeros as required. If the HLT key is activated during prompting, the prompting sequence shall be terminated and control shall be returned to the PARAMETER mode with no changes in the message.

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The text of each of the questions (for all three header formats) shall be as follows:

1. ENTER CHANNEL IDENTIFIER (3A)

2. ENTER ORIGINATOR ROUTING INDICATOR/PLAIN LANGUAGE ADDRESS (RI/PLA)

where the Routing Indicator/Plain Language Address (RI/PLA) shall be defined as:

RI/PLA CR

RI shall be 4 to 7 alpha characters and PLA shall be any amount of text necessary. Each line of PLA text must be terminated by an operator-entered carriage return. If a line of PLA text exceeds the current line length (69 characters or less) or a backspace (BS) is entered, an information message shall be printed, the RI/PLA shallnot be saved, and question 2 shall be repeated. The PLA text, and thus question 2, shall be terminated by entering a null line (CR). A total of 8 lines of RI/PLA may be entered. This paragraph applies to Originator, Action, Info, and exempt addresses.

3. ENTER MESSAGE PRECEDENCE (1A)

4. ENTER ORIGINATOR AND DESTINATION LANGUAGE AND MEDIA FORMAT (2A)

5. ENTER CLASSIFICATION (1A)

6. ENTER CONTENT INDICATOR/COMMUNICATION ACTION IDENTIFIER (4AN)

7. ENTER STATION SERIAL NUMBER (4N)

8. ENTER JULIAN DATE (3N)

9. ENTER TIME FILED (4N)

10. ENTER ACTION ROUTING INDICATOR/PLAIN LANGUAGE ADDRESS (RI/PLA)

Refer to question 2. Note multiple RI/PLA may be entered. Question 10 shall be repeated until a NULL RI/PLA is entered.

11. ENTER INFORMATION ROUTING INDICATOR/PLAIN LANGUAGE ADDRESS (RI/PLA)

Refer to question 10.

12. ENTER TIME OF DAY (ZULU TIME) (4N)

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13. ENTER DAY, MONTH AND YEAR (2N 3A 2N)

14. ENTER CHANNEL SEQUENCE NUMBER (3N)

For ACP-127 formats, the following additional questions shall be asked:

15. ENTER EXEMPT ROUTING INDICATOR/PLAIN LANGUAGE ADDRESS (RI/PLA)

where RI/PLA shall be as defined for question 10.

16. ENTER SECURITY PROSIGN (2A or NULL)

Two prosigns shall be recognized - UU and HH. A null input will be accepted in which case no security prosign shall be included in format line 1.

17. ENTER SECOND PRECEDENCE (1A or NULL)

For purposes of constructing the header, the answer to each question shall be considered a variable.

The variables and their corresponding number and kind of value are as follows: (Note that line one above corresponds to variable one (V1), line two to variable two (V2), etc.).

V1 - 3 alpha
V2 - RI/PLA
V3 - 1 alpha must be:
- Y - emergency command
- Z - flash
- O - immediate
- P - priority
- R - routine

V4 - 2 alpha
V5 - 1 alpha must be:
- U - unclas
- R - restricted
- E - unclas EFTO
- C - confidential
- S - secret
- T - top secret

V6 - 4 alpha/numeric
V7 - 4 numeric
V8 - 3 numeric
V9 - 4 numeric
V10 - RI/PLA (may have multiple inputs)
V11 - RI/PLA (may have multiple inputs)
V12 - 4 numeric
V13A - 2 numeric V13B - 3 alpha V13C - 2 numeric
V14 - 3 numeric
V15 - RI/PLA (may have multiple inputs)
V16 - 2 alpha V17 - 1 alpha

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The JANAP 128 PLAINDRESS header and trailer shall be formed as follows:

The header shall be formed as follows and inserted before the body of the message. Blanks are specified as b1 where one and only one blank is inserted.

V1 V14 CR

V3 V4 V5 V6 b1 (RI from V2) V7 b1 V8 V9 - (repeat V5 four times) -- (RIs from 10 and V11 separate d by spaces or CR as required). CR

ZN (if V5 = U then R else Y) b1 (repeat V5 five times) CR

V3b1V13AV12Zb1V13Bb1V13C CR

FMb1 V2 (PLA only) CR

TO b1 (V10 each entry shall be separated by a CR) CR

INFO b1 (V11 each entry shall be separated by a CR) CR

BT CR

(Based on V5, the classification shall be entered here and written out as follows:

U - UNCLAS CR
E - UNCLASSb1Eb1Fb1Tb10 CR
C - Cb10b1Nb1Fb1Ib1Db1Eb1Nb1Tb1Ib1Ab1L CR
S - Sb1Eb1Cb1Rb1Eb1 T CR
T - Tb10b1Pb1Sb1Eb1Cb1Rb1Eb1Tb1 CR

The trailer of the message shall be as follows and appended after the body of the message.

BT CR
V7 CR LF LF LF LF LF LF LF

The ACP-127 header shall be formed as follows and inserted before the body of the message.

V1V14b1b1b1b1b1V16 CR

V3V3b1(RIs from V10A and V11A separated by spaces or CR as required) CR

DEb1V2Ab1#V7b1V8V9 CR (Omit # if NATO Supp. 3)

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ZN (R OR Y) V5V5V5V5V5 CR (R if V5=U, Y otherwise)

V3b1V17b1V13AV12ZV13B CR (If V17 is null, omit V17b1.)

FMb1V2B CR

TOb1 (V10 Each entry shall be separated by a CR) CR

INFOb1 (V11 Each entry shall be separated by a CR) CR
XMTb1 V15 CR

BT CR

Security Word based on V5 shall be entered here and written out as follows:

USA Supp 1: V5	SECURITY WORDS
U	UNCLAS CR
E	UNCLASS E F T O CR
C	C O N F I D E N T I A L CR
S	S E C R E T CR
T	T O P S E C R E T CR

NATO Supp 3: V5	SECURITY WORDS
U	NATO UNCLAS CR
R	N A T O R E S T R I C T E D CR
C	N A T O C O N F I D E N T I A L CR
S	N A T O S E C R E T CR
T	N A T O T O P S E C R E T CR

The trailer of the message shall be as follows and shall be appended after the body of the message:

BT CR
* V7 CR
LF LF LF LF LF LF LF

If the system is operating in the BAUDOT mode, LTRS and FIGS codes shall be inserted into the message as appropriate for transmission only. In both BAUDOT and ASCII modes, end of line sequences (EOL) shall be substituted for CR as the message is transmitted. A * sign in BAUDOT is FIGS H (see character table, paragraph 3.7.5.3).

3.8.9.4.1.21 EXIT Subcommand. This subcommand shall cause the system to go to the System Command level from the PARAMETER mode. The system shall respond with the System Command prompt sequence.

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3.8.9.4.1.22 GO Subcommand. This subcommand shall allow the operator to restart the motor that operates the printer and ribbon feed. When this subcommand is executed, the system shall turn the printer on. If the PAPER subcommand was initiated by the operator, the system shall return to the PARAMETER command and print the Parameter prompt sequence. If the PAPER subcommand was automatically initiated by the system, the system shall return to the command level it was in before paper depletion and print the appropriate prompt sequence.

3.8.9.4.2 EDIT Command. The system shall allow the operator to edit a selected operator entered message at any time. The system shall enter the Edit mode and prompt with the Edit prompt sequence. If the operator does not enter a particular message number, the system shall assign one. The system shall assign these numbers sequentially, starting with 1 and ending with 9999. There shall never exist two messages with identical message numbers in the system. When 9999 is assigned to a message, the next number assigned will be the first free number starting back at 1. When the operator assigns a particular message number, the system shall check to see if this message exists and is a locally composed message in the system. If the requirements are met, an appropriate message stating the message number, followed by the EDIT prompt sequence shall be printed. If the requirements are not met, an error message shall be printed, followed by the System Command prompt sequence. Upon entering the EDIT mode, the line pointer shall point to the first line of the message and the EDIT subcommands shall be the only valid commands or subcommands until the EXIT subcommand is executed.

3.8.9.4.2.1 TOP Subcommand. The TOP subcommand shall set the line pointer to the first line of the current message being edited and that line shall be printed. If the message is empty an information message shall be printed.

3.8.9.4.2.2 BOTTOM Subcommand. The BOTTOM subcommand shall set the line pointer to the last line of the current message being edited and that line shall be printed. If the message is empty an information message shall be printed.

3.8.9.4.2.3 UP Subcommand. The UP subcommand shall move the line pointer the number of lines, as indicated by the operator, towards the beginning of the current message being edited and shall print the line at the new line pointer position. If the number of lines is not entered it shall be assumed to be 1. If an "*" is entered, the effect shall be the same as executing the TOP subcommand. If the number of lines entered is greater than 99 or not numeric, an error message shall be printed. If the number is greater than the number of lines from the current line pointer position to the first line of the message, the line pointer shall be moved to the first line and that line shall be printed. If the current message being edited is null, an information message shall be printed.

3.8.9.4.2.4 DOWN Subcommand. The DOWN subcommand shall move the line pointer the number of lines, as indicated by the operator, towards the end of the

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current message being edited and shall print the line at the new line pointer position. If the number of lines is not entered it shall be assumed to be 1. If an '*' is entered, the effect shall be the same as executing the BOTTOM subcommand. If the number entered is greater than 99 or not numeric, an error message shall be printed. If the number is greater than the number of lines from the current line pointer position to the last line of the message, the line pointer shall be set at the last line of the message and that line shall be printed. If the current message being edited is null, an information message shall be printed.

3.8.9.4.2.5 BEFORE Subcommand. The BEFORE subcommand shall allow the operator to insert a block of text before the current line. The process shall normally be terminated by depressing the HLT key. The line pointer shall remain at the original line. The BEFORE subcommand shall provide no prompt line. Lines shall be entered until the desired block has been entered. Depressing the HLT key shall then indicate that the desired text has been created and the subcommand shall be terminated. The actual insertion into the message shall be accomplished line by line. Each input line of text shall be entered into the message when a CR is entered or when the line reaches the current line length. In the latter case, a CR shall be automatically appended to the line. A nonempty line terminated by depressing the HLT key shall append a CR to the line, insert the line into the message, and terminate the BEFORE subcommand. Depressing the HLT key with an empty line shall cause the subcommand to be terminated. If there is not sufficient memory available to insert the current line, an error message shall be printed and the current input line shall not be inserted into the message, and the subcommand shall be automatically terminated. Within the system, blank lines, i.e., lines which consist of only a CR, shall always be inserted into the memory as a SPACE code followed by a CR code.

3.8.9.4.2.6 INSERT Subcommand. The INSERT subcommand shall allow the operator to insert a block of text after the current line. The command shall normally be terminated by depressing the HLT key. The line pointer shall be reset to point to the last inserted line of text. The INSERT subcommand shall provide no prompt line. Lines shall be inserted until the desired block of text has been entered. Depressing the HLT key shall indicate that the desired block of text has been inserted and the command shall be terminated. Each line of text shall be inserted into the message either when a CR is entered or when the line reaches the current line length. In the latter case, a CR shall be appended to the line and the line inserted after the current line. The line pointer shall be reset to point to the last line inserted. A nonempty line terminated with HLT shall append a CR to the line, shall insert the line into the message, shall set the line pointer to the inserted line, and shall terminate the command. Depressing the HLT key with an empty line shall cause the command to be terminated. If there is insufficient memory available to insert the current input line, an error message shall be printed; the line causing the condition is not inserted into the message; and the insert process shall be terminated. Within the system, lines which consist of only CRs shall be inserted into memory as a SPACE code followed by a CR code.

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3.8.9.4.2.7 DELETE Subcommand. This subcommand shall allow the operator to delete a specified number of lines from the currently edited message starting at the current line pointer. The number of lines to be deleted shall be specified by the operator. One line shall be assumed if nothing is specified. If an '*' is specified or the number of lines specified is greater than the number of lines remaining in the message, the remainder of the message shall be deleted. At the completion of a delete, the line pointer shall be set at either the next nondeleted line of the message or the new last line of the message. An information message shall be printed if the result of the deletion is an empty message. If the message is not null after the deletion, the line at the current line pointer shall be printed. If the number of lines specified is greater than 99 or not numeric, an error message shall be printed.

3.8.9.4.2.8 FIND Subcommand. This subcommand shall allow the operator to input a phrase of up to 20 characters to be found in the message text. The system shall search the subsequent lines as required to find the first occurrence of the phrase from this point in the message. The line pointer shall be set to the line in which the phrase is found. The line shall be printed and the subcommand terminated. Input of the phrase shall be such that any printable character may be used by the operator to delineate the characters of the phrase. The phrase may not contain a CR or the delimiter character and must contain at least one character or an error message shall be printed. If the phrase is not found, the line pointer shall not be changed, an error message shall be printed, and the subcommand terminated.

3.8.9.4.2.9 LIST Subcommand. This subcommand shall allow the operator to print a specified number of lines of the current message being edited, beginning at the current line pointer. The line pointer shall not be changed. If the number of lines to be printed is not entered, it shall be assumed to be 1. If an '*' is entered, the remainder of the message shall be printed beginning at the current line. If the number of lines is greater than 99 or is not numeric, an information message shall be printed. If the number of lines is greater than the number of lines from the current line to the end of the message, the remainder of the message shall be printed. Depressing the HLT key during printing shall cause the printing to stop and an information message shall be printed. If the message is empty, an error message shall be printed.

3.8.9.4.2.10 REPLACE Subcommand. The REPLACE subcommand shall allow the operator to find a phrase of 1 to 20 characters in the message text and to replace the phrase with a second phrase of 0 to 40 characters. The system shall search the current line and subsequent lines as required to find an occurrence of the first phrase. The line pointer shall be set to this line. The characters in the first phrase shall be replaced with the characters of the second phrase. If the replacement would cause a line length to be exceeded, a CR shall be inserted to maintain the designated length, thus creating a new line. The line pointer shall be set to the new line. The new line or lines shall be printed. If the first phrase cannot be found, an error

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message shall be printed and the subcommand terminated. Input of the phrase shall be such that any printable character may be used by the operator to delineate and separate the two phrases. Neither phrase may contain a CR or the delimiter character and the first phrase must contain at least one character or an error message shall be printed. If this subcommand causes an entire line to be deleted, i .e., all characters, the system shall delete the CR and the line pointer shall be set as if the lines were deleted by the DELETE subcommand. Printing shall also be as if the DELETE subcommand had been executed.

3.8.9.4.2.11 HEADER Subcommand. This Subcommand shall allow the operator to insert a previously defined JANAP-128, ACP-127 US Supp. 1 or ACP-127 NATO Supp. 3 header. The message header is composed of header information and trailer information. The header information shall be inserted prior to the first line of the current message. The trailer information shall be inserted following the last line of the current message. The selection of which header to insert shall be by operator input of '128', '1271' or '1273. If improperly specified, an error message shall be printed.

3.8.9.4.2.12 APPEND Subcommand. This subcommand shall allow the operator to append the contents of a designated message to the end of the current message. If the message number of the message to be appended is not a message in the system, or is the presently edited message an error message shall be printed and the command terminated. The pointer shall be set to the last line of the appended material and that line shall be printed. If the message to be appended contains characters with parity error, these error characters shall be converted to spaces as the append is carried out. If the APPEND Command causes memory to become full, an error message shall be printed ant the command shall not be carried out.

3.8.9.4.2.13 HELP Subcommand. This subcommand shall allow the operator to print a list of all the EDIT subcommands. The list shall be as follows:

THE EDIT PROMPT CHARACTER IS: *

DLC KEY DELETES THE LAST CHARACTER INPUT.
DLL KEY DELETES THE CURRENT INPUT LINE.
REV KEY REVIEWS (REPRINTS) THE CURRENT INPUT LINE.

EDIT SUBCOMMAND	COMMAND FORMATS	DESCRIPTION
TOP	TO	SETS LINE POINTER TO FIRST LINE OF THE MESSAGE.
BOTTOM	BO	SETS LINE POINTER TO LAST LINE IN THE MESSAGE.
UP	U	SETS LINE POINTER UP 1 LINE.
	U XX	SETS LINE POINTER UP REQUESTED NUMBER OF LINES BUT NOT PAST TOP OF MESSAGE.
	U *	SETS LINE POINTER TO TOP OF MESSAGE.

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DOWN	DO	SETS LINE POINTER DOWN 1 LINE.
	DO XX	SETS LINE POINTER DOWN REQUESTED NUMBER OF LINES, BUT NOT PAST BOTTOM OF MESSAGE.
	DO *	SETS LINE POINTER TO BOTTOM OF MESSAGE.
BEFORE	BE	INSERTS THE OPERATOR ENTERED BLOCK OF TEXT BEFORE THE CURRENT LINE AS DEFINED BY LINE POINTER. THE HLT KEY WILL TERMINATE THIS ACTIVITY.
INSERT	I	INSERTS THE OPERATOR ENTERED BLOCK OF TEXT AFTER THE CURRENT LINE AS DEFINED BY LINE POINTER. THE HLT KEY WILL TERMINATE THIS ACTIVITY.
DELETE	DE	DELETES CURRENT LINE.
	DE XX	DELETES THE SPECIFIED NUMBER OF LINES FROM CURRENT LINE POINTER DOWN, (INCLUDES CURRENT LINE).
	DE *	DELETES REMAINDER OF LINES (PHRASE).
FIND	FIDTTTTTD	SPECIFIES A GIVEN PIECE OF TEXT THAT IS TO BE LOCATED. (TTTTT) CAN BE UP TO 20 CHARACTERS WITH A CHARACTER DELIMITER (D) AT THE BEGINNING AND END.
LIST	LIS	LIST CURRENT LINE.
	LIS XX	LISTS REQUESTED NUMBER OF LINES, BEGINNING AT CURRENT LINE POINTER.
	LIS *	LISTS REMAINDER OF LINES. THE HLT KEY WILL TERMINATE ANY PRINT ACTIVITY.
REPLACE	REP DP1DP2D	WHERE (P1) INDICATES PHRASE TO BE REPLACED (UP TO 20 CHARACTERS), (P2) INDICATES REPLACEMENT PHRASE (UP TO 40 CHARACTERS) AND (D) INDICATES DELIMITER.
HEADER	HEADER 128	PLACES PREDEFINED JANAP-128 HEADER AND TRAILER AROUND MESSAGE.
	HEADER 1271	PLACES PREDEFINED ACP-127 USA SUPP. 1 HEADER AND TRAILER AROUND MESSAGE.
	HEADER 1273	PLACES PREDEFINED ACP-127 NATO SUPP. 3 HEADER AND TRAILER AROUND MESSAGE.
APPEND	AP XXXX	ADDS A COPY OF THE DESIGNATED MESSAGE TO THE END OF THE CURRENTLY EDITED MESSAGE.
EXIT	EX	RETURNS TO SYSTEM COMMAND LEVEL (@).
HELP	HEL	PRINTS LIST OF EDIT SUBCOMMANDS.

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NOTE: XXXX INDICATES THE sPECIFIC MESSAGE NUMBER. THE RANGE OF NUMBERS IS FROM 1 TO 9999.

NOTE: XX INDICATES NUMBERS OF LINES. THE RANGE OF NUMBERS IS FROM 1 TO 99.

3.8.9.4.2.14 EXIT Subcommand. This subcommand shall cause the system to go to the System Command level from the EDIT mode. The system shall respond with the System Command prompt sequence.

3.8.9.4.3 PRINT Command. This command shall allow the operator to specify the number of a message to be printed. If the number specified is not a message in the system, a message shall be printed and the command is complete.

If a valid message number is entered, the specified message shall be printed. The system shall execute the number of line feeds specified by the EOMLF parameter, print the message, mark the message as printed, and again execute the number of line feeds specified by the EOMLF parameter. The specified message shall not be deleted from the message memory when printing is completed.If printing is stopped, an information message shall be printed and the command is complete. The specified message shall be marked as printed only if it is completely printed.

If an "*" is entered, "ARE YOU SURE? (Y/N)" shall be printed and the system shall wait for the operator response.If the response is "N", no action shall be taken and the command shall be compete.If the response is "Y", all received messages shall be printed in order, first in-first out by precedence.The system shall look up the number of the oldest, highest precedence message, execute the number of line feeds specified by the EOMLF parameter, print the message and delete the message from message memory. This action shall be repeated until no received messages remain in the message memory, then the system shall execute the number of line feeds specified by the EOMLF parameter and the command shall be complete.

3.8.9.4.4 TRANSMIT Command. This command shall allow the operator to specify the number of a message to be transmitted. If the number specified is not a composed message in message memory, an error message shall be printed and the command shall be complete. If in the half-duplex mode of operation and the receive message process is saving a message, an information message shall be printed and the command shall be complete. If the number specified is the number of a composed message, the specified message shall be transmitted and printed at the some time. In the BAUDOT mode, the LTR's code shall be the first character transmitted in any message.The specified message shall be marked as being transmitted, and transmission shall be initiated. The system shall execute the number of LFs specified by the EOMLF parameter, print the message, mark the message as printed, execute the number of LFs specified by the EOMLF parameter and the command shall be complete. If a previous transmission has not been completed, an error message shall be printed and the command terminated. If printing is stopped by the HLT key, an information

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message shall be printed, control shall be returned to the operator and transmission shall continue until completion. If an empty message is specified an error message shall be printed and no transmission shall be done. If the 'TRANSMIT XXXX WITH NNN' option is used, the NNN shall replace characters 4, 5 and 6 in the first line of the message. If characters 4, 5 and 6 are not numeric or do not exist, an error message shall be printed and the command terminated. Upon completion of transmission, the message shall be marked as having been transmitted.

3.8.9.4.5 REMOVE Command. This command shall allow the operator to specify the number of a message to be removed from message memory. The command shall not be executed and an error/information message printed under the following conditions:

a. Specified message number is not a message in the system.

b. Specified message number is a received message which has not been printed.

c. Specified message number is a message which is in transmission.

This command shall remove the following message types:

a. Received messages which have been printed.

b. Composed messages which have been transmitted.

c. Composed messages which have not been transmitted. An information message shall be printed to notify the operator of this condition.

Before message removal, this command shall prompt 'ARE YOU SURE? (Y!N)'. If the response is 'N', the command shall be complete and no message shall be removed. If the response is 'Y', the specified message number shall be removed, an information message printed, and the command shall be complete.

3.8.9.4.6 JUSTIFY Command. This command shall allow the operator to specify the number of a message to be reformatted. If the specified number is not a message in the system, an error message shall be printed and the command shall be complete. If the specified message number is in the system, the prompt 'ARE YOU SURE? (Y/N)' shall be printed. If the response is 'N', the command shall be aborted. If the response is 'Y', the specified message shall be line justified. The specified message shall be accessed, all LFs shall be removed and all CRs replaced with spaces. CRs shall be inserted in place of spaces to form maximum length lines which are less than or equal to the current line length. The whole message shall be processed. No words shall be split, the CRs shall be inserted in place of spaces only. If any sequence of characters which is greater than the line length does not have a space to change to CR, the CR shall be inserted so as to form a line equal to the line length.

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3.8.9.4.7 QUERY Command. The system shall print out the following information in ASCII mode:

SOFTWARE SYSTEM CONFIGURATION = UGC-74B
SWITCH STATE = ICT, KSR or RO
OPERATIONAL STATE = ICT, KSR, or RO
MODE = ASCII
BAUD RATE = 75, 150, 300, 600, or 1200
STOP BITS = 1 or 2
PARITY OPTION = ODD, EVEN, or INHIBIT
DUPLEX MODE = HALF or FULL
END OF LINE OPTION = 0 TO 8 ASCII CHARACTERS REPRESENTING END OF LINE SEQUENCE
RECEIVE ENVELOPE OPTION = START RECEIVE ENVELOPE SEQUENCE (0 TO 8 ASCII CHARACTERS): STOP RECEIVE ENVELOPE SEQUENCE (0 TO 16 ASCII CHARACTERS)
TRANSMIT ENVELOPE OPTION = START TRANSMIT ENVELOPE SEQUENCE (0 TO 8 ASCII CHARACTERS): STOP TRANSMIT ENVELOPE SEQUENCE 0 TO 16 ASCII CHARACTERS)
SPACE OPTION = ON or OFF
CAPITAL LETTER OPTION = ON or OFF
LINE LENGTH = 40 to 132
LINE FEEDS = 1 or 2
END OF MESSAGE LINE FEEDS = 1 to 20
FONT = 1 ENG 64 CHAR, 2 ENG 96 CHAR or 3 ENG 128 CHAR
LINES PER PAGE = 1 to 66
NUMBER OF PRINT COPIES = 1 to 3

This command allows the operator to obtain a printout of the following information in Baudot mode:

SOfTWARE SYSTEM CONFIGURATION = UGC-74B
SWITCH STATE = ICT, KSR, or RO
OPERATIONAL STATE = ICT, KSR, or RO
MODE = BAUDOT
BAUD RATE = 45.5, 50, 75, 150, 300, 600 or 1200
STOP BITS = 1 or 2
BELL OPTION = FIGURES S or FIGURES J
DUPLEX MODE = HALF or FULL
END OF LINE OPTION = 0 TO 8 ASCII CHARACTERS REPRESENTING END OF LINE SEQUENCE
RECEIVE ENVELOPE OPTION = START RECEIVE ENVELOPE SEQUENCE (0 TO 8 ASCII CHARACTERS): STOP RECEIVE ENVELOPE SEQUENCE (0 TO 16 ASCII CHARACTERS)
TRANSMIT ENVELOPE OPTION = START TRANSMIT ENVELOPE SEQUENCE (0 to 8 ASCII CHARACTERS): STOP TRANSMIT ENVELOPE SEQUENCE (0 TO 16 ASCII CHARACTERS)
SPACE OPTION = ON
LINE LENGTH = 40 to 132

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LINE FEEDS = 1 or 2
END OF MESSAGE LINE FEEDS = 1 to 20
FONT SELECTED = 1 ENG 64 CHAR, 2 ENG 96 CHAR or 3 ENG 128 CHAR
LINES PER PAGE = 1 to 66
NUMBER OF PRINT COPIES = 1 to 3

3.8.9.4.8 STATUS Command. This command shall allow the operator to request the status of specified message(s). If an invalid message number is requested an error message shall be printed and the command shall be complete. If no message number is specified, the status of the latest received message shall be given. The status of the specified message(s) shall be printed and shall include: the message number, type of message (received or composed), precedence (as described below), number of memory blocks used by the message, and whether a received message has been printed or a composed message has been transmitted. If all messages are specified, "*" shall be entered as the operand and the status of the received messages shall be printed in order, first-in first-out by precedence. The status of the composed messages in order of message number shall then be printed and the command shall be complete. The HLT key shall stop the status printout and the command shall be complete.

The precedence shall be defined as:

Character	Precedence	Character Printed
Y	EMERGENCY COMMAND	Y
Z	FLASH	Z
O	IMMEDIATE	O
P	PRIORITY	P
R	ROUTINE	R
OTHER OR NOT DETERMINABLE	ROUTINE	R

For composed messages, the precedence shall be determined when exit from Edit mode is performed. The command shall be completed by printing:

"MEMORY NN % FULL."

where NN shall be a two digit number representing the percent of message memory in use.

3.8.9.4.9 TTY Command. This command shall cause a one line message to be transmitted. If the BAUDOT mode, the LTRs code is the first character transmitted in any message. When the command is given, the status of the transmitter shall be checked. If a composed message is being transmitted, an error message shall be printed and the command shall be terminated. If a previous TTY message is being transmitted, operator inputs shall not be accepted. If the transmitter is free, a CR/LF shall be issued and the operator may enter the text of the line. The text of the line is composed of

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all characters entered (with DLC, DLL and REV keys available) until a CR is entered. The line of text shall be transmitted with an end of line sequence appended and in an envelope. If there is no text, the command shall be terminated. The command shall then be complete.

3.8.9.4.10 HELP Command. This command shall allow the operator to print a list of all the system commands. The list shall be as follows:

SYSTEM PROMPT CHARACTER IS: @

DLC KEY	DELETES THE LAST CHARACTER INPUT.
DLL KEY	DELETES THE CURRENT INPUT LINE.
REV KEY	REVIEWS (REPRINTS) THE CURRENT INPUT LINE.

SYSTEM COMMAND	COMMAND FORMAT(S)	DESCRIPTION
PARAMETER	PAR	ALLOWS SYSTEM PARAMETERS TO BE SPECIFIED.
EDIT	ED XXXX	ALLOWS EDITING FUNCTIONS TO BE SPECIFIED. FOR MESSAGE XXXX.
	ED	NEW MESSAGE CREATION.
PRINT	PR XXXX	PRINTS MESSAGE XXXX.
	PR *	PRINTS ALL RECEIVED MESSAGES. THE HLT KEY WILL TERMINATE ANY PRINT ACTIVITY.
TRANSMIT	TR XXXX	TRANSMITS MESSAGE XXXX.
	TR XXXX WITH NNN	WITH NNN = A SPECIFIED CHANNEL SEQUENCE NUMBER (NNN = 1 TO 999). THE HLT KEY WILL TERMINATE ANY PRINT ACTIVITY.
REMOVE	REMO XXXX	REMOVES MESSAGE XXXX FROM STORAGE.
JUSTIFY	J XXXX	REFORMATS MESSAGE XXXX IN MESSAGE STORAGE.
QUERY	Q	PRINTS CURRENT SYSTEM STATUS.
STATUS	ST XXXX	PRINTS CURRENT STATUS OF MESSAGE XXXX.
	ST *	STATUS OF ALL MESSAGES.
	ST	LAST RECEIVED MESSAGE STATUS. THE HLT KEY WILL TERMINATE ANY PRINT ACTIVITY.
TTY	TT	IMMEDIATE TRANSMISSION OF A SINGLE LINE OF INPUT.
HELP	HEL	PRINTS LIST OF AVAILABLE SYSTEM COMMANDS.

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RESTART	RES	ALLOWS OPERATOR TO CHANGE UNIT SWITCH SETTINGS AND RESTART SYSTEM.

NOTE: XXXX INDICATES THE SPECIFIC MESSAGE NUMBER. THE RANGE OF NUMBERS IS FROM 1 TO 9999.

3.8.9.4.11 RESTART Command. This command shall provide the operator the ability to restart the system without turning the power off. All messages contained in message memory (ICT only) and all PARAMETER options (per paragraph 3.8.8.10) shall be retained during execution of this command. Warm restart shall abort all communication line activity. A message to this effect shall be printed and the "ARE YOU SURE? (Y/N)" query shall be issued. If the operator responds "N", this command shall terminate immediately. If the operator responds "Y", the prompt to "CHANGE SWITCH SETTINGS AND ENTER CR TO CONTINUE" shall be printed. The system shall then wait for the operator to respond with CR. Upon entry of "CR" by the operator, the switches shall be read and the state of the machine shall be changed (if required) to reflect the switch settings. The machine state shall be communicated to the operator via the printout described in the initialization requirement. Invalid switch options shall be as described in paragraph 3.8.8.3.

NOTE: Retention of PARAMETER options shall allow for selection of non default printer options followed by a switch to KSR or RO thus making these non default options available to these modes.

3.8.9.5 Full-Duplex Operation. The equipment shall be capable of full duplex operation (transmit and receive simultaneously) in a four wire system.

3.8.9.6 Half-Duplex Operation. The equipment shall be capable of half duplex operation (transmit and receive - one at a time) in a four-wire system. In half-duplex, receive shall not be initiated for one 1 +/- 0.25 second after completion of a transmission in order to prevent reception of a delayed echo.

3.8.10 Performance Requirements.

3.8.10.1 Response Times. The system shall accept and begin processing operator inputs within two seconds.

3.8.10.2 Storage Capacity. The equipment hardware shall provide a message storage capacity of 56000 characters segmented into 35 pages. Message memory block length shall not exceed 256 bytes.

3.8.10.2.1 Memory Retention. In the event of the loss of prime power, information in message memory shall be retained by providing back-up power from an external power source. The power source back-up shall be provided through a connector in the interface.

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3.8.10.3 Printing Speed. Printing throughput of the AN/UGC-74B(V) shall be as a minimum, 120 characters per second, (i.e., no data shall be lost when receiving a typical 69 through 132 characters per line English message at 1200 baud with all other terminal operations suspended during message reception).

3.8.10.4 Print Quality. When the line length is 40 to 80 inclusive, the characters shall be printed at 12 characters per horizontal inch (80 characters +/- 1/2 character per 6.67 inches and 6 lines per vertical inch (66 lines +/- 1 line per 11 vertical inches). When the line length is eighty-one (81) to one-hundred-thirty-two (132) inclusive, the characters shall be printed at approximately 20 characters per inch and 6 lines per vertical inch. Registration of printed characters shall be such that the centers of all characters are within +/- 0.01 inch of the nominal center of the line.

3.8.10.4.1 Print Registration. Registration of printed characters shall be such that the centers of all characters are within +/- 0.01 inch of the nominal center of the line. Horizontal shifting of printed characters shall not exceed one character space.

3.8.10.4.2 Printed Character Quality. Printed characters shall possess the following characteristics:

a. Printed characters shall have no missing dots.

b. Printed characters shall have no overprinting of dots from adjacent characters or lines.

c. Printed characters shall not exhibit compression, expansion or lightness to the extent that they become illegible.

d. The correct character shall be printed.

e. Margin shift shall not exceed one character space.

3.8.10.5 Print Legibility. All printable characters as specified in 3.8.5.1 shall be legible.

3.8.10.6 Automatic Time-out. The print carriage motor current shall be turned off if printing has not been required for 1 +/- 1/4 minute. When printing is required, there shall be no loss of printed characters.

3.8.11 Operator Prompting.

3.8.11.1 Error/Information Message List. The following information/error messages shall be printed in full with an optional identification number and English words.

a. NOT REMOVED, COMM PORT 'RECEIVE' MESSAGE NEVER PRINTED
b. MESSAGE NOT FOUND

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c. INTERNAL MESSAGE MEMORY FILLED
d. PHRASE NOT FOUND
e. DELIMITER/PHRASE IN ERROR
f. COMMAND NOT ACCEPTED, CURRENT LINE TOO LONG, CHANGE LINE PARAMETER
g. MESSAGE EMPTY
h. MAXIMUM NUMBER OF HEADER LINES EXCEEDED
i. USE 12 TABS MAXIMUM
j. INVALID OR MISSING COMMAND VARIABLE
k. COMMAND NOT ACCEPTED, COMMUNICATIONS PORT BUSY
l. COMMAND INPUT NOT RECOGNIZED
m. COMMAND NOT ACCEPTED. IMPROPER IN PRESENT COMMAND MODE
n. 'COMPOSED' MESSAGE HAS NOT BEEN TRANSMITTED
o. PRINTING TERMINATED
p. MESSAGE XXXX REMOVED FROM INTERNAL MESSAGE MEMORY
q. COMMAND NOT ACCEPTED, MESSAGE IS CURRENTLY IN TRANSMISSION ON COMM PORT
r. NO TABS SET
s. IMPROPER MESSAGE HEADER FORMAT
t. INPUT IGNORED, LINE LENGTH UNACCEPTABLE
u. NO HEADER DEFINED
v. INCORRECT TIME-OF-DAY, MUST BE 4 DIGITS
w. COMMAND NOT ACCEPTED, INCORRECT HEADER SELECTED
x. FONT NOT AVAILABLE
y. NO MORE ROOM FOR ORIGINATOR RI/PLA
z. NO MORE ROOM FOR ACTION RI/PLA
aa. NO MORE ROOM FOR INFORMATION RI/PLA
bb. NO MORE ROOM FOR EXEMPT RI/PLA
cc. NOT ENOUGH ROOM IN NEW HEADER VARIABLE FOR ALL OF EXISTING RI/PLA
dd. COMMAND NOT VALID FOR BAUDOT
ee. COMMAND NOT ACCEPTED, MESSAGE IN RECEIVE CATEGORY
ff. IMPROPER SWITCH SETTING - ASCII AT LESS THAN 75 BAUD
gg. IMPROPER SWITCH SETTING - DIO AT LESS THAN 300 BAUD OR IN BAUDOT
hh. IMPROPER SWITCH SETTING - DIO WITH EXTERNAL CLOCK
ii. WARM RESTART ABORTS ALL COMM LINE ACTIVITY
jj. BACKSPACE NOT ALLOWED

3.9 AN/UGC-74B(V) Signal Characteristics and Electrical Requirements.

3.9.1 Operating Modes. This equipment shall operate in the following modes one at a time.

3.9.1.1 ASCII Code Interface.

a. Start/stop, balanced low level digital interface using internal clock.

b. Start/stop, balanced low level digital interface using an external clock.

c. Start/stop, high level digital interface using internal clock.

d. Start/stop, high level digital interface using an external clock.

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3.9.1.1.1 Signal Speeds (ASCII). The equipment shall operate with signals at speeds (bps) as follows:

a. ASCII 10.0 unit, start/stop at 75, 150, 300, 600, and 1200 (Balanced low level).

b. ASCII 10.0 unit, start/stop at 75 and 150 (High level).

c. ASCII 11.0 unit, start/stop at 75, 150, 300, 600, and 1200 (Balanced low level).

d. ASCII 11.0 unit, start/stop at 75 and 150 (High level).

3.9.1.1.2 External Clock Mode. This equipment shall operate up to 1200 bps (Balanced low level digital interface); 75 and 150 bps (High level digital interface).

3.9.1.2 BAUDOT Code Interface.

a. Start/stop, balanced low level digital interface using internal clock.

b. Start/stop, balanced low level digital interface using external clock.

c. Start/stop, high level digital interface using internal clock.

d. Start/stop, high level digital interface using an external clock.

3.9.1.2.1 Signal Speeds (BAUDOT). In the internal and external clock modes, this equipment shall operate with signals at speeds (bps) as follows:

a. BAUDOT 7.0 unit, start/stop at 45.5, 50, 75, 150, 300, 600, and 1200 (Balanced low level).

b. BAUDOT 8.0 unit, start/stop at 45.5, 50, 75, 150, 300, 600, and 1200 (Balanced low level).

c. BAUDOT 7.0 unit, start/stop at 45.5, 50, 75 and 150 (High level).

d. BAUDOT 8.0 unit, start/stop at 45.5, 50, 75 and 150 (High level).

3.9.1.3 Speed Accuracy. The equipment shall transmit at the specified speed +/- 0.1 percent.

3.9.2 Clock INT/EXT/KG-30 Switch. With the INT/EXT/KG-30 switch in the INT position, the unit shall operate at the signal speed selected by the BAUD RATE switch. With the INT/EXT/KG-30 switch in the EXT position, the unit shall operate at a bit rate equal to the external clock frequency being received. With the INT/EXT/KG-30 switch in the KG-30 position, separate external clocks shall be used for transmit and receive functions. The maximum data rate for any of these modes shall be 1200 bps.

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3.9.2.1 Internal Clock Mode - Transmit Clock. In the internal clock mode, the internal clock shall be made available to other units at the communications connector at all times. This transmit clock shall be a symmetrically shaped signal having a frequency equal to the bit rate selected by the BAUD RATE switch. With the clock switches positioned for noninverted internal clock, the transmitted data transitions shall correspond to the positive going slope of the transmit clock. With the clock switch in the inverted internal clock position, the transmitted data transition shall correspond to the negative going slope of the transmit clock.

3.9.2.2 External Clock/Data Relationship. The external clock shall be used with nonreturn to zero data only. That is, the external clock shall not be used in conjunction with conditioned diphase signals.

3.9.2.2.1 External Clock/Transmitted Data Relationship. With the clock switches positioned for noninverted external clock, the transmitted data transitions shall correspond to the positive-going slope of the received clock signal as shown in Figure 1. With the clock switches in the inverted external clock position, the transmitted data transitions shall correspond to the negative-going slope of the received clock signal.

3.9.2.2.2 External Clock/Received Data Relationship. The relationship of external clock to received data shall be as shown in Figure 1.

3.9.2.3 Data Transmission. For transmission the delay between data and clock signals (internal and external) shall not exceed that shown in Figure 1.

3.9.3 Detailed Requirements for Digital Interface Circuits.

3.9.3.1 Signals. The equipment shall provide nonreturn to zero (NRZ) and conditioned diphase signals in accordance with Figure 2. The type of signal shall be selectable by means of an internal switch.

3.9.3.1.1 Conditioned Diphase. The conditioned diphase signal shall be provided only for data rates of 300 bps, 600 bps and 1200 bps.

3.9.3.2 Low Level Digital Interface. The low-level digital interface shall meet the requirements specified in paragraph 5.1 of MIL-STD-188-114, dated 24 March 1976 with the following exceptions:

a. Paragraph 5.1.1.3, Test Termination Measurement shall not be applicable.

b. Paragraph 5.1.2.2, Receive Input Sensitivity Measurement differential voltage shall be "1500 mV".

c. Paragraph 5.1.2.3, Receive Input Balance Measurement differential voltage shall be "1500 mV".

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3.9.3.3 High Level Digital Signal Interface.

3.9.3.3.1 Idle Transmit Condition. The transmit interface shall provide a closed circuit for mark and an open circuit for space for a 130V (+/- 2%), 60 or 20 mA signal line.

3.9.3.3.2 Transmitter Source Impedance. The source impedance of the transmit interface shall not exceed 150 ohms.

3.9.3.3.3 Receive Interface (130 Volt Operation). The receive interface shall be insensitive to the polarity of the receive signal in the 130V (+/- 2%), 60 or 20mA signal line. A change from 32 mA to 28 mA on a 60 mA neutral signal and from 12 mA to 8 mA on a 20 mA neutral signal shall cause correct operation of the receive interface.

3.9.3.4 Normal Input Keying (NIK) Interface. The equipment shall be compatible with the LOOP IN and LOOP OUT circuits of COMSEC equipments such as the KW-7. The RECEIVE circuit shall simulate the characteristics of a teletypewriter selector magnet when the driving element produces an output current of 20 mA and open circuit voltage of 48V. The SEND circuit shall provide a switch closure for mark and an open for space. The mark loop current supplied by the COMSEC equipment shall be approximately 70 microamperes. The maximum value of shunt capacitance to ground at the COMSEC equipments LOOP INPUT is defined to be 0.07 microfarads, and the mark to space decision is made at an input voltage of + 1.5V. This voltage shall be defined as the point where the NIK wave-shape is sampled for all distortion measurements. The maximum allowable distortion under these conditions shall be 35 percent for marking and spacing end or characteristic distortion (see 6.4.2) and 40 percent for marking or spacing bias distortion. These distortion limits supersede the requirements of 3.9.4.2 for NIK operation only.

3.9.4 Signal Distortion.

3.9.4.1 Receive Equipment. When internal clock is selected, the receiving equipment shall operate error-free when the received data signal is degraded as described in 3.9.4.1.1 or 3.9.4.1.2. The equipment shall operate error-free when these degradations occur simultaneously, when neither one degradation is more than 50 percent of its indicated maximum.

3.9.4.1.1 Digital Distortion. The maximum bias or end distortion of the received low-level data signal will be 42 percent of the shortest duration pulse encountered in the normal signal. That is, maximum digital distortion for nonreturn to zero formatted data shall be 42 percent of a bit period and for conditioned diphase formatted data shall be 21 percent of a bit period. The maximum bias or end distortion of the received current-loop data signal shall be 42 percent of the shortest duration pulse encountered in the nominal signal.

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3.9.4.1.2 Spurious Pulses. The received data-signal may contain spurious pulses having a time duration less than 25 +/- 1 percent of the shortest duration pulse encountered in the nominal signal. Rejection of individual spurious pulses separated by less than 10 percent of a nominal bit period shall not be required if their combined pulse width exceeds the 25 percent requirements. This requirement shall not apply when using external clock.

3.9.4.2 Transmit Equipment. The transmitting equipment output data signals shall have less than two percent bias or end distortion and shall be at the specified bit rate +/- 0.1 percent.

3.9.5 Isolation of Chassis Ground, Signal, and Clock Lines. The signal and clock return lines shall be isolated from chassis ground by a minimum of 100 kohms.

3.9.6 System Interface. The equipment shall operate satisfactorily with the KW-7 (interface defined in 3.9.3.4) and with equipment utilizing a low level digital interface (defined in 3.9.3.2) and a high level digital interface (defined in 3.9.3.3).

3.9.6.1 Pin Assignments.

a. Pin assignments for the Communication Connector are specified in Figure 3.

b. Pin assignments for the Power Connector are specified in Figure 4.

c. Pin assignments for the Battery Back-Up Connector are specified in Figure 5.

3.9.7 Power.

3.9.7.1 Power Consumption. The steady state power drain shall not exceed 110 watts.

3.9.7.2 Sources. The equipment shall be capable of operation from the following power sources by only changing external power cords:

a. 22 to 30 VDC vehicular electrical system (nominal 26 +/- 4 VDC).

b. 115/230 V, 50/60/400 cycle, single phase, AC. AC voltage tolerances of +/- 15 percent and frequency tolerance of +/- 5 percent.

3.9.7.3 Overvoltage and Transients.

3.9.7.3.1 AC Overload. The equipment shall be capable of withstanding an AC input voltage 20 percent greater than the specified high line voltage for a five minute period, with no component degradation. The AC overvoltage shall not cause a loss or degradation of data.

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3.9.7.3.2 DC Overvoltage and Transient Voltage Protection. The equipment shall operate without degradation of equipment or data during overvoltage and transient voltages in vehicular power supplies as follows:

a. Overvoltage 36 VDC input for a 1-hour maximum.

b. Overvoltage transient +/- 250 V peak spike induced on the normal input voltage having a rise time not exceeding 50 ns, a frequency of oscillation greater than 100 kHz and less than 600 kHz and an energy content of 15 millijoules maximum. Data degradation is permissible during each transient.

c. Undervoltage transients, 15 VDC input for a period of at least 75ms, at a repetition rate of 10 pulses per minute for one hour. Data degradation shall be permissible during each transient.

3.9.7.4 Isolation of Chassis Ground, Power Lines. The grounded circuit conductor (power neutral) shall be isolated and insulated from the equipment grounding conductor (chassis ground). A separate line shall be used to carry the equipment grounding conductor. The only exception to the preceding is that the equipment grounding conductor shall be connected to the grounded circuit conductor at the power source. Minimum isolation resistance shall be 100k Ohms.

3.9.7.5 Battery Back-up. Battery Back-up power drain shall not exceed 2 amps at 12 +/- 1.2 VDC.

3.10* Keyboard TT-745/UGC-74A(V) Requirements.

3.10.1 Operating Modes. The keyboard is a dependent device requiring a Teleprinter TT-744/UGC-74B(V) for operation.

3.10.2 Keyboard Arrangement. The keyboard shall consist of the 59 standard keys required by MIL-STD-1280 for type 1, Class 1 keyboards plus four additional keys for basic editing control together with the supporting structure for these 63 keys.

3.10.3 Two Key Rollover. The two key rollover feature shall prevent loss or incorrect generation of keyboard code output when a second key is depressed before the release of a first key. The code of the second key shall be generated upon the release of the first key.

3.10.3.1 Key Pressure. The pressure necessary to operate any key shall not be less than 3 ounces or more than 6 ounces. The required striking force shall be the same at any speed and for any equipment tilted up to 20 degrees from normal (10 percent sampling).

3.10.4 Keyboard.

3.10.4.1 Lower Case Characters. The lower characters shown on the key-tops

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shall be the characters processed with keyboard in nornal state, i.e., neither the shift nor control key operated.

3.10.4.2 Shift Key. The upper characters shown on the key-tops shall be the characters processed with the shift key operated.

3.10.4.3 Shift Lock Key. A depression of the LOC key (Shift Lock key) shall lock the SHIFT key in its operating mode. Momentary depression of the SHIFT key shall disengage the LOC key and return the keyboard to its nornal state. An indication shall be given whenever the keyboard is in the shift position.

3.10.4.4 Machine Functions. The characters indicated on the following key-tops shall be processed, regardless of control and/or shift key positions: CR (carriage return), LF (line feed), BS (back space), DEL (delete), TAB (horizontal tab), and SPACE.

3.10.4.5 Control Codes. The two and three character groups (control characters) shown on the left side of the key tops shall be processed when the control key is operated regardless of the position of the shift key.

3.10.4.6 Control Keys. The control key shall have no effect on the codes generated from keys with no control character groups on the key tops.

3.10.4.7 Repeat Key. When the RPT key is operated, the code corresponding to that of the last coded key operated shall be generated continuously until the RPT key is released. The character rate for the repeat operation shall be five characters per second.

3.10.4.8 TAB Key. When the TAB key is pressed, the print position shall be moved to the smallest tab position value which is greater than the print position. The next character shall be printed in that position. If the present print position is greater than or equal to the largest tab position, then the tab shall be replaced by a CR. If the space option is set, then the tab function shall be carried out by inserting the proper number of spaces into the buffer instead of the tab character. If the space option is not set, then the tab character is put in the buffer and the print position shall be set.

3.10.4.9 BS Key. In the ASCII mode, when the BS key is pressed, the print position shall be moved one position to the left for each time the BS key is pressed. Subsequent printable characters shall overstrike any text already printed. A minimmn of 256 characters/line shall be accommodated for original print, BS, and overprint characters. An error message shall be printed if backspacing (BS) is attanpted during input of header/PLA data (see 3.8.9.4.1.20). In the BAUDOT mode, when the BS key is pressed, a FIGS code shall be entered into menory.

3.10.4.10 Message Processing Keyboard. The LTRS and FIGS codes shall be individually entered by depressing the DEL or BS key respectively (Applicable

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only in BAUDOT mode). The following four special keys on the keyboard shall be used for basic editing of all lines of input into the terminal (except in the KSR Character mode), command or otherwise:

Key	Description
DLC	Shall delete the last character input. The system response shall be a backward slash over the character to be deleted to indicate the character has been deleted. This key shall only be valid over the current line. If the current line is empty a CR/LF shall be output.
DLL	Shall delete the current input line. The system response shall be a carriage return/line feed, to indicate that the current input line has been deleted. This key shall only be valid over the current input line before a CR is entered. If the current line is empty a CR/LF shall be output.
REV	Shall review or reprint the current input line. The system shall respond with a carriage return/line feed, and a reprint of the current input line. The next character entered shall be printed immediately following the last character of the reviewed line. This key shall only be valid over the current line. If the current line is empty a CR/LF shall be output.
HLT	Shall halt the operator initiated printing currently being performed. The system shall respond by terminating the printing and follow with CR/LF, "PRINT ABORTED", CR/LF and the proper prompt sequence. The HLT key shall also be used to terminate the INSERT and BEFORE subcommands.

3.11 TEMPEST Requirements. The TEMPEST characteristics of the equipnents shall be in accordance with the requirements of MIL-T-49261.

3.12 Acoustic Noise. Acoustic noise shall be sufficiently low in intensity to comply with the noise level requirements of MIL-STD-1474 as modified in Table IV.

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TABLE IV. ACOUSTIC NOISE LEVELS

OCTAVE BAND 1/ CENTER FREQ (Hz)	OCTAVE BAND CENTER FREQ (Hz)	NOISE LEVEL IN DB 2/
53	63	79
106	125	73
212	250	71
425	500	69
850	1000	67
1700	2000	65
3400	4000	62
6400	8000	60

1/ THE REASON FOR THE TWO OCTAVE BANDS IS TO PERMIT THE USE OF AVAILABLE TEST EQUIPMENT WHICH WOULD NORMALLY FALL INTO ONE OF THE TWO OCTAVE BANDS.

2/ REFERENCED TO 0.0002 DYNES/CM2

3.13 Electromagnetic Compatibility. The equipment shall comply with the following requirements of MIL-STD-461:

a. First Article Group D Inspection.

CE01	CS01	RE02 2/	RS02
CE02	CS02	RE02.1 1/	RS03 3/
CE03	CS06	RS03.1 1/ 3/	
CE04 2/			
CE05			

1/ The upper frequency limit for radiated requirements RE02.1 and RS03.1 shall be 1,000 MHz.

2/ Transient interference due to initial on/off power transients and line break are exempt from meeting the requirenents.

3/ The E-Field test signal shall be amplitude modulated 50 percent with 1 kHz tone as follows:

Frequency (Hz)	E-Field
10k - 2M	1 volt per meter
2M - 30M	5 volts per meter
30M - 1,000M	10 volts per meter

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b. Subsequent Group D Inspection.

CE04 2/	RE02 1/ 2/
CE05	RE02.1 1/

1/ The upper frequency limit for RE02 and RE02.1 shall be 400 MHz.

2/ Transient interference due to initial on/off power transients and line break are exempt from meeting the requirements.

3.14 System Nuclear Survivability. All contractor furnished mechanical configurations, electronic assemblages, electronic equipment, electronic circuit and electronic pieceparts shall withstand the nuclear environnent as specified in Communications-Electronics Command drawing A3152457 (Blast and thermal not applicable). The equipment shall meet the operational requirements of Section 3 herein, after being subjected to the specified nuclear environments. This recovery in operational effectiveness may include automatic or manual correction of nuclear environment-induced changes in the system but not the replacement of pieceparts or components. The equipnent shall be energized and the Contractor Furnished Equipment (CFE) and Government Furnished Equipment (GFE) configured for the worst case system response during exposure. The contractor shall ensure that the GFE is not degraded by nuclear enviromnent-induced signals coupled from the CFE; and vice versa, that the CFE is not degraded by nuclear environment-induced signals coupled from the GFE.

3.15 Environmental Service Conditions. The equipment shall be constructed so that no fixed part shall become loose, cracked, naked, pitted or blistered; no movable part or control shall be shifted in setting, position or adjusiment, and there shall be no permanent degradation in perfonnance under the following environmental conditions.

3.15.1 Temperature.

3.15.1.1 Operating Temperatures. The ambient temperature range is -25 F (-32 C) to +155 F (+68 C). In the low temperature range between -25 F and +20 F, the following is permissible:

a. Special ribbons

b. Operator assistance for paper motion during the first two minutes of operation

3.15.1.2 Nonoperating Temperatures. The nonoperating exposure range is -65 F (-54 C) to +155 F (+68 C).

3.15.2 Humidity. The equipment shall be operable without pennanent degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

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3.15.3 Altitude. The equipment shall be operable without permanent degradation in specified performance at altitudes up to 15,000 feet above sea level and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.15.4 Immersion. The equipment shall be immersed to a covering depth of three feet of water for a period of two hours without evidence of water leakage in excess of 2 cubic centimeters.

3.15.5 Vibration. The equipment shall be designed for the ground mobile environnent. The ground mobile environnent consists largely of broadband random vibration resulting from the interaction of vehicle suspension and structures with road and surface discontinuities. The nature of the terrain, vehicle speed, vehicle dynamic characteristics, and suspension loading all affect vibration responses.

3.15.5.1 Bounce, Loose Cargo. The equipment, including necessary paper supply (full roll), shall withstand the shocks normally induced during field transportation as loose cargo.

3.15.6 Shock.

3.15.6.1 Drop. The equipment shall withstand shocks nonnally encountered during loading and unloading of equipments. After nonnal setup procedures, the equipment must meet all requirements.

3.15.6.2 Bench Handling. The equipment shall withstand shock encountered in servicing.

3.15.7 Ballistic Shock. The equipment as mounted for operation with power supplied shall withstand simulated ballistic shocks without any part becoming separated from the whole. Both hardware and software integrity shall be maintained. The equipment shall be completely operational following each shock.

3.15.8 Dust. The equipment shall operate after being in the operating configuration in blowing dust, as encountered in the desert.

3.15.9 Rain. The equipment shall be operable without degradation in specified performance and sustain no physical damage during periods of heavy precipitation with intermittent wind. It is allowable to replace paper and/or ink ribbons after the test. Operator assistance as required is allowed for paper movement during operation of equipnent.

3.15.10 Salt Fog. The equipment shall be resistant to the corrosive effects of salt fog atmosphere.

3.15.11 Orientation. The equipment, bolted in place to the mounting surface, shall operate in any orientation up to 20 degrees from normal operating position (that is forward, backward, left or right).

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3.15.12 Fungus. The equipment, in the assembled and ready for delivery condition, shall provide no nutrients in material, coating, or contaminant form or support fungal growth when tested in accordance with paragraph 4.15.12. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.15.13 Leakage Air Seal Test. The equipment shall lose no more than 0.01 pound per square inch in one minute when a vacuum of 1 pound per square inch is provided in the equipnent.

3.16 Reliability. The quantitative requirements of the AN/UGC-74B(V) with keyboard shall have a specified (θ_o) mean-time-between-failure (MTBF), as defined in MIL-STD-781, of at least 1800 hours.

3.17 Systems Safety Engineering.

3.17.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract.

3.17.2 Edge Rounding Exposed. Exposed edges shall be rounded sufficiently to minimize laceration puncture hazards, the goal being a minimum radius of 0.04 inch (1 mm), and exposed corners to a minlmum of 0.5 inch (13 mm).

3.17.3 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.17.4 Overload Protection. Electrical overload protection shall be in accordance with Requirements 1 and 8 of MIL-STD-454.

3.18 Maintainability.

3.18.1 Scheduled Maintenance. The equipment shall be capable of operation for at least 500 hours between lubrications or before adjustments are required to compensate for degradation of performance. It shall be capable of operation for at least 1000 hours, observing prescribed maintenance and lubrication schedules before replacement of any part is required.

3.18.2 Consumables. The time required to replace consumables, assuming no repairs are required, shall not exceed five minutes. Paper and ink cartridge/ ribbon shall be consumable items.

3.18.3 Mean-Time-To-Repair. Mean downtime allowable to return the equipment to service shall not exceed four hours for scheduled maintenance servicing. Mean-time-to-repair (MTTR) is defined as the total corrective maintenance time divided by the total number of corrective, maintenance actions during a given

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period of time. MTTR applies to the replacement of all modules (see 6.4.6), subassemblies and chassis mounted piece parts. It does not include the repair of modules and subassemblies. The MTTR (Mc) shall be 30 minutes at the direct support level and 2.5 hours at the general and depot support levels.

3.19 Weight. The weight of the equipment including front cover and copy holder shall not exceed 103 pounds.

3.20 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268

Plastic material and parts

Wiring and cabling, including:

Protection
Clearance
Splicing and stretching
Grounding cable shielding
Wire and cable shielding
Cleaning
Soldering and brazing
Welding
Controls

b. In this specification:

3.2 Construction

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of Inspection. Inspection shall be classified as follows:

a. First Article Inspection (See 4.3).

b. Inspection Covered by Subsidiary Documents (See 4.4).

c. Quality Conformance Inspection of Equipment Before Packaging (See 4.5).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units. The quantities shall be as cited in the bid request and contract.

4.3.2 First Article Inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for software verification, group A, group B, group C and group D (see tables V through XIII).

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TABLE V. Software Verification Inspection. 1/

REQUIREMENT	REQ'T PARA.
Keyboard/Print Codes	3.8.5.1
Coded BAUDOT Printer Functions	3.8.5.3
Carriage Return	3.8.5.3.1
Line Feed	3.8.5.3.2
Letters	3.8.5.3.3
Figures	3.8.5.3.4
Parity	3.8.5.3.5
TAB	3.8.5.4.1
CR	3.8.5.4.2
LF	3.8.5.4.3
BS	3.8.5.4.4
Parity Error	3.8.5.4.5
Automatic Carriage Return	3.8.5.5
Automatic Line Feed	3.8.5.6
Print Pin Pressure Selection	3.8.5.7.1.2
Line Length	3.8.5.7.2
Compressed Text Mode	3.8.5.7.2.1
Multiple Character Fonts	3.8.5.7.3
Character Font ROM Capacity	3.8.5.7.3.1
Character Font Selection	3.8.5.7.3.2
Standard Character Font Symbol Sets	3.8.5.7.3.3
Operation Validation/Detennination State	3.8.8.3
ICT State	3.8.8.4
RO State	3 8.8.5
KSR State	3.8.8.6
Self Test State	3.8.8.8
Factory Selftest State	3.8.8.9
Warm Restart State	3.8.8.10
Message Processing	3.8.9
Message Storage	3.8.9.1
Memory Full	3.8.9.1.1
Operator Input	3.8.9.1.1.1
Incoming Message	3.8.9.1.1.2
Message Receipt	3.8.9.2
Receive Message Process	3.8.9.2.1
Transmit Message Process	3.8.9.3
System Commands	3.8.9.4
PARAMETER Command	3.8.9.4.1
LINE Subcommand	3.8.9.4.1.1
SETAB Subcommand	3.8.9.4.1.2
CLRTAB Subcommand	3.8.9.4.1.3
DISPLAY Subcommand	3.8.9.4.1.4
CAPIT Subcommand	3.8.9.4.1.5

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TABLE V. Software Verification Inspection - Continued. 1/

REQUIREMENT	REQ'T PARA.
REMIND Subcommand	3.8.9.4.1.6
EOL Subcommand	3.8.9.4.1.7
LF Subcommand	3.8.9.4.1.8
SPACE Subcommand	3.8.9.4.1.9
PAPER Subcommand	3.8.9.4.1.10
RENVELOPE Subcommand	3.8.9.4.1.11
TENVELOPE Subcommand	3.8.9.4.1.12
EOMLF Subcommand	3.8.9.4.1.13
FONT Subcommand	3.8.9.4.1.14
COPIES Subcommand	3.8.9.4.1.15
DUPLEX Subcommand	3.8.9.4.1.16
HEADERS Subcommand	3.8.9.4.1.17
HELP Subcommand	3.8.9.4.1.18
PAGE Subcommand	3.8.9.4.1.19
SETUP Subcommand	3.8.9.4.1.20
EXIT Subcommand	3.8.9.4.1.21
GO Subcommand	3.8.9.4.1.22
EDIT Command	3.8.9.4.2
TOP Subcommand	3.8.9.4.2.1
BOTTOM Subcommand	3.8.9.4.2.2
UP Subcommand	3.8.9.4.2.3
DOWN Subcommand	3.8.9.4.2.4
BEFORE Subcommand	3.8.9.4.2.5
INSERT Subcommand	3.8.9.4.2.6
DELETE Subcommand	3.8.9.4.2.7
FIND Subcommand	3.8.9.4.2.8
LIST Subcommand	3.8.9.4.2.9
REPLACE Subcommand	3.8.9.4.2.10
HEADER Subcommand	3.8.9.4.2.11
APPEND Subcommand	3.8.9.4.2.12
HELP Subcommand	3.8.9.4.2.13
EXIT Subcommand	3.8.9.4.2.14
PRINT Command	3.8.9.4.3
TRANSMIT command	3.8.9.4.4
REMOVE Command	3.8.9.4.5
JUSTIFY Command	3.8.9.4.6
QUERY Command	3.8.9.4.7
STATUS Command	3.8.9.4.8
TAW Command	3.8.9.4.9
HELP Command	3.8.9.4.10
RESTART Command	3.8.9.4.11
Half Duplex Operation	3.8.9.6
Response Times	3.8.10.1
Storage Capacity	3.8.10.2
Error/Information Message List	3.8.11.1
Battery Back-up	3.9.7.5

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1/ This document does not outline the sequence of tests necessary to demonstrate compliance with the requirements of the table.

TABLE VI. Group A Inspection - AN/UGC-74B(V).

INSPECTION	REQUIREMENT	REQ'T PARA.
Visual & Mechanical	Keyboard Arrangement	3.10.2 3/
	Workmanship	3.20 3/
	Air Seal	3.15.13 3/
Electro-mechanical	Visibility of Copy	3.8.2 4/
	Copy Illumination	3.8.3 4/
	Power Indicator	3.8.6.2 4/
	Self Test	3.8.7.1.4 4/
	External Power Switch	3.8.7.2.3 4/
	Cold Start State	3.8.8.2 4/
	Operation Validation/ State Determination State	3.8.8.3 4/
	KSR State	3.8.8.6 4/
	Self Test State	3.8.8.8 4/
	High Level Digital Signal Interface	3.9.3.3 1/ 4/
	Low Level Digital Signal Interface	3.9.3.2 1/ 4/
Print Quality	Print Registration	3.8.10.4.1 2/ 4/
	Printed Character Quality	3.8.10.4.1 2/ 4/

1/ Functional Loop Back only.

2/ Note: Special test condition as follows: Print 36 consecutive 80 character lines, with no character errors allowed. In the event of a print quality failure, adjustment of the print head assembly, line feed assembly, and carriage assembly is permissible without reburn-in (see 6.4.8).

3/ Use Sampling Plan 1 for this requirement (see TABLE VII).

4/ Use Sampling Plan 2 for this requirement (see TABLE VIII).

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TABLE VII. Sampling Plan 1. 2/

Lot Size	Sample Size
2 to 8	* 1/
9 to 15	13
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102

1/ A '*' indicates that the entire lot must be inspected.

2/ Acceptance number in all cases is zero.

TABLE VIII. Sampling Plan 2. 1/

Lot Size	Sample Size
2 to 8	5
9 to 15	5
16 to 25	5
26 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40

1/ Acceptance number in all cases is zero.

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TABLE IX. GROUP B INSPECTION - AN/UGC-74B(V).

REQUIREMENT	REQ'T PARA.
ASCII/BAUDOT Control	3.8.4 3/
Local Printer Function	3.8.5.2 3/
Parity Error (ASCII)	3.8.5.4.5 3/
Paper/Out	3.8.5.7.4 3/
Audible Alarm Adjustment	3.8.6.1.1 3/
Audible Alarm Actuation	3.8.6.1.2 3/
Battery Backup Indicator	3.8.6.2.2 3/
Paper Low Indicator	3.8.6.2.3 3/
Line Indicator	3.8.6.2.4 3/
Received Message Indicator	3.8.6.2.5 3/
Transmit Indicator	3.8.6.2.6 3/
Memory Full Warning Indicator	3.8.6.2.7 3/
Parity Indicator	3.8.6.2.8 3/
End of Line Indicator	3.8.6.2.9 3/
Switches	3.8.7 3/
PARITY Switch	3.8.7.1.1 3/
STOP BITS Switch	3.8.7.1.2 3/
STATE Switch	3.8.7.1.3 3/
SELF TEST Switch	3.8.7.1.4 3/
MODE Switch	3.8.7.1.5 3/
FIGURES Switch	3.8.7.1.6 3/
CLOCK (+/-) Switch	3.8.7.1.7 3/
BAUD RATE Switch	3.8.7.1.8 3/
REC MODE Switch	3.8.7.1.9 3/
XMIT MODE Switch	3.8.7.1.10 3/
SIGNALS Switch	3.8.7.1.11 3/
CLOCK (INT/KG-30/EXT) Switch	3.8.7.1.12 3/
ABORT Switch	3.8.7.2.1 3/
Audio Alarm Reset	3.8.7.2.2 3/
EXTERNAL POWER Switch	3.8.7.2.3 3/
Illumination control	3.8.7.2.4 3/
TRANSFER Switch	3.8.7.2.5 3/
PARITY RESET Switch	3.8.7.2.6 3/
LINE FEED Switch	3.8.7.2.7 3/
Audio Control	3.8.7.2.8 3/
Operation Validation/State Determination State	3.8.8.3 3/
Battery Backup State	3.8.8.7 2/
Self Test State	3.8.8.8 2/
Full Duplex Operation	3.8.9.5 3/
Memory Retention	3.8.10.2.1 3/
Printing Speed	3.8.10.3 3/
Print Quality	3.8.10.4 3/
Print Legibility	3.8.10.5 3/

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TABLE IX. GROUP B INSPECTION - AN/UGC-74B(V) - Continued.

REQUIREMENT	REQ'T PARA.
Automatic Timeout	3.8.10.6 3/
ASCII Code Interface	3.9.1.1 3/
Signal Speeds (ASCII)	3.9.1.1.1 3/
External Clock Mode	3.9.1.1.2 3/
BAUDOT Code Interface	3.9.1.2 3/
Signal Speeds (BAUDOT)	3.9.1.2.1 3/
Speed Accuracy	3.9.1.3 3/
Clock INT/EXT/KG-30 Switch	3.9.2 3/
Internal Clock Mode - Transmit Clock	3.9.2.1 3/
External Clock/Transmitted Data Relationship	3.9.2.2.1 3/
External Clock/Received Data Relationship	3.9.2.2.2 3/
Data Transmission	3.9.2.3 3/
Detailed Requirements for Digital Interface Circuits	3.9.3 1/ 3/
Signals	3.9.3.1 3/
Conditioned Diphase	3.9.3.1.1 3/
Low Level Digital Interface	3.9.3.2 3/
Idle Transmit Condition	3.9.3.3.1 3/
Transmitter Source Impedance	3.9.3.3.2 3/
• Receive Interface (130 Volt Operation)	3.9.3.3.3 3/
Normal Input Keying (NIK) Interface	3.9.3.4 3/
Receive Equipment	3.9.4.1 3/
Digital Distortion	3.9.4.1.1 3/
Spurious Pulses	3.9.4.1.2 3/
Transmit Equipment	3.9.4.2 3/
Isolation of Chassis Ground, Signal, and Clock Lines	3.9.5 2/
System Interface	3.9.6 3/
Pin Assignment	3.9.6.1 3/
Isolation of Chassis Ground, Power Lines	3.9.7.4 2/
Two Key Rollover	3.10.3 3/
Key Pressure	3.10.3.1 3/
Lower Case Characters	3.10.4.1 3/
Shift Key	3.10.4.2 3/
Shift Lock Key	3.10.4.3 3/
Machine Functions	3.10.4.4 3/
Control Codes	3.10.4.5 3/
Control Keys	3.10.4.6 3/
Repeat Key	3.10.4.7 3/
TAB Key	3.10.4.8 3/
BS Key	3.10.4.9 3/
Message Processing Keyboard	3.10.4.10 3/
Safety	3.17 through 3.17.4 1/ 2/

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1/ First Article test only.

2/ Use Sampling Plan 2 for this requirement (see TABLE VIII).

3/ Use Sampling Plan 3 for this requirement (see TABLE X).

TABLE X. Sampling Plan 3. 1/

Lot Size	Sample Size
2 to 8	3
9 to 15	3
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29

1/ Acceptance number in all cases is zero.

TABLE XI. First Article Group C Environmental Inspection.

	SAMPLE UNIT 1/												
INSPECTION	1,2,3	4	5	6	7	8	9,10	11,12	13				
Immersion	6,8												
AC Overload	3												
Power Consumption	1												
Sources	2												
DC Overload & Transient	4												
Shock, Drop	7												
Vibration	5												
Temperature High		1					1						
Temperature Low		2					2						
Altitude		3											
Bench Handling		4											
Bounce (Loose Cargo)		5					3						
Ballistic Shock		6					4						
Humidity 2/			1					1					
Rain			2					2					

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TABLE XI. First Article Group C Environmental Inspection. - Continued.

	SAMPLE UNIT 1/								
INSPECTION	1,2,3	4	5	6	7	8	9,10	11,12	13
Orientation			3						
Acoustic Noise			4					3	
Paper Usage			5					4	
Fungus 2/				1					
Salt Atmosphere 2/					1				
Maintainability Demonstration									1
Systems Interface						1			
Dust 2/						2			

1/ The numbers in the unit column specify the order of inspections for the indicated unit.

2/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection.

TABLE XII. GROUP C INSPECTION - AN/UGC-74B(V).

REQUIREMENT	REQ'T PARA.
Subgroup I	
Immersion	3.15.4
Subgroup II	
Unit 1	
Temperature High	3.15.1
Temperature Low	3.15.1
Altitude	3.15.3
Bounce, Loose Cargo	3.15.5.1
Bench Handling	3.15.6.2
Ballistic Shock	3.15.7
Unit 2	
Power Consumption	3.9.7.1
Sources	3.9.7.2
Overvoltage & Transients	3.9.7.3.2
AC Overload	3.9.7.3.1
Vibration	3.15.5
Immersion	3.15.4
Drop Shock	3.15.6.1
Immersion	3.15.4

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TABLE XII. GROUP C INSPECTION - AN/UGC-74B(V) - Continued.

REQUIREMENT	REQ'T PARA.
Subgroup III	
Humidity	3.15.2
Rain	3.15.9
Orientation	3.15.11
Acoustic Noise	3.12
Paper Usage	3.8.1
Subgroup IV	
Unit 1	
Fungus	3.15.12
Unit 2	
Salt Atmosphere	3.15.10
Unit 3	
Dust	3.15.8
Subgroup V	
System Interface	3.9.6
Subgroup VI	
Software Verification	(See Table V)
Subgroup VII	
Interchangeability	3.6

TABLE XIII. GROUP D INSPECTION - AN/UGC-74B(V).

REQUIREMENT	REQ'T PARA.
Undesired signal data transmissions (TEMPEST)	3.11
Electromagnetic compatibility	3.13
Nuclear	3.14
Reliability	3.16

4.3.2.1 First Article Software Verification Inspection. Two units shall be subjected to the software verification inspections in accordance with Table V. These units shall be selected from the first lot of first article units. These units shall have passed group A and group B inspections.

4.3.2.2 First Article Group A Inspection. All first article units shall be subjected to the group A inspection (see Tables VI through VIII).

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4.3.2.3 First Article Group B Inspection. The First Article Group B inspection (see Tables VIII through X) shall be performed on units from the First Article lot.

4.3.2.4 First Article Group C Inspection. First article units shall be subjected to group C inspection in accordance with Table XI. The selected units shall have passed group A and group B inspections.

4.3.2.5 First Article Group D Inspection. First article units shall be subjected to group D inspection (see Table XIII). The selected units shall have passed group A and group B inspections. The reliability of the equipment shall be demonstrated in accordance with Appendix B, Category 1, 50.1 of MIL-STD-781. The number of randomly selected equipment to be tested shall be eight (8). The specified MTBF (0) to be used is specified in 3.16.

4.4 Inspection Covered By Subsidiary Documents. The following shall be inspected under applicable subsidiary documents as part of the inspection before packaging:

Item	Where Required
Marking	3.5
Parts, Materials & Processes: General	3.3
Finish	3.4

4.5 Quality Conformation Inspection of Equipment Before Packaging. Each unit of each lot of equipment shall be subjected to bounce preconditioning (see 4.6.1) and burn-in (4.6.2) in that order prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government may, at its option, review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements (see 6.3). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table VII. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection in accordance with the appropriate Sampling Plan that is noted in TABLE VI. Lots in which samples exhibit any failures shall be screened for that failure prior to being subjected to Group B inspection.

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4.5.1.1 Order of Inspection Within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B Inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to Table IX and the appropriate Sampling Plan that is noted in TABLE IX.

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in Table XII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling For Group C Inspection.

4.5.3.1.1 Subgroup I. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every month. Subgroup I tests shall not be run when Subgroup II is specified.

4.5.3.1.2 Subgroup II. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, two units shall be selected and tested in the sixth month, in the tenth month, and every six months thereafter. Immersion test shall be performed both before and after conduction of the drop shock tests.

4.5.3.1.3 Subgroup III. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected and tested at six month intervals.

4.5.3.1.4 Subgroup IV. For this subgroup, three units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected and tested for fungus one time at midpoint of the production cycle.

4.5.3.1.5 Subgroup V. For this subgroup, two units shall be selected from the initial production lot.

4.5.3.1.6 Subgroup VI. For this subgroup, two units shall be selected from the first production lot.

4.5.3.1.7 Subgroup VII. For this subgroup, one unit shall be selected from each month's production units.

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in an order which is satisfactory to the Government.

4.5.3.3 Group C Failures. Actions required relative to group C failures shall be as specified in the contract or purchase order.

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4.5.4 Group D Inspection. This inspection shall consist of the tests specified in Table XIII and shall be performed on sample units from lots that have been subjected to and met group A and group B inspection.

4.5.5 Reinspection of Conforming Group C and Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and pass group C or D inspection, as appropriate, may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Preconditioning.

4.6.1 Bounce Preconditioning. Each equipment produced shall with shock mounts (if any) removed or blocked, be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of three inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute.

4.6.2 Burn-in Debugging.

4.6.2.1 Test Conditions. Every equipment shall be subjected to a 24 hour burn-in period. The last 12 hours of burn-in shall be failure free. Failure criteria is to be determined by 4.16.9. Each unit which fails during the 12 hours failure free period shall be repaired and returned to test until it successfully survives a 12 hour period without failure. In addition, where practical, all spare modules produced shall be subjected to the same test conditions (above). The test environment utilized during burn-in shall be IAW Appendix B, Category 1, 50.1 of MIL-STD-781.

4.6.2.2 Compliance. Units passing burn-in debugging shall be submitted to group A, B and either C or D inspection in that order without benefit of repair, cleaning, realignment, or other changes, unless specified herein. If after successful completion of burn-in debugging the equipment fails either the group A or B tests, the equipment will be reworked and resubmitted to burn-in debugging.

4.7 Visual and Mechanical Inspection of Equipment. Equipment shall be examined for the defects listed in MIL-STD-252. (see 3.20).

4.8 Inspection for Dimensional Interchangeability. The dimensions listed below shall be gauged to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

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b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separate parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts on the plug-in assembly or maior unit.

4.9 Functional Requirements for Terminal, Communications AN/UGC-74B(V). The requirements identified within the subparagraphs of paragraph 3.8 of this specification shall be verified during First Article or the Quality Conformation Inspections.

4.10 AN/UGC-74B(V) Siganl Characteristics and Electrical Requirements Test. The requirements identified within the subparagraphs of paragraph 3.9 of this specification shall be verified during First Article and the Quality Confirmation Inspections. The test requirements will be modified as follows:

4.10.1. Low Level Digital Interface Tests. The equipment shall be tested for compliance with 3.9.3.2. The test procedure shall be in accordance with MIL-STD-188-114, dated 24 March 1976, Section 5.1 (exclusions and changes in 3.9.3.2 shall be applicable).

4.10.2 Overvoltage and Transients Test. The equipment shall be stabilized by operating it for an hour before performing any test. All tests required to verify compliance with 3.9.7.3.1 and 3.9.7.3.2 shall be performed at 77 degrees +/- 5 F.

4.11 Keyboard Transmitter TT-745/UGC-74B(V) Requirements Test. The requirements identified within the subparagraphs of paragraph 3.10 of this specification shall be verified during First Article and the Quality Conformation Inspections.

4.12 TEMPEST. The equipment shall be subiected to the First Article and Group D tests specified in MIL-T-49261. The sampling rate shall be one unit tested per 600 produced under the contract. Samples shall be selected and tested from the initial production lot and at the beginning of each successive lot.

4.13 Electromagnetic Compatibility. The equipment shall be tested for compliance with the requirements of 3.13. Test procedures shall be in accordance with the test methods of MIL-STD-462 as implemented by a contractor-prepared, Government approved test plan. The subsequent Group D sampling shall be one unit from each 600 produced under the contract.

4.14 Nuclear Hardness Validation. The following nuclear tests shall be performed: EMP, gamma dose-rate, neutron fluence, and gamma total dose. The tests shall be in accordance with 4.14.1 through 4.14.3 on one first article

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unit. All items delivered after first article testing will require certification from the contractor that all active and passive components are equivalent to those delivered with the first article equipment. The contractor shall also certify that there has been no change in material or thickness of the combination cases, and that workmanship of the cases for EMP survival is at least equal to that of the acceptable first article equipment.

4.14.1 Tests and Analyses. Tests and analyses shall be performed on the equipment in its intended operating configuration to assure the system nuclear survivability as specified in 3.14. Specific areas of consideration are discussed in paragraphs 4.14.2 through 4.14.3 inclusive, as required in the contract. Calibration certification of the test facilities to be used are required by the Contracting Officer Representative COTR, 45 days prior to nuclear test. Verification of the wave shape of the test simulator is required. Calibration and wave shape verification shall be supplied to the COTR by the contractor.

4.14.2 Electromagnetic Pulse (EMP). The equipment shall be subjected to the EMP environments as required in 3.14. The equipment shall be placed within a ground-effect electromagnetic pulse facility in the following orientation:

a. Connecting cables extended parallel to the electric field and perpendicular to the magnetic field; all individual units must be in the test volume but widely separated.

The equipment shall be exposed at threat level in the above orientation and in each operating mode of the equipment. The equipment shall meet the operational requirements of Section 3 upon completion of threat level tests. Where necessary, test data to satisfy the EMP requirements may be obtained by extrapolation of diagnostic test levels to threat level and injection pulsingat the calculated level.

4.14.3 Initial Nuclear Radiation. For the purpose of design, but not validation testing, electronic pieceparts and circuits for which nuclear test data are available or for which an acceptable level of confidence has been established through mathematical analysis need not be subjected to further nuclear design testing. Acceptance of these electronic pieceparts, circuits and mathematical analyses will be determined by the procuring agency from data supplied to the Government by the contractor.

4.14.3.1 Neutron Fluence. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Neutron Fluence required in 3.14. The equipment shall then perform as specified in Section 3.

4.14.3.2 Gamma Total Dose. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Gamma Total Dose required in 3.14. The equipment shall be energized during exposure and shall then perform as specified in Section 3.

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4.14.3.3 Peak Gamma Dose Rate. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the peak Gamma dose rate required in 3.14. The equipment shall be energized during exposure and shall then recover and perform as specified in 3.14.

4.15 Environmental. The following tests shall be performed on the AN/UGC-74B(V) (see 3.15). Equipment operation shall be verified in accordance with 4.19.

4.15.1 Temperature.

4.15.1.1 High Temperature. The equipment shall be subjected to the test of Method 501.2, Procedure II of MIL-STD-810 for cycling exposure. In Step 12, raise the internal chamber temperature to 120 F (49 C). Maintain internal chamber temperature for 6 hours at 120 F (49 C). Raise the internal chamber temperature to 155 F (68 C) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to 120 F (49 C) within a time period of one hour. Repeat the above steps two additional times to make a total of three 12-hour cycles. Step 14 shall be done as follows: For all units, adjust the internal chamber temperature to 155 F (68 C) and maintain this temperature until the test item stabilizes. Operate the test item and obtain data in accordance with the approved test plan.

4.15.1.2 Low Temperature. The equipment shall be subjected to Test Method 502.2, Procedure I of MIL-STD-810. The storage temperature in step 2 shall be -65 F (-54 C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (step 5) shall be -25 F (-32 C) and the verification test in accordance with 4.19 shall be performed (see 3.15.1). A special low temperature ribbon may be usedduring the low temperature test (see 3.15.1).

4.15.2 Humidity. The equipment with case open shall be subjected to the test Method 507.2, Procedure III, Aggravated of MIL-STD-810 with the following modifications:

a. Step 5 shall be increased to 32 hours

b. Step 6 shall repeat steps 2, 3, 4 & 5 for a total of 5 cycles (not less than 240 hours)

The equipment shall be operated once in every 24 hour period (4.19) and during last 5 hours of the final cycle of step 5. This test shall be performed with the required roll of paper installed. However, if the condition of the paper is such that it will interfere with proper operation of the equipment, then changing the roll of paper is permitted. After the humidity test, the verification test in accordance with 4.19 shall be performed (see 3.15.2).

4.15.3 Altitude. The equipment shall be subjected to test Method 500.2, Procedure II of MIL-STD-810. The chamber internal pressure of step 2 shall be 15,000 feet above sea level. The test procedure shall include a 40,000 feet

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altitude nonoperational exposure for not less than 1 hour previous to the 15,000 feet altitude operational portion of step 2. The test item operation is to be verified during the elevation phase of step 2 at 15.000 feet above sea level in accordance with test requirements of 4.19 (see 3.15.3).

4.15.4 Immersion. The equipment shall be subjected to the test of Method 512.2, Procedure I of MIL-STD-810 (see 3.15.4).

4.15.5 Vibration.

4.15.5.1 Tracked Vehicle Configuration. The equipment shall be subjected to the test of MIL-STD-810, Method 514.3, Procedure 1, Category 8 (Ground Mobile). The vibration level shall be in accordance with the attached test schedules (DWG-00071 through 00079). See Figures 6 through 14. Test time shall be 2 hours in each of three mutually perpendicular directions for a total of 6 hours. The equipment shall be mounted on isolators.

In addition, isolated equipment shall also be vibrated with the isolators removed or blocked out as follows:

Transverse Axis	0.0125 g2/Hz (5-500 Hz)
Longitudinal Axis	0.0075 g2/Hz (5-500 Hz)
Vertical Axis	0.005 g2/Hz (5-500 Hz)

With the isolators removed or blocked, the equipment shall be vibrated for 30 minutes in each of three mutually perpendicular directions for a total 1-1/2 hours.

Prior to, during (10 minutes of each hour), and following these tests, theequipment shall be operated in accordance with MIL-STD-810 general test performance guidance paragraph 4.5. The equipment shall not have any degradation in electronic performance and shall not suffer any mechanical damage.

4.15.5.2 Wheeled Vehicle Configuration. The equipment shall be subjected to the test of MIL-STD-810, Method 514.3, Procedure 1, Category 8 (Ground Mobile). The vibration level shall be in accordance with Figures 514.3-7 through 514.3-9 of MIL-STD-810 with the following exceptions:

a. The first breakpoint for the vertical axis (Figure 514.3-7) and the longitudinal axis (Figure 514.3-9) shall be 6 Hz.

b. The first breakpoint for the transverse axis (Figure 514.3-8) shall be at 6 Hz and the corresponding power spectral density shall be .18724 g2/Hz.

Prior to, during (10 minutes of each hour), and following these tests, the equipment shall be operated in accordance with MIL-STD-810 general test performance guidance paragraph 4.5. The equipment shall not have any degradation in electronic performance and shall not suffer any mechanical damage. Test time shall be 4 hours in each of three mutually perpendicular directions for a total of 12 hours. The equipment shall not be mounted on isolators.

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4.15.5.3 Bounce, Loose Cargo. The equipment in its field transport mode complete with full paper supply roll and cover-stored items (copy holder, AC power cable, and loopback plug) shall be subjected to the test of Method 514.3, Procedure II, Category 3, of MIL-STD-810. After each 1/2 hour cycle, the verification test in accordance with 4.19 should be performed (see 3.15.5.1). The fences of the plywood package tester shall be positioned so that the horizontal motion of the test item is limited to not more than two inches in a direction parallel to the axis of the shafts. In a direction perpendicular to the axis of the shafts, the fences shall be positioned at a distance sufficient to ensure that the unit will not rebound from fence to fence.

4.15.6 Shock. Proper operation of each equipment after each shock test shall be checked by performing the verification test in accordance with 4.19 (see 3.15.6).

4.15.6.1 Drop. The equipment shall be subjected to the test of Method 516.3, Procedure IV of MIL-STD-810 for equipment weighing over 100 pounds. Minor damage (see 6.4.1) is permitted (see 3.15.6.1).

4.15.6.2 Bench Handling. The equipment shall be subjected to the test of Method 516.3, Procedure VI of MIL-STD-810 (see 3.15.6.2).

4.15.7 Ballistic Shock. (See 3.15.7.)

a. Use the shock testing machine for lightweight equipment as shown inFigure 1 of MIL-S-901.

b. Mount the unit under test on a shock mount. The use of CECOM shock mount (drawing SC-D972467) is recommended but not mandatory. Fasten the shock mounted unit under test to the test plate in its normal operating position.

c. Establish compliance with the operational requirements under standard ambient environmental conditions.

d. Maintenance of software integrity shall be verified for each blow via the following procedure:

(1) Set the unit under test to the ICT operational mode and apply power.

(2) Fill the message memory of the unit under test.

(3) Subject the unit under test to the shock blow.

(4) The unit under test shall accurately print the received message both before and after each blow.

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e. Subject the unit under test as mounted and in accordance with the operational requirements of paragraph 4.15.7d above to a total of nine blows, one each of the following: 1 foot blow, 3 foot blow and 5 foot blow on the back, side and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. Minor damage (see 6.4.1) is permitted.

4.15.8 Dust. The equipment shall be subject to the test of Method 510.2, Procedure I of MIL-STD-810. The equipment shall be in operating configuration during this test. After the dust test, the verification test shall be performed in accordance with 4.19. Penetration of dust into the equipment without causing an operational problem shall not be regarded as a failure (see 3.15.8).

4.15.9 Rain. The equipment shall be subjected to the test of method 506.2, Procedure I of MIL-STD-810. The equipment shall be powered on during the rain test, however, printing/feeding of paper shall not be required during rainfall. The paper shall be torn off against the tear bar after initialization and will remain in this position during the rain test.

After the rain test, the verification test in accordance with 4.19 shall be performed. Penetration of rain into the equipment without causing an operational problem shall not be regarded as a failure (see 3.15.9). Operator assistance as required for paper movement during operation is allowed.

4.15.10 Salt Fog. The equipment shall be subjected to the salt fog test specified in Method 509.2, Procedure I of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. After the salt fog test, the verification test shall be performed in accordance with 4.19 (see 3.15.10).

4.15.11 Orientation. The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left and right) to an angle of 20 degrees, plus 3, minus 1 degree. During inclination in each plane, the equipment shall be operated and the verification test in accordance with 4.19 shall be performed (see 3.15.11).

4.15.12 Fungus. The equipment model in the assembled and As Ready For Delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to test Method 508.3 of MIL-STD-810 except that the temperature humidity conditions of step II-3.2.2a shall be maintained at a constant 84 C (29 F) and 95% RH (minimum) during the life of the test. The equipment test model or samples shall not be cleaned for 72 hours prior to the fungus test. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 28 days, based on visual examination, the equipment assembly or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other

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than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall fail if it shows more than the growth specified above.

4.15.13 Air Seal Test. The equipment shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter the equipment, as field transported, shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the combination case. The vacuum then shall be valved off and the interior pressure measured during the ensuring period of one minute. During this one minute period the decrease in the vacuum shall not exceed 0.01 pound. The gauge used for measurement of this vacuum shall be of such accuracy that a difference of 0.01 pound can be determined accurately (see 3.15.13).

4.16 Reliability Testing. The reliability acceptance test shall be conducted every other month, starting in the first month of production and using Test Plan VIIC of MIL-STD-781. The minimum number of randomly selected equipment to be tested shall be eight and the maximum number shall be twelve from each two months production. The sample (8 minimum, 12 maximum) shall be divided approximately even between units randomly selected from each of the two consecutive months production. The specified MTBF (O) shall be as specified in 3:16.

4.16.1 Test Level. All reliability testing shall be conducted under the environment of Category 1, 50.1 (Appendix B) of MIL-STD-781 except that the keyboard shall be tested at ambient level room temperature during itsoperational cycle. During test, all equipment under test will be mounted without shock mounts in the normal attitude. Ten minutes out of every hour, the equipment shall be vibrated vertically with a sinusoidal motion having a frequency between 20 and 60 Hz. Throughout the test, the ambient will be maintained at 122 degrees plus nine degrees minus zero degrees Fahrenheit. Humidity will not be controlled except that it will not be allowed to go below 5%. The chamber temperature and equipment vibration shall be monitored and recorded in the test log.

4.16.2 Duty Cycle. Each equipment shall have a 100% duty cycle. The keyboard shall have 50 percent duty cycle operating at a 50 words-per-minute typing rate. Keyboards shall be alternated on a once a week basis between the exerciser and operating modes.

4.16.2.1 Exerciser. The keyboard shall be mounted at room temperature on a keyboard exerciser and each of the keys shall be depressed sequentially at a five character per second rate. All operating input voltages shall be applied to each keyboard while it is being exercised. The keyboards shall continue to be exercised for 168 hours.

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4.16.2.2 Operating Mode. Each keyboard shall be mounted on its printer in the normal operating mode. Each equipment shall continue being subjected to the vibration and temperature environments described previously. After 24 hours of testing, each key switch shall be operated sequentially 5 times (in the temperature environment) with the printer monitoring the keyboard. The keyboards shall continue to be operated for 168 hours.

4.16.3 Speed, Code and Interface. During first article and production reliability testing, two samples shall operate at 75 baud, 7 unit BAUDOT code, with a high-level interface, and the remaining samples at the highest speed, 10 unit ASCII code with a low-level interface.

4.16.4 Input Voltage. Operating time of each system shall be equally divided between 24 Vdc and 115V, 60 Hz with a tolerance of +5%, -2%. Half of the systems shall be operated in each condition and alternated on a once a week basis.

4.16.5 Preventive Maintenance. The equipment shall be capable of operation for at least 500 hours between lubrication periods or before adjustments are required to compensate for degradation of performance.

4.16.6 Operator Maintenance. Operator maintenance during this test shall consist of replacing the consumables. All operator maintenance will be recorded in the test log.

4.16.7 Test Time. The minimum test time per system shall be in accordance with MIL-STD-781, Test Plan VIIC. For reliability testing, test time shall include only time when the systems are operating. Off or stand-by time shall not be included in the calculation of the system's reliability.

4.16.8 Sample Selection. The equipment used for reliability testing shall be randomly selected by an authorized Government representative.

4.16.9 Classification of Events.

4.16.9.1 Failure. Any electrical, mechanical, or adjustment malfunction which causes the equipments to print, transmit or receive improperly. Failure will be classified in accordance with paragraph 5.6.1 of MIL-STD-781.

4.16.9.1.1 Nonrelevant Failures. The following failures will be classified as nonrelevant:

a. Failures caused by operator maintenance.

b. Failures caused by operator error.

c. Normal replacement of consumable items.

d. Failures of test equipment or facilities.

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e. Failures caused by operation of the equipment outside of its specified design.

f. Failures caused by act of God.

4.16.9.1.2 Relevant Failures. All failures except those listed above are relevant failures unless determined otherwise by the procuring activity.

4.16.9.1.2.1 Print Quality Failures. Print samples of any 30 consecutive full lines containing 4 or more character errors (see 6.4.8) shall be a relevant failure.

4.16.10 Errors. An error is an unrelated, nonrepeated failure of undetermined cause.

4.16.10.1 Error Rate. The AN/UGC-74B(V) shall not make more than one error in processing one million characters averaged over the test.

4.16.10.2 Chain of Errors. In the system test to be conducted, an error caused by a unit will cause errors to appear in the other units of the system. Because the system is closed looped these errors will repeat until the error has been eliminated. Only the initial error of the system errors will be counted as one error.

4.16.11 Failure Actions. Paragraph 5.6.2 of MIL-STD-781 shall apply. Entries shall be made on the data sheets for each failure and failed equipment shall be removed from test and repaired. When an intermittent failure occurs the equipment may be allowed to continue on test until the frequency of the occurrence becomes great enough to allow determination of the cause. Until a repair is made, these occurrences will be considered as one failure unless anunrelated new symptom occurs which may then be considered an additional failure. Verification of the repair may be accomplished by operating the equipment up to four hours only if specified in advance (in the test environment, if necessary). This verification time shall not be included as part of the statistical "on time." Failures that occur during this time will not be counted as part of the accept-reject criteria even if they are not related to the initial failure. They will, however, be recorded and reported to the procuring agency. Determination of compliance with the accept/reject criteria shall be in accordance with paragraph 5.7 inclusive of MIL-STD-781.

4.16.12 Conditions Precedent to Reliability Testing. Prior to performing any reliability acceptance testing, the following actions shall be completed in order as follows:

4.16.12.1 Group A and B Testing. Reliability samples shall be selected from lots which have passed group A and B testing.

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4.16.13 Functions to be Performed. During the reliability tests the transmit, receive and print functions shall be performed. Details of the configuration and monitoring required shall be specified in the contractor's reliability test procedure.

4.16.14 Post Test Verification. After completing each reliability test, all test samples shall be resubjected to the group A tests. The results of group A testing shall be recorded and shall become part of the reliability test.

4.16.15 Sample Log Forms. Test records and logs shall be similar to and contain all information as the forms in Figures 3 through 8 of MIL-STD-781.

4.16.16 Test Results. The results of the reliability demonstration testing shall be analyzed as follows:

a. Analyzed in accordance with paragraph 5.13 of MIL-STD-781.

b. Recommendations for improvement of equipment system reliability.

4.17 Acoustics. Tests shall be conducted to measure compliance with the minimum noise levels (during operation) specified in MIL-STD-1474 (modified by Table IV) (see 3.12).

4.18 Verification Test - Terminal, Communications AN/UGC-74B(V).

4.18.1 Purpose. The purpose of this test is to determine if any degradation of the equipment occurred during or after any environmental test to which the equipment was subjected.

4.18.2 Equipment. Communications terminal under test.

4.18.3 Verification Test.

4.18.3.1 Pre-Test Verification.

a. Visually inspect equipment for damaged parts or assembly corrosion and loose fasteners.

b. Turn power on (if necessary).

c. Perform Self Test per 3.8.8.8.

d. Verify correct operation per 3.8.8.8.

4.18.3.2 Verification During Environmental Exposure.

a. Turn power on (if necessary).

b. Perform Factory Self Test per 3.8.8.9.

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c. Verify correct operation per 3.8.8.9.

4.18.3.3 Post Test Verification.

a. Visually inspect equipment for damaged parts or assembly corrosion and loose fasteners.

b. Turn power on (if necessary).

c. Perform Self Test per 3.8.8.8.

d. Verify correct operation per 3.8.8.8.

4.18.4 Acceptance. The criterion of successful completion is that the equipment passes each test above and meets the requirements of Section 3, Paragraph 3.2.6, MIL-STD-810 upon completion of tests.

4.18.4.1 Print Quality Acceptance. Print samples of any 8 full lines containing less than 4 character errors (see 6.4.8) shall be considered acceptable.

4.18.4.1.1 Vibration Print Quality Acceptance Exceptions. During exposure to vibration (see 4.15.5.1 and 4.15.5.2) and performance of 4.19.3.2, the following exceptions to 6.4.8 shall apply:

a. Margin shift of greater than 1 character shall be acceptable if the message is legible or repeated.

b. Multiple Line Feed and Reprint shall be acceptable provided the unit prints within 8 seconds.

c. For hard mount tracked vehicle, print quality failures shall be non-relevant.

4.19 Maintainability Demonstration. The maintainability requirements shall be demonstrated in accordance with MIL-STD-471, Test Method 9. The sample size (nc) of corrective maintenance tasks to be performed for the demonstration of the mean corrective maintenance time (uc) shall be fifty (50) at each specified maintenance level. The specified mean corrective maintenance time (uc) shall be 30 minutes at the organizational level and 2.5 hours at the direct support level.

a. The consumers risk (B) for demonstration of uc shall be 10%. The equipment will be considered to have met the requirements when the specified parameters have received accept decisions. If the specified parameters fail to meet the accept decisions, then design corrective action shall be initiated at contractor's expense. When proper design corrective action kas been completed, the equipment shall be retested at those levels of maintenance that previously failed. The same accept-reject criteria shall apply in the retest phase.

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b. The contractor shall be responsible for data collection during the maintainability demonstration. This shall be accomplished by proper use of an accurate time recorder.

The time shall be recorded on appropriate data sheets. These data sheets shall have provisions for the recording of the following times on each task to be demonstrated:

(1) Preparation
(2) Localization
(3) Isolation
(4) Disassembly
(5) Interchange
(6) Reassembly
(7) Alignment
(8) Verification Checkout

c. While initiating corrective action during the demonstration, the maintenance technician shall follow procedures outlined in the equipment technical manual and none other, except where approved by the procuring activity.

d. The demonstration test team shall consist of:

(1) A contractor director - introduces faults within the equipment

(2) A contractor maintenance technician - performs maintenance tasks

(3) A contractor data collector

(4) Government witnesses

e. Maintenance tasks to be simulated shall be selected by Government representatives from a pool of proposed maintenance tasks submitted by the contractor at least two weeks prior to the actual demonstration. Task selection shall be in accordance with MIL-STD-471, Appendix A. Also, the pool of proposed maintenance tasks shall be subject to Governnent approval.

f. Fault simulation shall be accomplished in the following manner:

(1) Introducing an internally shorted or opened component.

(2) Introducing a similar component with a different value or rating.

(3) Introducing misalignment by detuning.

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(4) Other contractor submitted and approved methods.

4.20 Weight. Weigh the equipment to verify conformance to 3.19.

4.21 Systems Safety Test. A visual and mechanical inspection shall be performed to verify compliance with those portions of 3.17 which can be determined visually.

5. PACKAGING

5.1 Packaging Requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.1).

6. NOTES

6.1 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendnent thereto.

b. Type required.

c. Date of DODISS (see 2.1.1).

d. The number of 1st article samples (see 4.3.1).

e. Technical literature required.

f. Quantity and kind of running spare parts required.

g. List of cable assemblies to be furnished by the Government.

h. Nuclear test calibration certification (see 4.14.1).

i. Levels of preservation and packing (see section 5).

6.2 Cables. The following cables (one each are required) are used with, but not part of the AN/UGC-74B(V). If not furnished as GFP, the contractor shall fabricate cables as required for testing purposes.

a. Cable Assy, Power, Electrical, 26 VDC, Drawing SM-D-764480

b. Cable Assy, Power, Electrical, 115 VAC, Drawing SM-D-764481

c. Cable Assy, Power, Electrical, 230 VAC, Drawing SM-D-764482

d. Cable Assy, XMIT/RCV High Level, Drawing SM-D-915889

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e. Cable Assy, Lo Level Data, XMIT/RCV, Drawing SM-D-915896

f. Cable Assy, Lo Level XMIT/RCV Clock/Data, Drawing, SM-D-915897

g. Cable Assy, Battery Back-Up, Drawing SM-D-915890

h. Cable Assy, KW-7, Drawing SM-D-964513

i. Cable Assy, KY-57, Drawing SM-D-964514

j. Cable Assy, KG-84, Drawing A3049250

6.3 Verification Inspection. Verification by the governnent will be limited to the amount deemed necessary to detennine compliance with the contract and will be limited to severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.4 Definitions.

6.4.1 Minor Damage. Minor damage is defined as chips and scratches of the paint finish and dents in the combination case that do not interfere with:

a. Sliding the equipnent chassis in and out of its combination case to replenish inking ribbon or paper.

b. Compliance with the requirements of the Verification Test, paragraph 4.19.

c. Compliance with the requirements of the Imnersion Test, paragraph 4.15.4.

6.4.2 Characteristic Distortion. Distortion caused by transient or ripple voltage deviations occurring after some specified unit pulse interval. Only deviations occurring after 35% of the space signal time interval that deviate more than +0.15 or -0.075V from the nominal +1.5V space signal are to be considered as not meeting requirements (NIK Only).

6.4.3 Error. An error is an unrelated, nonrepeated event of undetermined cause.

6.4.4 Failure. A failure is defined as any electrical, mechanical or adjustment malfunction which causes the equipment to operate improperly.

6.4.5 Performance Requirements of the Equipment. Whenever referenced in this specification, the 'performance requirements of the equipment' is to be understood as a reference to all the electrical and mechanical performance specified and implied by this equipment specification.

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6.4.6 Module. The smallest subassembly that is replaced during a repair activity. A module is an integrated set of components (see 6.4.7) that is removed and replaced as a whole. An example of a module is a printed wiring card and its electronic components. A module may be either repairable or nonrepairable.

6.4.7 Component. The smallest part that is replaced during a repair activity. All components are considered to be nonrepairable throwaway items.

6.4.8 Character Errors. Each occurrence of any of the following items shall be considered a "character error":

a. Missing dots in a character.

b. Overprinting of dots of adjacent characters and lines (dots touching adjacent characters and lines is acceptable, overlapping dots are unacceptable).

c. Illegible characters (may be caused by compression or stretching of characters or light print).

d. Incorrect character printed.

e. Margin shift of greater than one character space.

6.5 Nomenclature. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Color. The color chip furnished by the procuring agency shall be CARC Forest Green or 383 Green per MIL-C-46168, and may be obtained upon request to the Contracting Officer.

6.7 Government Furnished Equipment (GFE).

6.7.1 System Interface Test. The following GFE will be provided to verify the system interface requirements of 3.9.6; KW-7 (With both BAUDOT and ASCII capability).

6.8 System Configurations. The AN/UGC-74B(V) communications terminal configurations are shown in Table XIV.

6.9 Subject Term (Keyword) Listing.

Communication Terminal
Dot Matrix Printer
Printer
Teletypewriter

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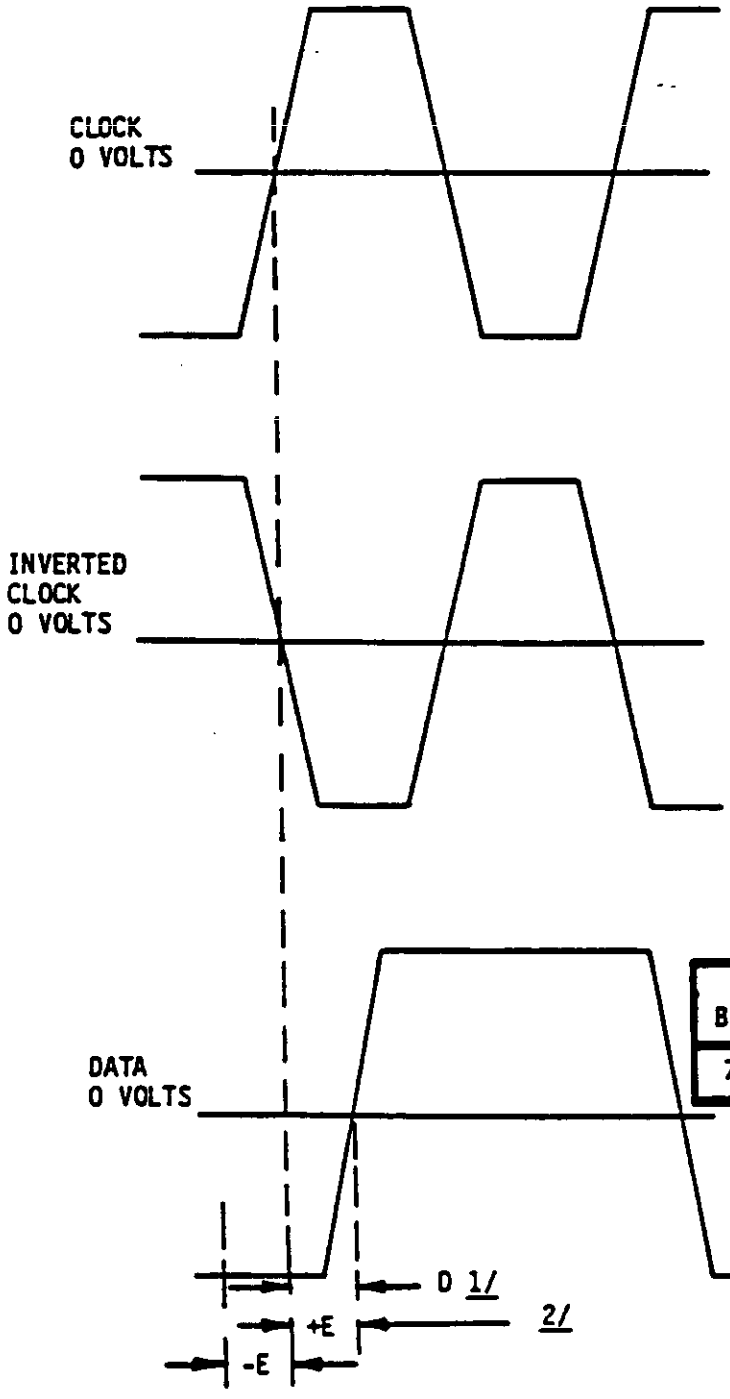
6.10 Changes From Previous Issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerance Percent Defective) from military specifications.

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TABLE XIV. TERMINAL, COMMUNICATION AN/UGC-74B(V) CONFIGURATIONS.

NOMENCLATURE	TELEPRINTER TT-744A/ UGC-74B(V)	KEYBOARD TT-745/ UGC-74A(V)	CASE, TELEPRINTER CY-7735/ UGC-74A(V)	COVER, TELEPRINTER MX-10038/ UGC-74A(V)
TERMINAL, COMMUNCATIONS AN/UGC-74B(V)1	X		X	X
TERMINAL, COMMUNICATIONS AN/UGC-74B(V)2	X		X	
TERMINAL, COMMUNICATION AN/UGC-74B(V)3	X	X	X	X
TERMINAL, COMMUNICATION AN/UGC-74B(V)4	X	X	X	

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BPS	E=4.5% TR CLOCK	D=12% REC CLOCK
75	.6 ms	1.6 ms

1/ D is a maximum of 12% (REC CLOCK) of bit length
2/ E is a maximum of 4.5% (XMIT CLOCK) of bit length

Figure 1. Relationship of Transmitted Data to Clock

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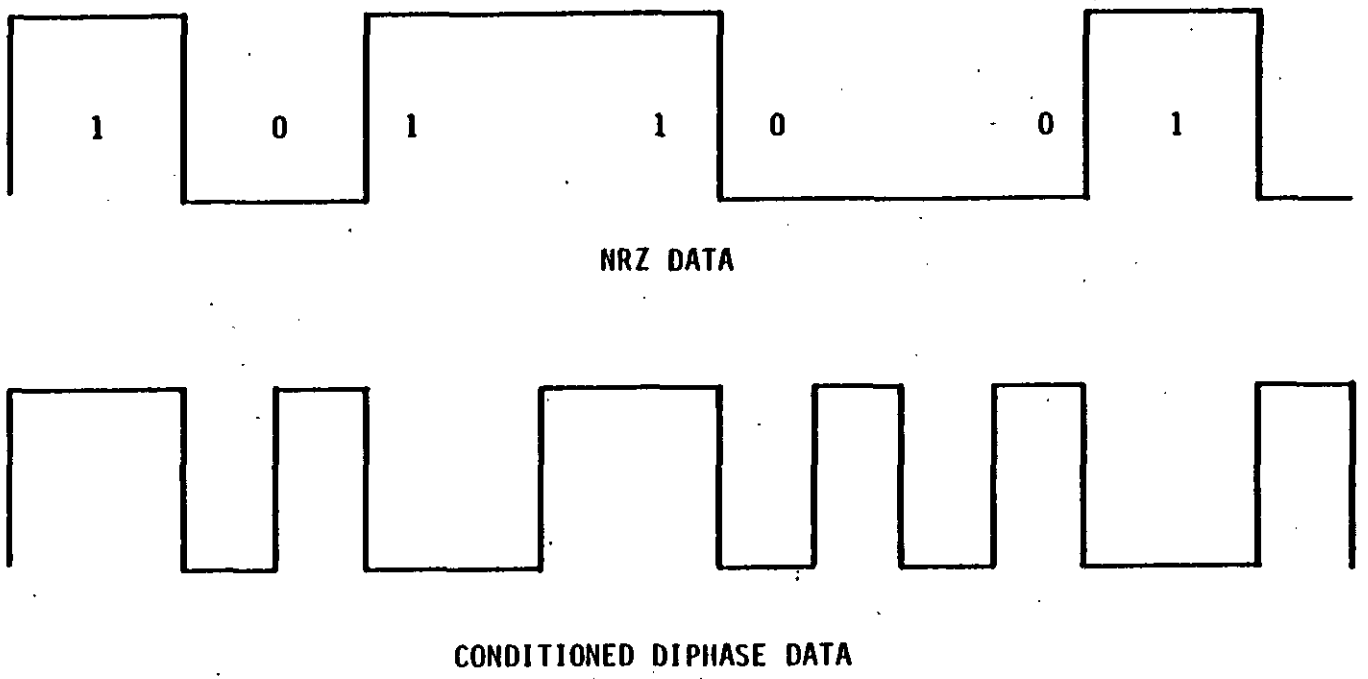
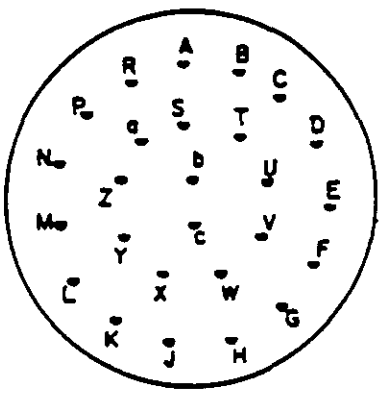


FIGURE 2. CONDITIONED DIPHASE MODULATED OUTPUT

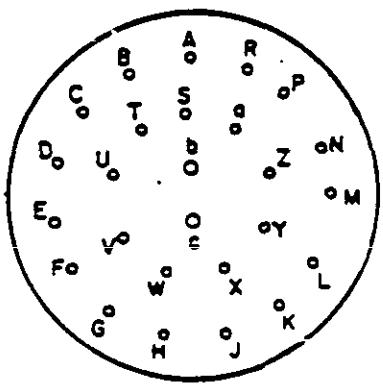
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Figure 3 Communication Connector Pin Assignment

Name	Connector	Description
Receive Clock Hi	A	External Baud Rate Clock Hi
Receive Clock Lo	B	External Baud Rate Clock Lo
Receive Gated Clock Hi	D	Receive Baud Rate Clock Hi
Receive Gated Clock Lo	E	Receive Baud Rate Clock Lo
Transmit Clock Hi	P	Transmit Baud Rate Clock Hi
Transmit Clock Lo	N	Transmit Baud Rate Clock Lo
Receive Data Hi	G	Serial Data In
Receive Data Lo	H	Serial Data In
Transmit Data Hi	K	Serial Data Out
Transmit Data Lo	L	Serial Data Out
Signal Return	S	Signal Return
Shield Ground	T	Shield Ground
Transfer	R	Provides a method for keying radio transmitters; switched to signal ground by the transfer switch of the AN/UGC-74 front panel.



A. CONNECTOR SIDE

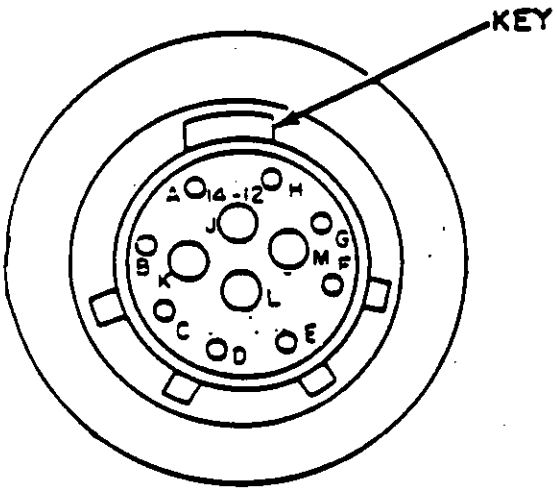


B. SOLDER SIDE

MS 3112E16-26S

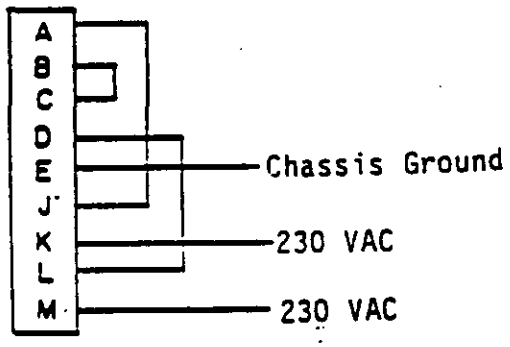
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Figure 4. Pin Assignments for Power Connectors

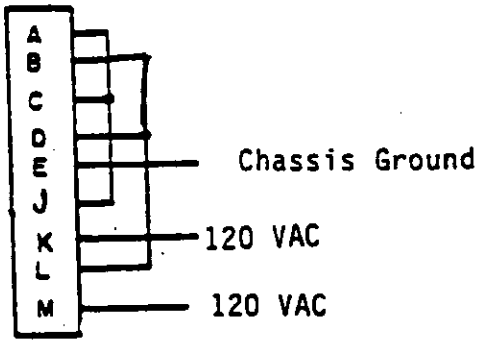


CONNECTOR SIDE

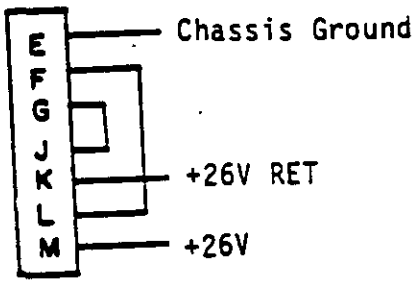
MS3112E14-12P



230 VAC Cable



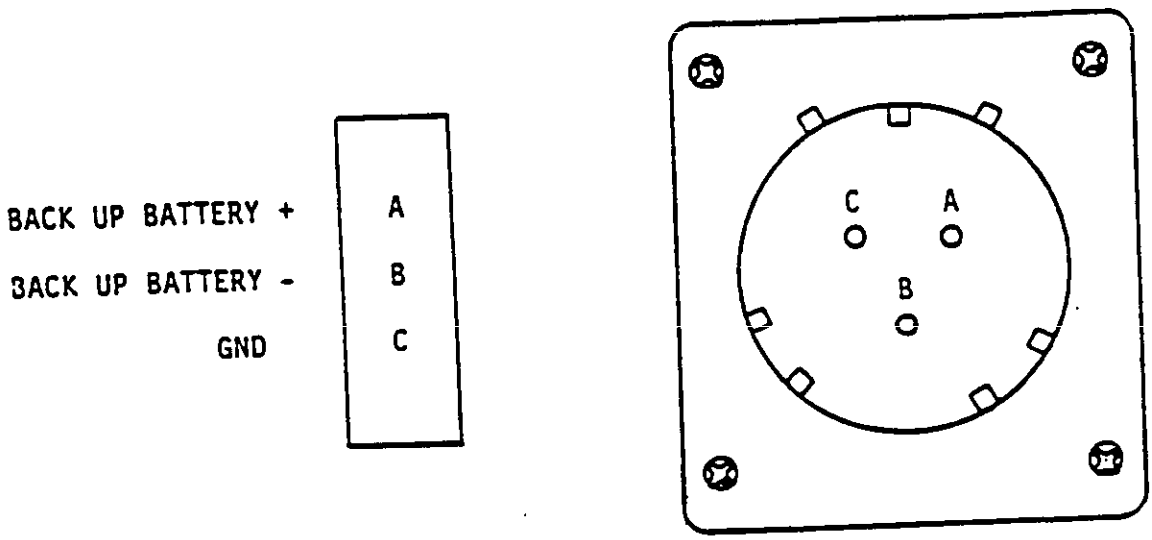
120 VAC Cable



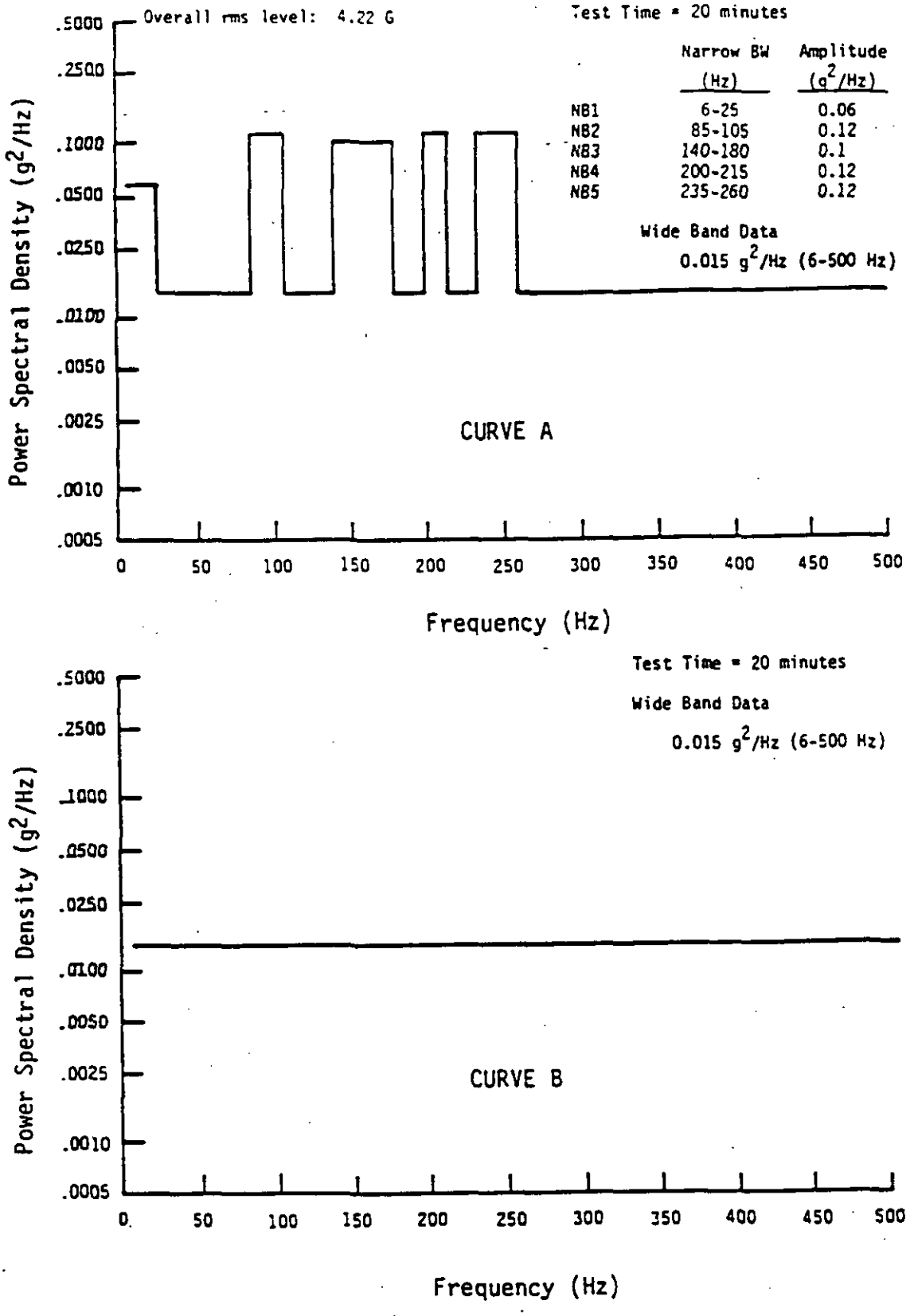
26 VDC Cable

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Figure 5. Battery Back-Up Connector Pin Assignment.



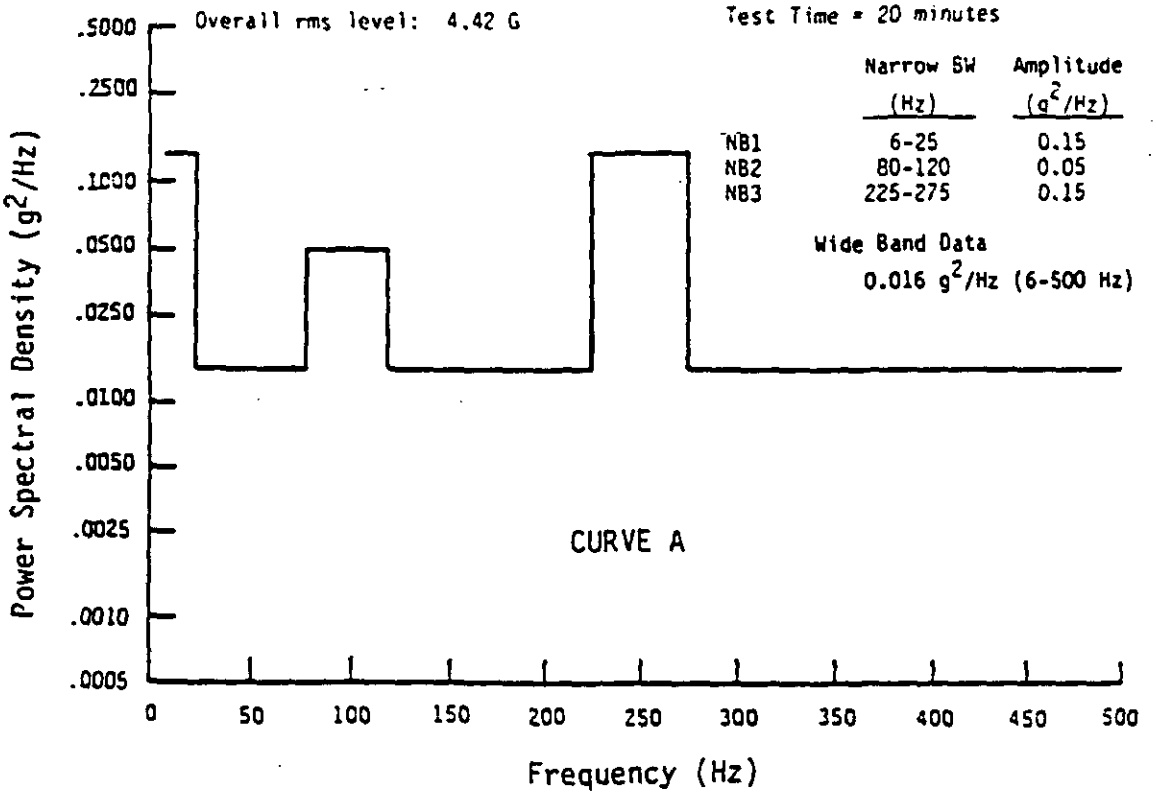
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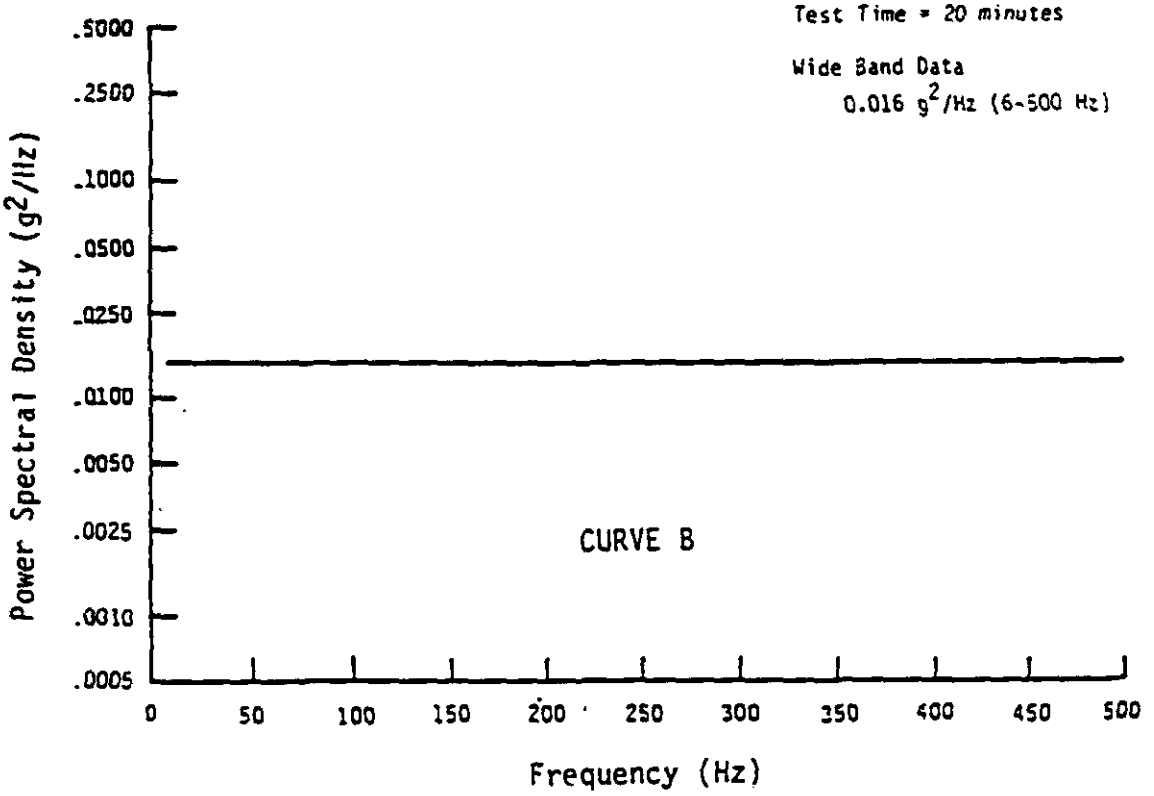


Vibration Level for AN/UGC-74B (M-113 Installation), Transverse Axis, Test Phase 1

Figure 6. DWG-00071

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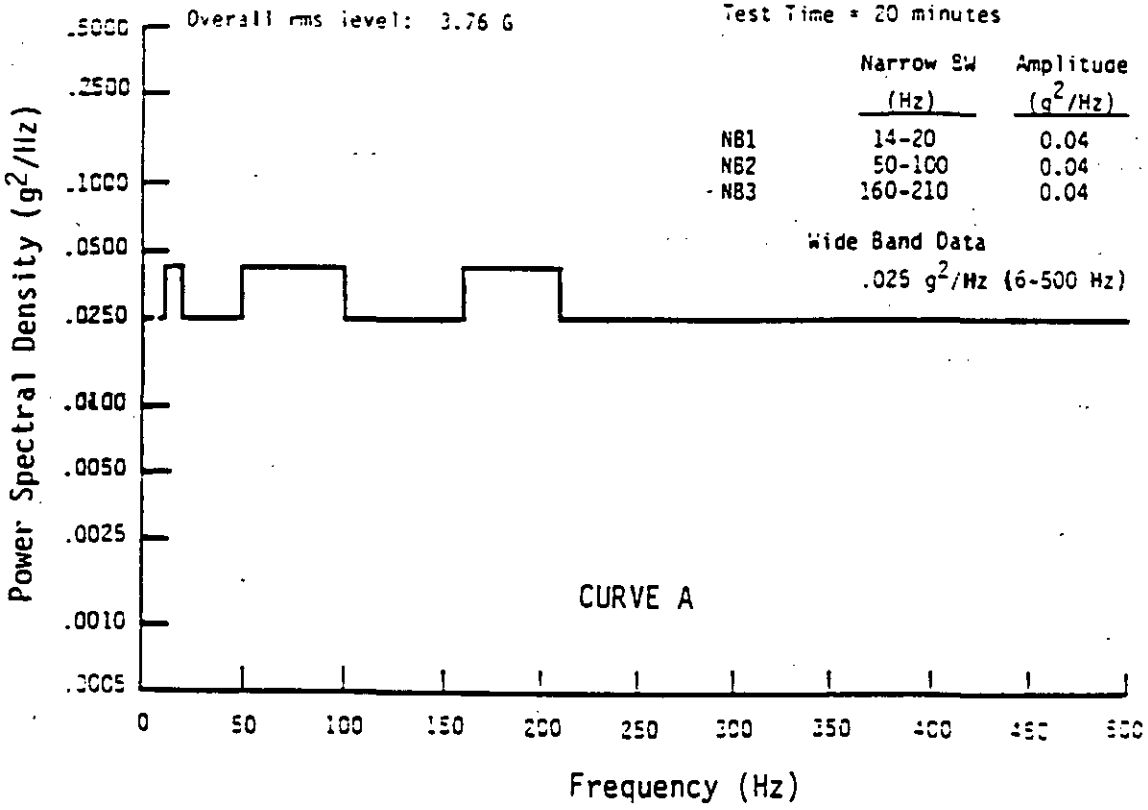


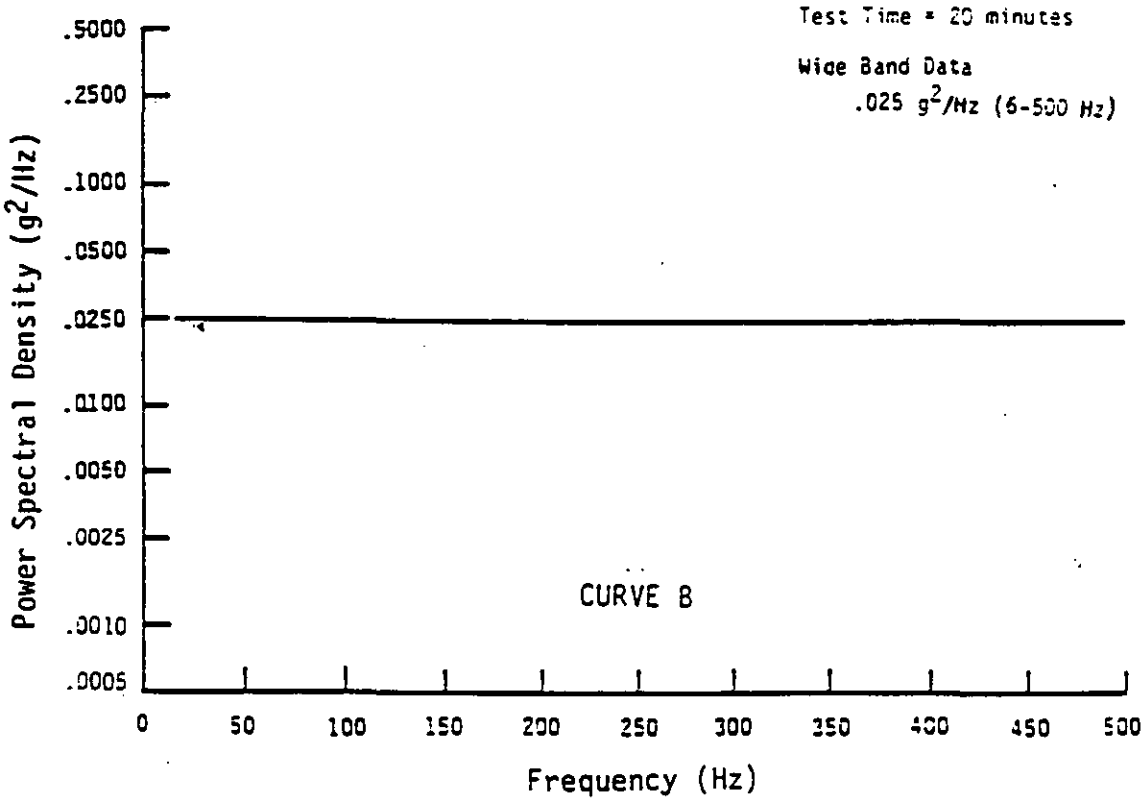


Vibration Level for AN/UGC-74B (M-113 Installation), Transverse Axis, Test Phase 2

Figure 7. DWG-00072

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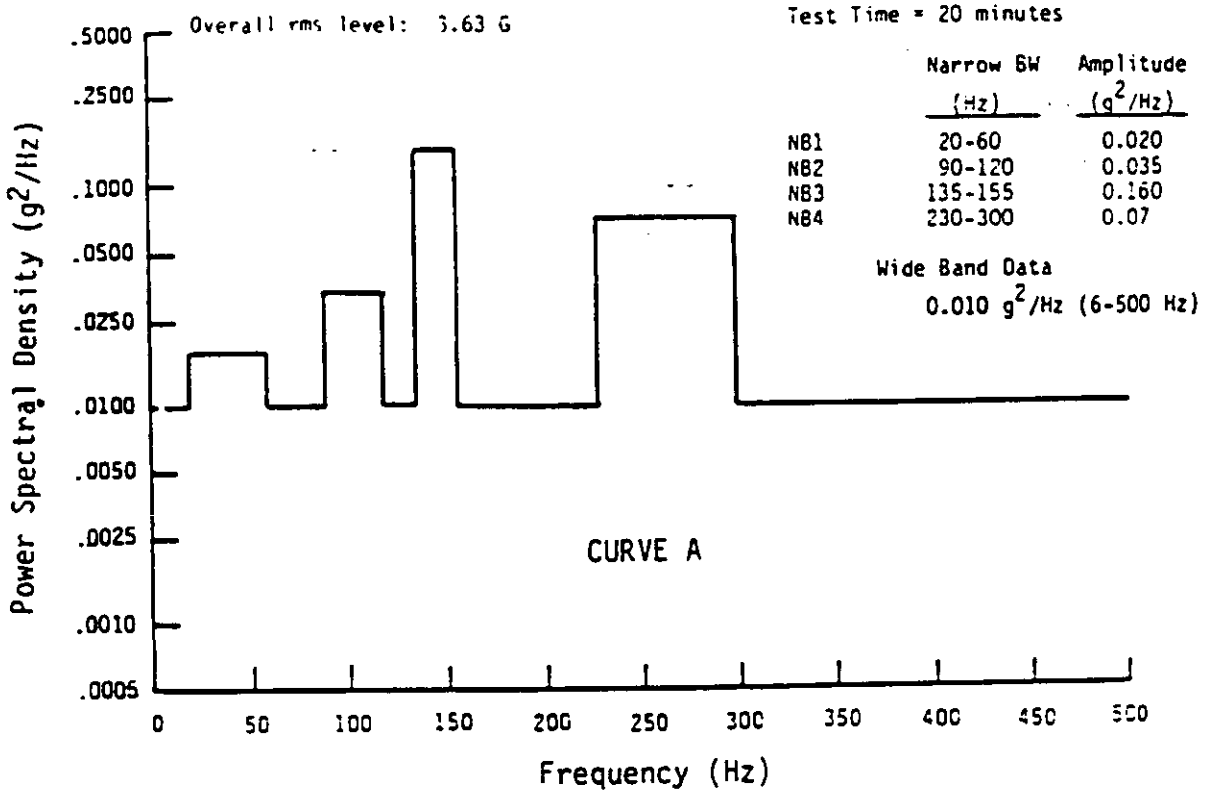


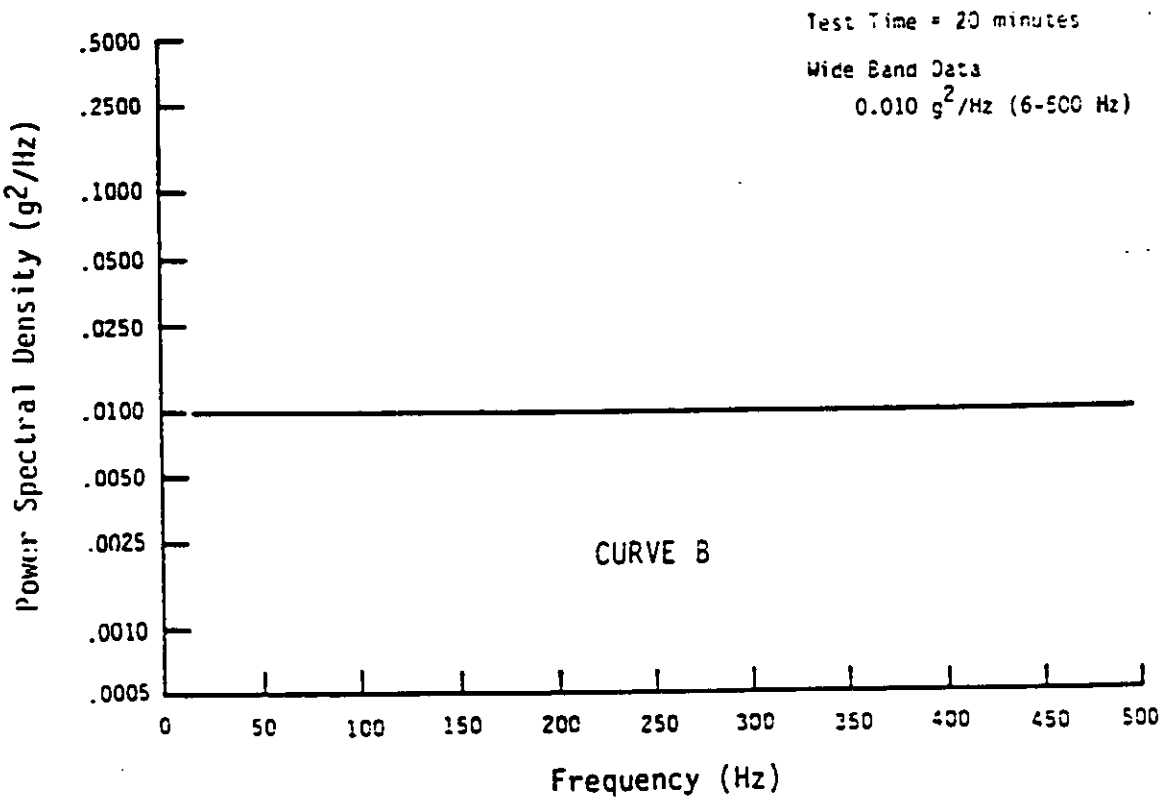


Vibration Level for AN/UGC-74B (M-113 Installation),
Transverse Axis, Test Phase 3

Figure 8. DWG-00073

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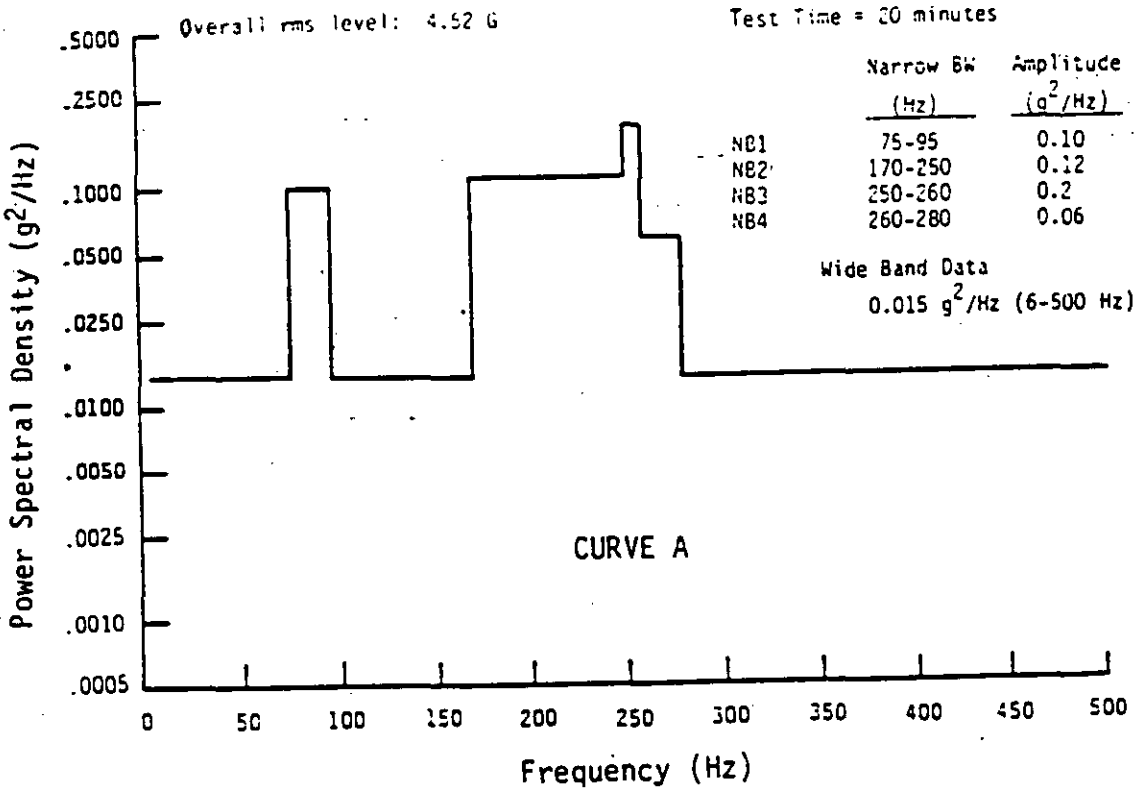


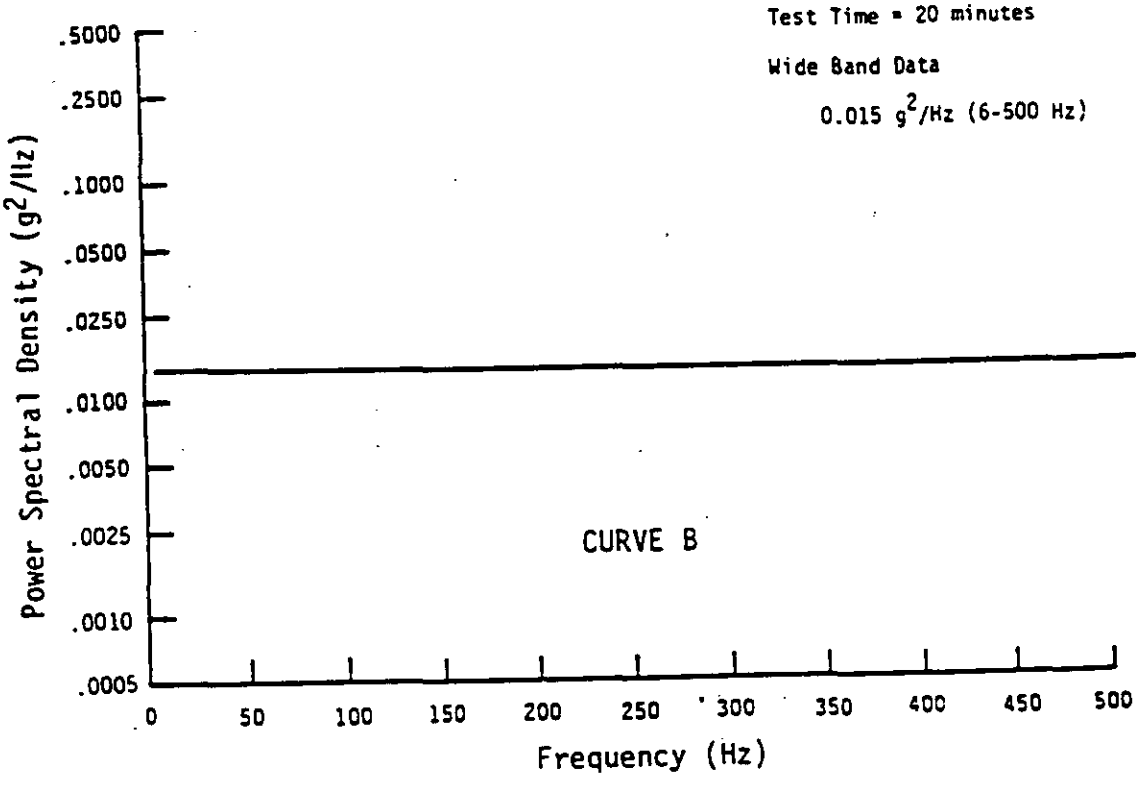


Vibration Level for AN/UGC-74B (M-113 Installation), Longitudinal Axis, Test Phase 1

Figure 9. DWG-00074

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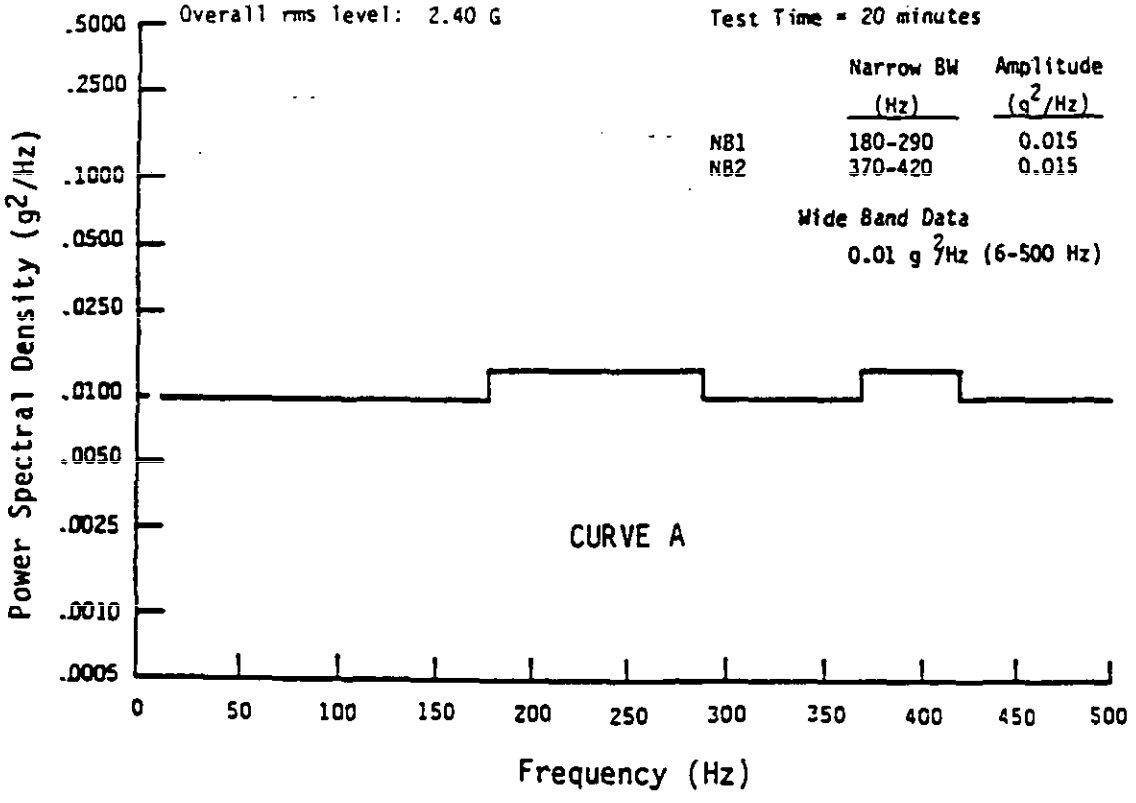


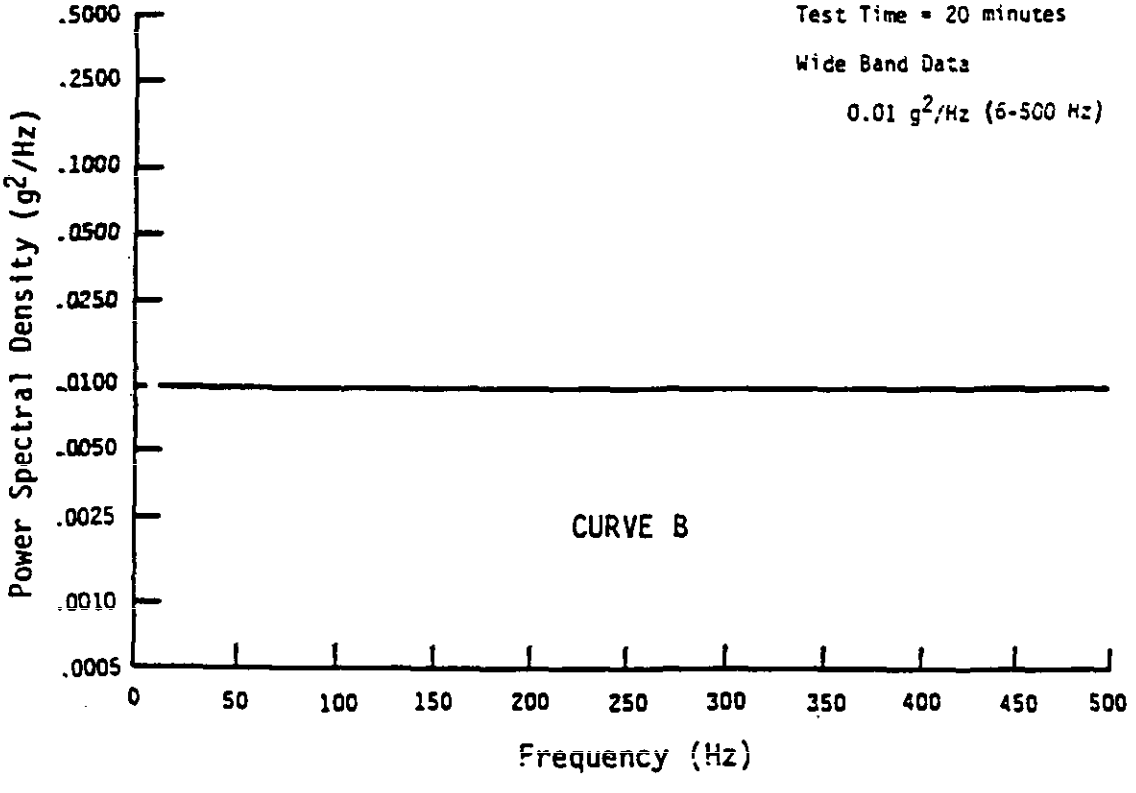


Vibration Level for AN/UGC-74B (M-113 Installation), Longitudinal Axis, Test Phase 2

Figure 10. DWG-00075

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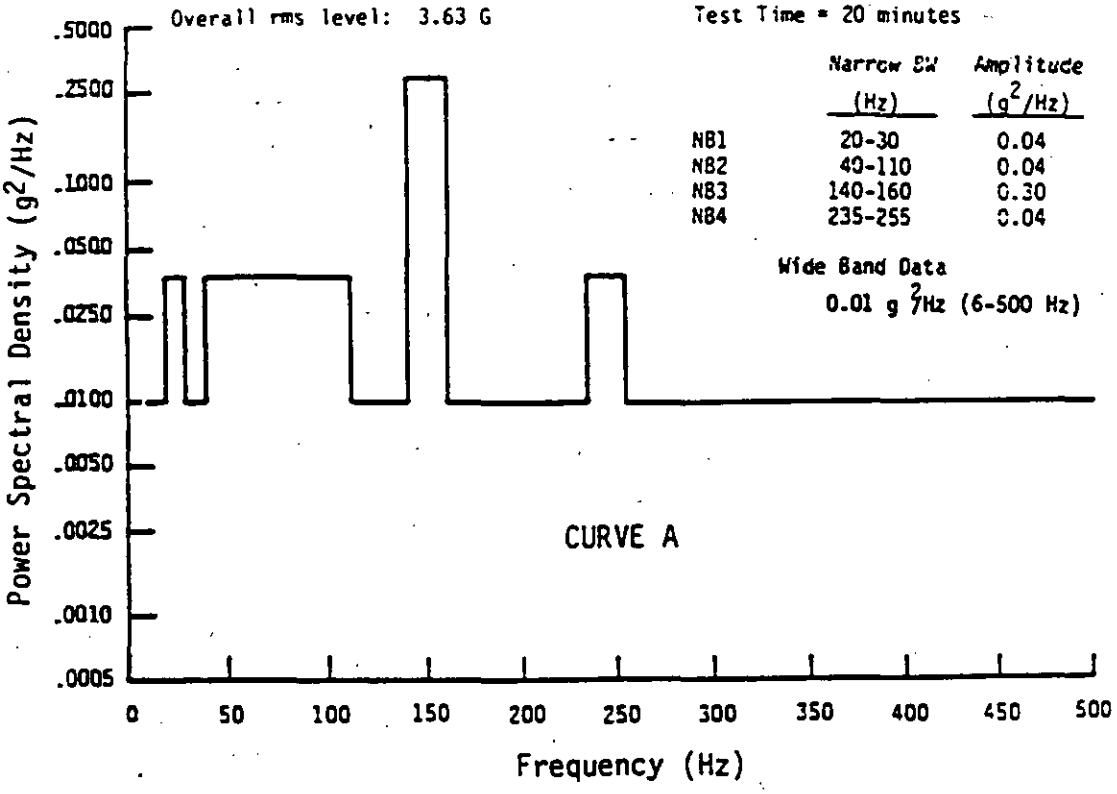


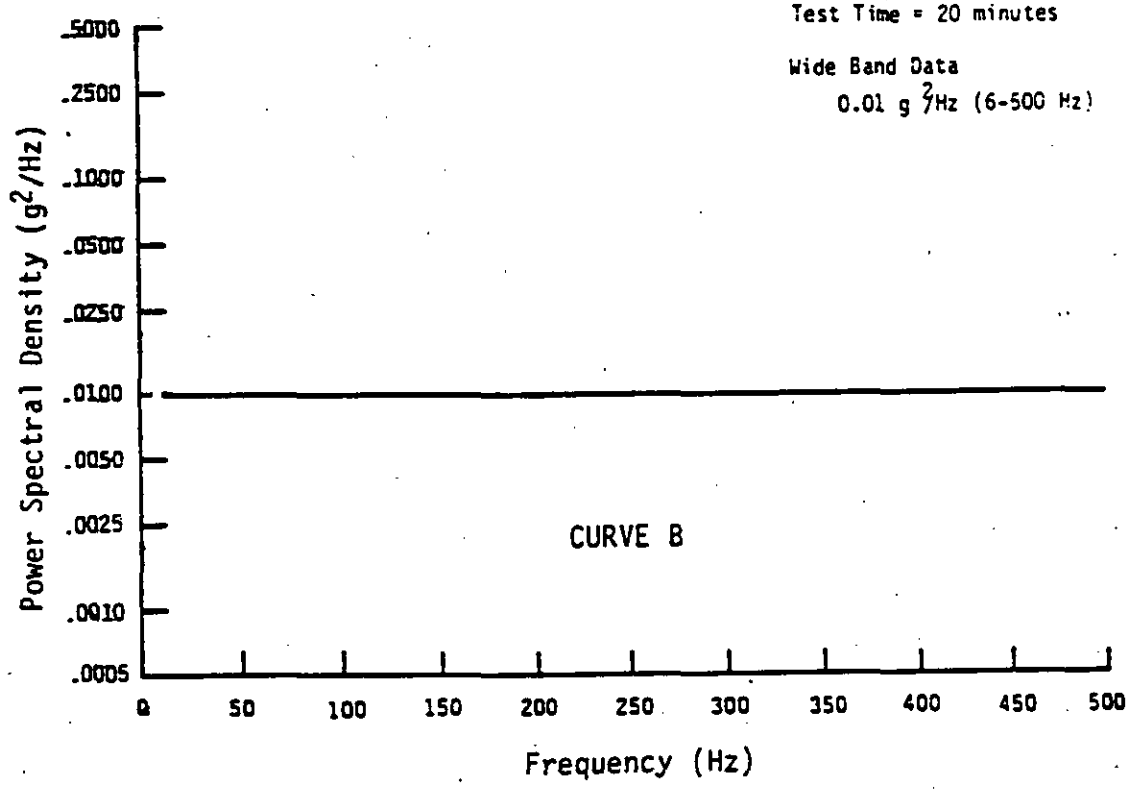


Vibration Level for AN/UGC-74B (M-113 Installation), Longitudinal Axis, Test Phase 3

Figure 11. DWG-00076

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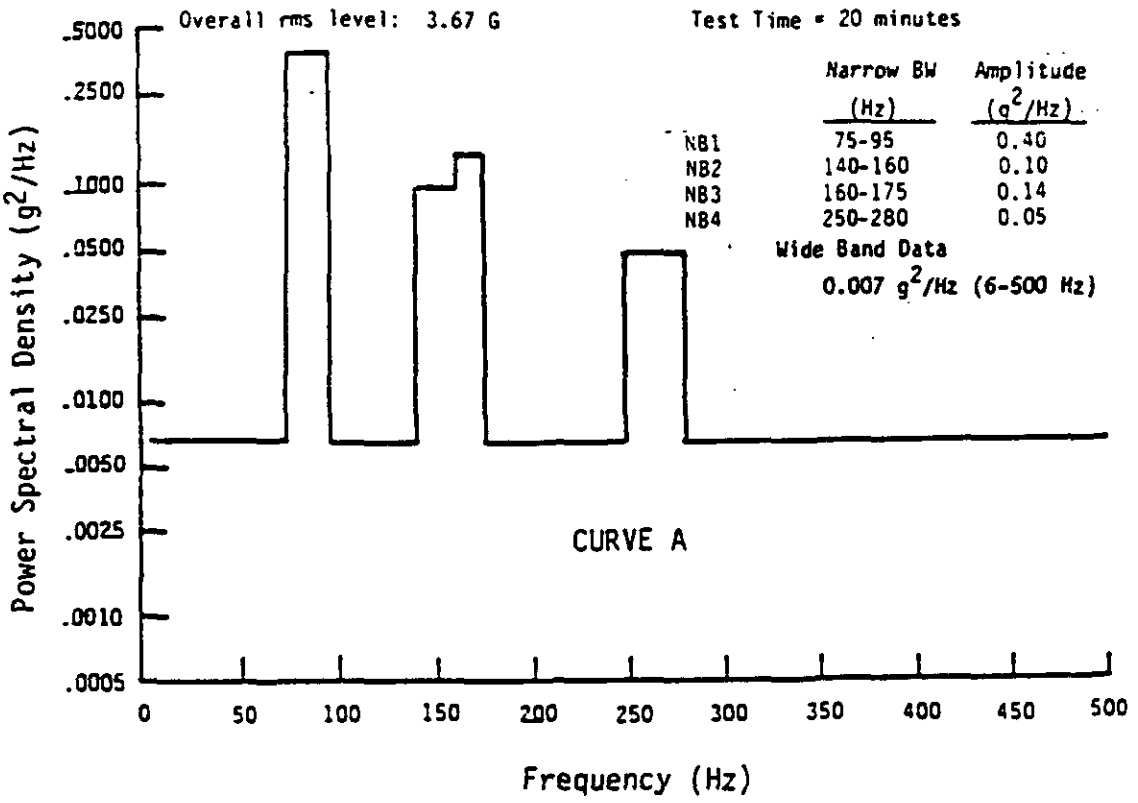


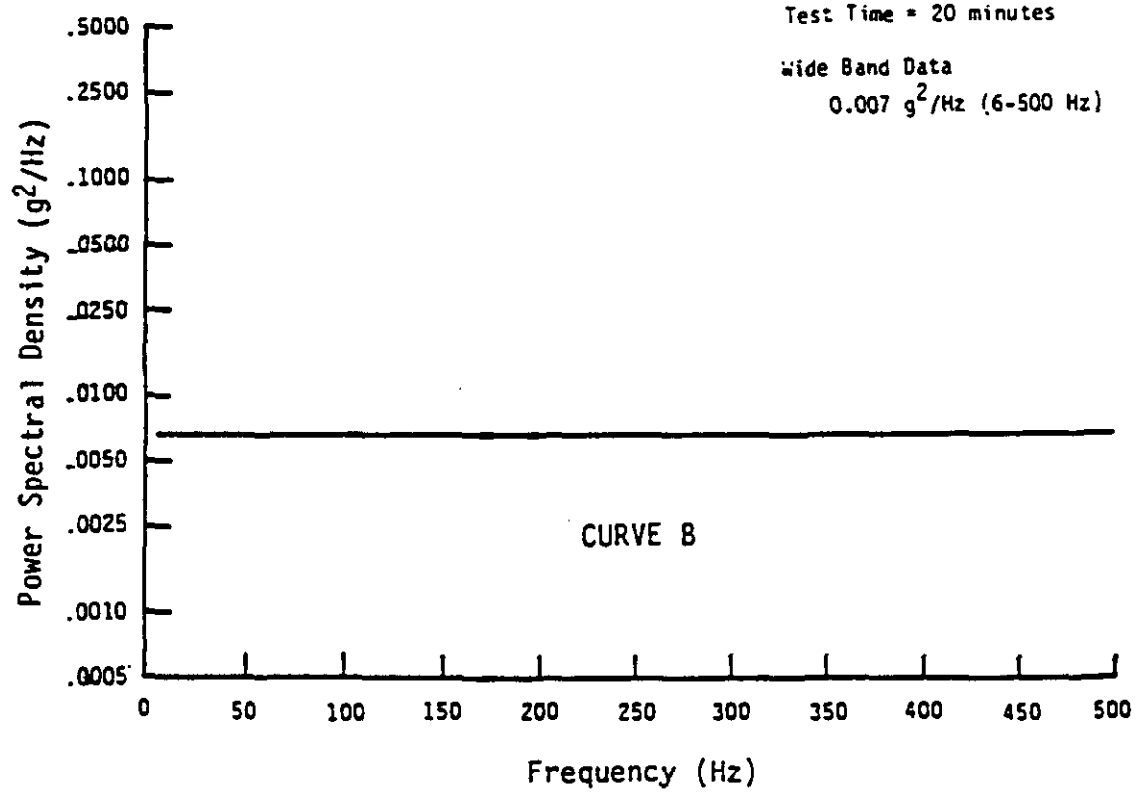


Vibration Level for AN/UGC-74B (M-113 Installation),
Vertical Axis, Test Phase 1

Figure 12. DWG-00077

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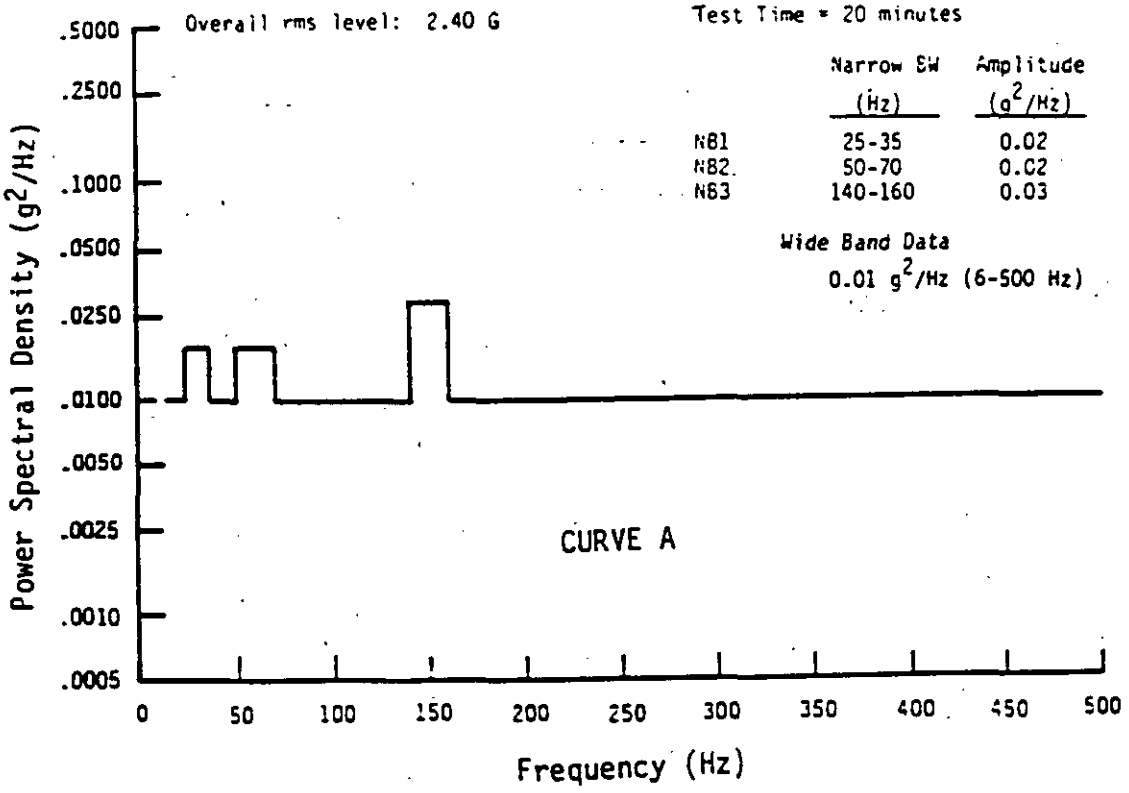


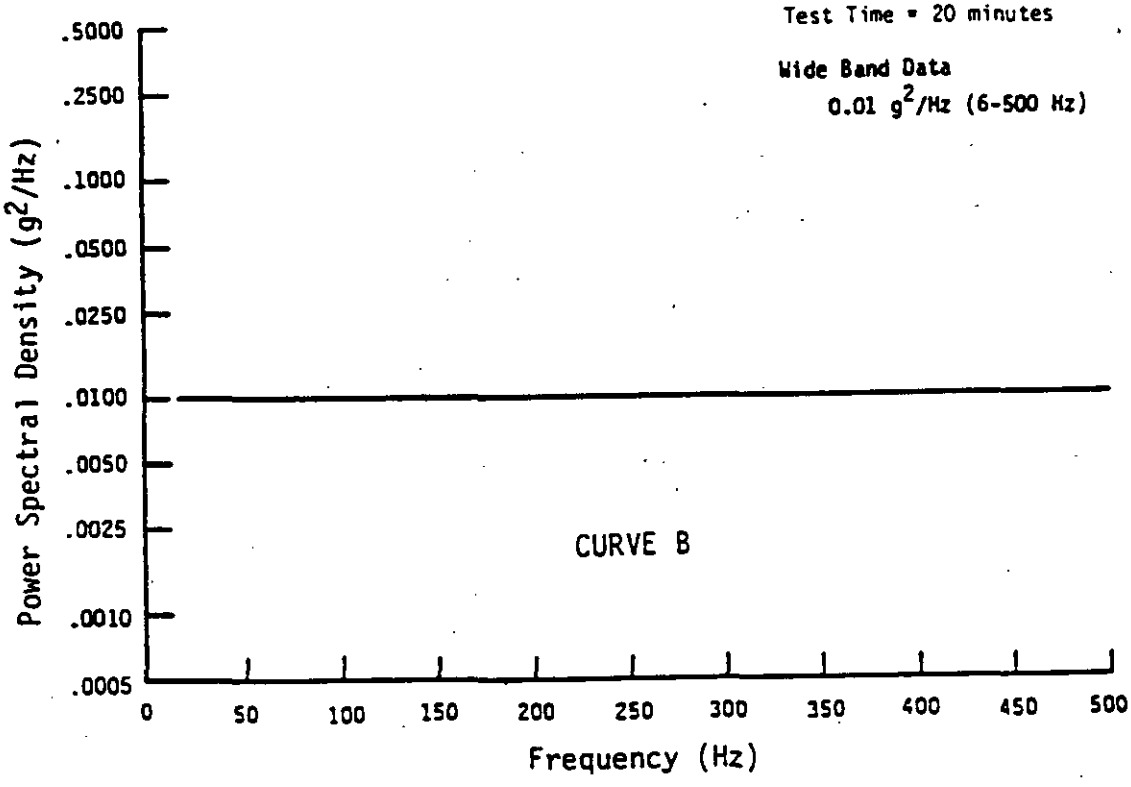


Vibration Level for AN/UGC-74B (M-113 Installation),
Vertical Axis, Test Phase 2

Figure 13. DWG-00078

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Vibration Level for AN/UGC-74B (M-113 Installation),
Vertical Axis, Test Phase 3

Figure 14. DWG-00079

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APPENDIX

TERMINAL, COMMUNICATIONS AN/UGC-74C(V)

10. SCOPE

10.1 Scope. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance. This appendix covers the Terminal, Communications AN/UGC74C(V), which is composed of Teleprinter, TT-744B/UGC-74C(V); Keyboard, TT-745A/UGC-74C(V); Case, CY-7735/UGC-74A(V); Cover MX-10038/UGC-74A(V); and Auxiliary Memory Module, MU-856/UGC-74C(V) (see 60.8).

20. APPLICABLE DOCUMENTS

20.1 Government Documents.

20.1.1 Specifications and Standards. Same as 2.1.1.

20.1.2 Other Government Documents, Drawings, and Publications. Same as 2.1.2 except:

DRAWINGS

A3041600	Teletypewriter AN/UGC-74C(V)

20.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

STANDARDS

Electronic Industry Association Standard RS422	Electrical Characteristics of Balanced Voltage Digital Interface Circuits, April 1975
International Standards Organization ISO 3309-1976 (E)	Data Communications - High Level Data Link Control Procedures - Frame Structure

20.3 Order of Precedence. In the event of conflict between the text of this appendix and the references cited herein, the text of this appendix shall take precedence.

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APPENDIX

30. REQUIREMENTS

30.1 First Article. When specified in the contractor's purchase order, the contractor shall furnish First Article units in accordance with 40.3.

30.2 Construction. This equipment shall be constructed in accordance with the requirements of this appendix and of Drawing A3041600 (including the requirements for parts, materials, and processes therein).

30.3 Parts,Materials,and Processes: General. Same as 3.3.

30.4 Finish. Same as 3.4.

30.5 Marking. Same as 3.5.

30.6 Interchangeability. Same as 3.6.

30.7 Preconditioning. The equipment shall be capable of meeting the requirements herein after subjection to the preconditioning of 40.6.1 and 40.6.2

30.8 Functional Requirements for Terminal, Communications, AN/UGC-74C(V). The AN/UGC-74C(V) Terminal, Conmunications shall meet the requirements of 3.8 and subparagraphs with the exceptions and additions specified in the following subparagraphs.

30.8.1 Paper Usage. Same as 3.8.1.

30.8.2 Visibility of Copy. Same as 3.8.2.

30.8.3 Copy Illumination. Same as 3.8.3

30.8.4 ASCII/Baudot Control. Same as 3.8.4.

30.8.5 Printer Functions. Same as 3.8.5 and subparagraphs.

30.8.6 Alarms and Indicators.

30.8.6.1 Audible Alarm.

30.8.6.1.1 Audible Alarm Adjustment. Same as 3.8.6.1.1.

30.8.6.1.2 Audible Alarm Actuation. Same as 3.8.6.1.2 with the addition of the relay port as follows:

a. A steady-state alarm is actuated when a message, with a precedence of IMMEDIATE or higher, has been received over the relay port. The alarm is turned off when the message is printed or accessed.

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30.8.6.2 Indicators. Same as 3.8.6.2 with the exceptions and additions specified in the following subparagraphs.

30.8.6.2.1 Memory Full Lamp. The MEM FULL indicator lamp shall light in a steady state when less than 15% of UGC-74 message memory or when less than 5% of the Auxiliary Memory Module (AMM) remains to be allocated. The MEM FULL indicator lamp shall flash when less than 1% of the AMM remains to be allocated.

30.8.6.2.2 Received Message Indicator. The MSG RCVD indicator lamp shall be activated (low precedence - steady state; high precedence - flash) when a message has been received through the relay port and stored in the AMM or through the communication line and stored in the UGC-74 message memory.

30.8.7 Switches. Internal and external switches shall operate as specified in 3.8.7 and subparagraphs except that the ABORT Switch shall have no effect on the message relay operation.

30.8.8 System States.

30.8.8.1 OFF State. Same as 3.8.8.1.

30.8.8.2 Cold Start State. Same as 3.8.8.2.

30.8.8.3 Operator Validation/State Determination State. Same as 3.8.8.3 except that a test shall be perfonned to determine if an operational AM/RF is present. If an operational AM/RF is present, the system configuration shall be UGC-74C and the system initialization printout shall contain the following:

```
SOFTWARE SYSTEM CONFIGRUATION = UGC-74C
          AUTO SAVE OPTION = ON
        AUTO PRINT OPTION = OFF
            NOFORM OPTION = ON
```

30.8.8.4 ICT State. Same as 3.8.8.4.

30.8.8.5 RO State. Same as 3.8.8.5. The AM/RF related functions shall be nonoperational.

30.8.8.6 KSR State. Same as 3.8.8.6. The AM/RF related functions shall be nonoperational.

30.8.8.7 Battery Back-up State. Same as 3.8.8.7. The AMM contents shall be retained; however, any data being used for current processing by the AM/RF subsystem may be lost.

30.8.8.8 Self Test State. Same as 3.8.8.8 except that the additional AM/RF components shall be tested and the faults isolated to the optimum replaceable subassembly. During Self Test, all AMM data shall be preserved.

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30.8.8.9 Factory Self Test State. Same as 3.8.8.9 except that the factory Self Test shall be functionally equivalent to the operational Self Test as specified by 10.3.8.8.8.

30.8.8.10 Warm Restart State. Same as 3.8.8.10.

30.8.9 Message Processing. Same as 3.8.9.

30.8.9.1 Message Storage. Same as 3.8.9.1.

30.8.9.1.1 Memory Full.

30.8.9.1.1.1 Operator Input.

30.8.9.1.1.1.1 Operator Input Message Memory. Same as 3.8.9.1.1.1.

30.8.9.1.1.1.2 Operator Input Auxiliary Memory. When auxiliary memory is 1% available or less when the operator initiates an auxiliary memory save, the message shall not enter auxiliary memory storage and an information message shall be printed. If this condition does not exist on initiation and auxiliary memory becomes 100% full during the save, an information message shall be printed and the partial message deleted from the auxiliary memory. To alleviate the auxiliary memory full condition, the operator shall be able to clear memory by deleting messages via a processing command.

30.8.9.1.1.2 Incoming Messages.

30.8.9.1.1.2.1 Incoming Messages Communication Line. Same as 3.8.9.1.1.2 with the addition of the AMM auto message store capability. An auto save option shall be provided such that messages received over the communication line are automatically stored both in the UGC-74 message memory and auxiliary memory. The auto save operation shall be terminated if emergency print is activated. All messages waiting to be auto saved shall be printed and deleted from message memory without being saved on the AMM. If the message name is a duplicate, a sequence letter shall be appended and the operator notified via printed message. When auxiliary memory becomes 1% available or less, an information message shall be printed and auto save disabled. If auxiliary memory becomes 100% full during the auto save process, an information message shall be printed, the auto save process shall be disabled, and the partial message shall be deleted from auxiliary memory. To alleviate this condition, the operator shall be able to attach a new AMM or delete auxiliary memory messages via a processing command.

30.8.9.1.1.2.2 Incoming Messages Relay Port. All incoming relay port messages shall be stored in the auxiliary memory. When auxiliary memory becomes 1% available or less, an information message shall be printed. If auxiliary memory becomes 100% full during a relay message save, the partial message shall be deleted and an information message shall be printed both at the receiver and sender of the relay message.

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30.8.9.1.1.2.3 Format and Naming Conventions. Setting of the NOFORM options shall determine whether or not unfonmatted messages may be stored on the AMM (see para 10.3.8.9.4.16). The test for a formatted message shall be as follows:

a. If the first line beglns with three letters followed by three numbers and the second line begins with a Y, Z, 0, P, or R, the message shall be considered formatted.

b. If the fourth character of the first line is not numeric, then the message shall be considered formatted if a proper precedence character is incharacter position 1, the Julian date/time is in character positions 22 through 28, and a dash is in character position 29 in line 1 of the Header.

Messages stored on the AMM shall be named (for CATALOG) citing information (precedence, day, time of day, month, year, and originator PLA) from format lines 5 and 6 if these lines can be located. If these lines cannot be located, the message shall be given a dummy name. Catalog formats shall be as follows:

a. blPblddhhmmZblmonblyrblblRECEIVEDblACCESSEDblFMbloriginator

b. SblPblddhhmmZblmonblblblblblCOMPOSEDblACCESSED (CR)
(originator not found)

c. SblPblddhhmmZblmonblyrblRELAYblNOTblACCESSEDblFMbl
originator (CR)

d. SblPblddhhmmZblblblblblblblblblblbl
RELAYblACCESSED (CR) (originator not found)

e. SblRbl778899Zblblblblblblblblblbl

30.8.9.2.1 Receive Message Process. The operation of the AN/UGC-74C(V) in the receive mode shall be identical to 3.8.9.2 and subparagraphs and shall also provide the capability to automatically transfer the received message to the AMM (auto save) without operator intervention. The operator shall be provided with the means to enable/disable the auto save capability. An auto save/auto print function shall be provided, which when enabled (operatoroption), shall print each received message after that message has been auto saved on the AMM (auto save option must be "ON"), and remove it from UGC-74 message memory without operator intervention.

30.8.9.3 Transmit Message Process. The Transmit message process for the communication line shall be as specified in 3.8.9.3. The message to be transmitted may be either composed in message memory or stored in the AMM. If the message is in message memory, transmission shall be identical to 3.8.9.3. Transmission of a message stored in the AMM shall require the operator to first move the message from the AMM to message memory via a processing command. Transmission shall now be identical to 3.8.9.3.

30.8.9.3.1 Relay Message Process. The relay message process shall provide the operator with the capability to send messages from any terminal on the relay network to any combination of other terminals on that local relay network. Relay shall be accomplished by entering a command followed by the message name and the channel numbers which are to receive the relayed message. Upon successful or unsuccessful reception of an entire message, the transmitting operator shall be notified by an information message. Upon successful reception, the receiving operator shall be notified as specified in 30.8.6 via the received message indicator and/or audible alarm. Only messages stored in the AMM shall be capable of relay. Relay of composed or received communication line messages shall require that the message, first, be saved on the AMM.

30.8.9.4 System Commands. The AN/UGC-74C(V) system commands shall operate as specified in 3.8.9.4 and subparagraphs with the following exceptions.

30.8.9.4.1 STATUS Command. This command shall print the following additional data:

a. NNN% AMM MEMORY AVAILABLE

where NNN shall be a three digit number representing the percentage of memory available on the AMM.

30.8.9.4.2 QUERY Command. This command shall print the following additional information:

a. AUTO SAVE OPTION = ON or OFF

b. AUTO PRINT OPTION = ON or OFF

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c. NOFORM OPTION = ON or OFF

30.8.9.4.3 AUTOS Subcommand. This PARAMETER subcomnand shall allow the operator to set the Auto Save option. When "ON," all messages received on the communications line shall be saved in UGC-74 message memory and on the AMM, when available, without operator intervention. When "OFF", received messages shall be saved in UGC-74 message memory only. If the software system configuration is not UGC-74C and this subcommand is entered, an information message shall be printed to inform the operator the subcommand is not available. If "ON" or "OFF" is not entered, an error message shall be printedand no change made. On power-up initialization, the auto save option shall be "ON".

30.8.9.4.4 AUTOP Subcommand. This PARAMETER subcommand shall allow the operator to set the Auto Print option. This option shall have meaning only if the auto save option is on. When "ON", once an entire message is auto saved on the AMM, it shall be printed and deleted from UGC-74 message memory. When off, no action shall be taken. If the software system configuration is not UGC-74C and this subcommand is entered, an information message shall be printed to inform the operator the subcommand is not available. If "ON" or "OFF" is not entered, an error message shall be printed and no change made. On power-up initialization, the auto print option shall be "OFF".

30.8.9.4.5 CATALOG Command. This command shall allow the operator to retrieve and print the catalog (i.e., a list by name) of messages stored on the AMM. If no parameter is specified, the catalog shall be printed in a Last-In-First-Out (LIFO) order. If an S is specified in the parameter field (i.e., CATALOG S), the catalog shall be sorted and then printed. The sort shall be performed based on message header time/date code. If duplicates exist, they shall be additionally sorted by sequence letter. If anything other than an "S" is entered in the parameter field, an error message shall be printed and the catalog shall not be printed. The catalog shall specify whether a message received over the relay network has been accessed by the operator or not. The catalog shall contain three types of messages: relay (received over the relay network), received (always considered accessed - received over the communications line and saved) and composed (always considered accessed - composed by operator and saved). Depressing the HLT key during printing shall cause the printing to stop and an information message shall be printed. If the AMM is not available or the catalog is empty, the operator shall be notified via a printed information message. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.6 GET Command. This command shall allow the operator to retrieve a specified message from the AMM and store it in UGC-74 message memory. If the name is incorrectly specified or not resident in the AMM catalog, an error message shall be printed. If the name is correctly specified and resident in the AMM catalog, it shall be retrieved, assigned a message number and stored

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into UGC-74 message memory. The operator shall then be notified, via printed message, of the message name and newly assigned message number. If the UGC-74 message memory is full (less than 1024 characters) or becomes full before the entire message is saved in message memory, the partial message shall be deleted and the operator shall be notified via printed message. If the AMM is not available, the operator shall be notified via a printed information message. Once retrieved, composed and relay messages shall be considered composed. Once retrieved, received messages shall be considered received, printed messages. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.7 NEXT Command. This command shall allow the operator to retrieve the oldest (date/time group), highest precedence, nonaccessed message from the AMM and store it in UGC-74 message memory. If there are no nonaccessed messages on the AMM, an information message shall be printed and no message retrieved. Once the appropriate message is identified, operation is identical to the GET conmand (see 30.8.9.4.6). If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not availalble.

30.8.9.4.8 SAVE Command. This command shall allow the operator to save a specified message contained in UGC-74 message memory on the AMM. If the number specified is not a message contained in UGC-74 message memory, an error message shall be printed and no message is saved. If the message number is acceptable, the message shall be retrieved, stored on the AMM, and the AMM catalog updated. The operator shall then be notified, via printed message, of the message name (extracted from the message header) and the message number. If the message name is a duplicate, a sequence letter shall be appended and the operator notified via printed message. If the AMM is not available or full (less than 1% left), the operator shal] be notified via a printed information message. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.9 CLEAR Command. This command shall allow the operator to clear a named file from the AMM. For security purposes, the operator shall be given the ability to clear the entire AMM (CLEAR *). No trace of the contents shall be retained after execution of the * option. Before execution of this command, the prompt 'ARE YOU SURE? (Y/N)' shall be printed to prevent the inadvertent erasure of messages on the AMM. If the operator responds with a 'N', no messages shall be cleared. If the operator responds with a 'Y', the specified message or the entire AMM shall be cleared. If the name is incorrectly specified or not resident on the AMM, an error message shall be printed. If the AMM is not available, the operator shall be notified via a printed information message. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

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30.8.9.4.10 RENAME Command. This command shall allow the operator to change the name of an existing AMM catalog entry. If either name is incorrectly specified, an error message shall be printed and no change made. If the first name is not resident in the AMM catalog or the second name already exists in the AMN catalog, an error message shall be printed and no change made. If both names are valid, the name contained in the AMM catalog shall be replaced with the new requested name. The name as it appears in the message header is not altered. If the AMM is not available, the operator shall be informed via a printed information message. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.11 CHANNEL Command. This command shall have four forms. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.11.1 CHANNEL M. This command shall designate the unit as the master. To assign a unit as the master, the CHANNEL M command shall be entered before assigning a channel number. If this command is entered and the unit has previously been assigned a channel number (see 30.8.9.4.11.2) which has not been canceled (see 30.8.9.4.11.3), the operator shall be informed, via printed message, that the master designation cannot be accepted. The unit shall retain the previously assigned channel number. This command shall be entered before but shall not take effect on the local relay network until a channel number is assigned to the unit.

30.8.9.4.11.2 CHANNEL XX. This command shall allow the operator to assign a relay network channel number to the unit. If the channel number entered is out of range (XX = 1 to 16), an error message shall be printed and no channel number assigned. If an attempt is made to assign or reassign a channel number while the relay is active, an error message shall be printed and no change made. If the channel number requested is already assigned to another unit on the local network, an information message shall be printed and no change made. If a CHANNEL M command was previously entered and there is already a master on the network, the operator shall be informed, via printed message, no channel number shall be assigned, and the master designation shall be canceled. If the channel number is accepted, that number shall be assigned to the unit and the operator shall be informed, via printed message, that the channel is assigned.

30.8.9.4.11.3 CHANNEL 0. This command shall allow the operator to cancel the master designation and/or the channel number. The operator shall be advised via printed message when the activity is complete. If the unit has not been designated as the master and/or there is no assigned channel number, this command shall have no effect.

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30.8.9.4.11.4 CHANNEL. If no parameter is entered, the currently assigned channel number and/or the master designator shall be reported to the operator. The operator shall be notified, via printed message, if the channel number has not been assigned or is not active.

30.8.9.4.12 RELAY Command. This command shall allow the operator to retrieve the specified message from the AMM and transmit it using the relay port to the specified channels on the local relay network. If the name is incorrectly specified or not resident on the AMM, an error message shall be printed and nothing transmitted. If any of the channel numbers are incorrectly specified or out of range, an error message shall be printed and nothing transmitted. If a channel number has not been assigned for the unit, an information message shall be printed and nothing transmitted. If a valid name and channel numbers are entered, the message shall be retrieved from the AMM, formatted to be sent over the relay network and the operator shall be notified, via printed message, that a relay has been initiated. If the AMM is not available, the operator shall be notified, via printed message, and the relay shall not be initiated. Upon receipt of the relay initiated message, the operator shall be able to enter all commands except CHANNEL M, CHANNEL XX, RELAY, or TST commands. Upon completion of the relay to all specified channels, the transmitting station operator shall be notified that the relay is complete along with all channel numbers that did not receive the message, if any. If a receiving unit is unable to save the incoming message (AMM not available, AMM full), its operator shall be notified, via printed message, of the condition. The receiving operator shall also be advised of the relay message number (generated by transmitting station and sent with the message) and channel number of station attempting to transmit. The transmitting station's operator shall be notified, via printed message, that the relayed message was not saved by the receiving station. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.13 CANCEL Command. This command shall allow the operator to abort the Relay Transmit process. The Relay Receive process shall not be affected by this command. If a RELAY command had not previously been entered, this command shall have no effect. Two system conditions may exist upon acceptance of this command:

a. Message in system waiting for access to relay network

b. Message being sent over the relay network

When the first condition exists, the operator shall be informed, via printed message, that the relay has been canceled and the relay message number. The receiving station shall give no indication of the cancel. When the second condition exists, the operator shall be informed, via printed message, that the relay was in progress when cancelled and the relay message number. The receiving station shall report that the relay message number from the transmitting unit number was not saved. If the software system configuration

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is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.14 LS Command. This command shall allow the operator to get a printed status of the dual purpose lamps: MEM FULL Lamp and MSG RCVD Lamp (see paragraph 30.8.6.2). The following information shall be printed:

a. Amount of AMM available (MEM FULL Lamp)

b. Amount of message memory in use (MEM FULL Lamp)

c. Presence or absence of low precedence non-printed received messages in message memory (MSG RCVD Lamp)

d. Presence or absence of high precedence non-printed received messages in message memory (MSG RCVD Lamp)

e. Presence or absence of nonaccessed low precedence messages on the AMM (MSG RCVD Lamp)

f. Presence or absence of nonaccessed high precedence messages on the AMM (MSG RCVD Lamp)

If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.15 TST Command. This command shall allow the operator to initiate the AM/RF self-test.

If a relay channel has been assigned and is active upon acceptance of this command, the operator shall be notified, via printed message, that this command cannot be accepted at this time. This test shall preserve both UGC-74 message memory and the AMM contents. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available.

30.8.9.4.16 NOFORM Command. This command shall allow the operator to set or reset the NOFORM option. If the option is on, all messages (formatted or not) shall be saved on the AMM. If the option is off, unformatted messages shall not be saved on the AMM (this shall include messages received by the AM/RF from both the UGC-74 and the relay port). If "ON" or "OFF" is not inserted in the parameter field, an error message shall be printed and no change made. If the software system configuration is not UGC-74C and this command is entered, an information message shall be printed to inform the operator the command is not available. On power-up initialization the NOFORM option shall be "ON."

30.8.9.4.17 HELP Command. This command shall list the additional system commands for the AM/RF.

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APPENDIX

SYSTEM COMMAND	COMMAND FORMAT(S)	DESCRIPTION
GET	GE DDTTTT,L	Retrieves a message from Auxiliary Memory Module (AMM) and saves it in message memory.
NEXT	N	Retrieves oldest highest precedence message from AMM to message memory.
SAVE	SA XXXX	Saves a message which is in message memory into AMM.
CATALOG	CAT	Displays catalog of message names stored in AMM; in last-in, first-out (LIFO) order.
	CATS	Displays catalog of message names stored in AMM: sorted by date, time, sequence letter. The HLT key will terminate any print activity.
CLEAR	CLE DDTTTT,L	Delete a message from AMM.
	CLE *	Clears entire AMM to a known state.
RENAME	RENA DDTTTT,L/ DDTTTT,L	Renames an entry in AMM catalog.
CHANNEL	CH XX	Sets Relay Port channel number (XX = 1 TO 16).
	CH M	Designates unit as the master.
	CH 0	Cancels Master Designation and/or assigned channel master.
	CH	Prints Master Designation and/or assigned channel number.
RELAY	REL DDTTTT,L/ (XX,XX,XX,...)	Retrieves a message from AMM and transmits it using the Relay Port to listed channels. (XX,XX,XX...) is a list of channel members, for example, (2, 3, 12, 15).
TST	TS	Initiates AMM Self Test and reports results.
CANCEL	CAN	Cancels relay transmission in progress.

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LS	LS	Prints status of dual purpose lamps (MEM FULL, MSG RCVD).

NOTE: DDTTTT,L indicates catalog entry. DD = day of month, TTTT = Military time of day, L = sequence letter or space.

30.8.9.4.18 HELP Subcommand (Parameter). This command shall list the additional parameter subcommands for the AM/RF.

Parameter Subcommand	Command Format(s)	Description
AUTOS	AUTOS ON	Sets Auto Save option.
	AUTOS OFF	Resets Auto Save Option.
AUTOP	AUTOP ON	Sets Auto Print option.
	AUTOP OFF	Resets Auto Print option.
NOFORM	NOF ON	Sets NOFORM option.
	NOF OFF	Resets NOFORM option.

30.8.9.5 Full-Duplex Operation (Communication Line). Same as 3.8.9.5.

30.8.9.6 Half-Duplex Operation.

30.8.9.6.1 Communication Line Half-Duplex Operation. Same as 3.8.9.6.

30.8.9.6.2 Relay Port Half-Duplex Operation. The Relay port shall be capable of half-duplex operation (transmit and receive one at a time) in a two-wire system.

30.8.10 Performance Requirements.

30.8.10.1 Response Times. Same as 3.8.10.1 except for the following conditions:

a. Clear Command Response. The CLEAR command shall be capable of clearing the entire Auxiliary memory in less than 60 seconds.

b. Relay Message Response. The system response time between operator input of the Relay command and return of control to the operator shall be as specified in 3.8.10.1. The Relay command shall initiate the relay message process.

The relay message process response time shall be the time between operator input of the Relay conmand and the completion of the relay to all specified channels. The relay message response time of a relay network carrying the maximum traffic load of typical messages as specified in 30.8.10.7 shall be five (5) minutes maximum under normal operating conditions.

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30.8.10.2 Storage Capacity. The UGC-74 message memory storage capacity shall be as specified in 3.8.10.2. The Auxiliary Memory Module (AMM) storage capacity shall be capable of storing a minimum of 120,000 characters in nonvolatile memory. The AMM block length shall not exceed 128 Bytes. The AMM shall be a removable media which can be replaced without affecting other terminal operations.

30.8.10.2.1 Memory Retention. Message memory retention shall be as specified in 3.8.10.2.1. The AMM contents shall be in nonvolatile memory which shall be retained without the use of an external power source.

30.8.10.2.2 AMM Attachment and Removal. The AMM shall be attachable and removable at any time during the UGC-74C processing cycle without loss or alteration of previously retained complete messages.

30.8.10.3 Printing Speed. Same as 3.8.10.3.

30.8.10.4 Print Quality. Same as 3.8.10.4.

30.8.10.5 Print Legibility. Same as 3.8.10.5.

30.8.10.6 Automatic Time-out. Same as 3.8.10.6.

30.8.10.7 Relay Network Capacity. The relay port shall operate in a network that shall support up to 16 AN/UGC-74C(V) terminals. The relay network shall be capable of handling up to 10 typical messages (1200 characters) per minute while satisfying the relay message response time as specified in 30.8.10.1.

30.8.11 Operator Prompting. Same as 3.8.11 except the following additional information/error messages pertaining to the AM/RF shall also be printed in full with an optional identification number and English words.

a. COMMAND NOT ACCEPTED, AUTO SAVE MUST BE ON
b. CANNOT RELAY, NO CHANNEL ASSIGNED
c. NOT ASSIGNED, CHANNEL MUST BE IN 1-16 RANGE
d. RELAY MESSAGE #_____ CHANNEL #_____ NOT SAVED
e. AUXILIARY MEMORY MODULE NOT ACCESSIBLE
f. AUXILIARY MEMORY MODULE ERROR
g. CANNOT DESIGNATE AS MASTER, CHANNEL ALREADY ASSIGNED
h. INSUFFICIENT AUXILIARY MEMORY TO SAVE MESSAGE
i. MESSAGE NOT FOUND, CHECK NAME
j. CATALOG EMPTY
k CANNOT RELAY TO CHANNEL #_____
l. OLD MESSAGE NAME NOT FOUND
m. NEW MESSAGE NAME ALREADY EXISTS
n. AUXILIARY MEMORY MODULE FULL
o. MESSAGE NOT SAVED; NO MORE SEQUENCE INDICATORS AVAILABLE FOR DUPLICATE NAME

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p. NOT ASSIGNED, ANOTHER CHANNEL IS MASTER
q. NOT ASSIGNED, CHANNEL #_____ ALREADY USED
r. COMMAND NOT ACCEPTED, MESSAGE RELAY IN PROGRESS
s. TEST NOT ALLOWED, CHANNEL ACTIVE
t. MAINTENANCE NOTE: AMRF ROM TEST FAILED (A2A4)
u. MAINTENANCE NOTE: AMRF TEST FAILED (A2A4)
v. MAINTENANCE NOTE: AMF RELAY TEST FAILED (A2A4)
w. MAINTENANCE NOTE: AMRF RTC TEST FAILED (A2A4)
x. COMMAND NOT ACCEPTED, AUX MEMORY OR RELAY BUSY
y. AUX MEMORY AND RELAY FUNCTION NOT OPERATING
z. AM/RF INTERFACE FAILURE -- COMMAND PROCESS AND/OR AUTO SAVE ABORTED

(1) Enter TST command to allow AM/RF interface to clear (ignore response). Enter second TST command to obtain valid results.

(2) The following two lines are also printed if the auto save option is ON:

AUTO SAVE SUSPENDED - TO REENABLE AUTO SAVE ENTER PARAMETER SUBMODE, TURN AUTO SAVE ON AND EXIT PARAMETER SUBMODE

aa. NO RELAY MESSAGE TO CANCEL
bb. AMM COMMAND ERROR
cc. RELAY ADDRESS ERROR -- RAM ERROR
dd. RELAY MESSAGE SIZE LIMIT EXCEEDED: RE-EDIT
ee. CANNOT ACTIVATE MASTER CHANNEL, RAM ERROR
ff. CANNOT SORT, TOO MANY ENTRIES IN CATALOG
gg. AMRF SYSTEM ERROR: ILLEGAL OPERATION
hh. NO CHANNEL ASSIGNED
ii. INVALID ADDRESS; CHANNEL MUST BE IN 1-16 RANGE
jj. MAINTENANCE NOTE: AMRF AMM TEST FAILED (A2A4)
kk. INVALID ADDRESS; CANNOT BE SENDER
ll. INVALID ADDRESS; TOO MANY CHANNELS IN LIST
mm. RELAY MESSAGE #_____ NOT SAVED BY CHANNEL #_____
nn. UNFORMATTED MESSAGE; NOT SAVED ON AMM
oo. SORT NOT ALLOWED; RELAY CHANNEL ACTIVE
pp. AMRF RAM ERROR
qq. CANNOT RELAY, CHANNEL NOT YET ACTIVE

30.9 AN/UGC-74C(V) Signal Characteristics and Electrical Requirements.

30.9.1 Operating Modes.

30.9.1.1 Communication Line Operating Modes. The AN/UGC-74C(V) communication line shall operate as specified in 3.9.1 and subparagraphs.

30.9.1.2 Relay Port Operating Modes.

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30.9.1.2.1 Relay Network Control. Each relay network shall be controlled by a single terminal known as the master. There shall be one and only one master permitted on a given relay network. All other terminals on the network shall be referred to as slaves. The operator shall designate via a processing command which unit is the master. The operator shall also designate a relay network channel number for each terminal on the network. The master terminal shall be responsible for performing the network polling sequence. The polling sequence shall provide active slaves the option to relay a message and inactive slaves the option to become active. In addition to performing the polling sequence, the master shall operate and have the same capabilities as the slave terminals.

30.9.1.2.2 Relay Communication Formats. Data transferred over the relay network shall be one of two basic types: control or message data. Control data shall be organized into command packets used for establishing and managing the logical connection between units. Message data shall consist of variable length data packets which comprise the message to be transferred. All transmissions of data and control packets on the network shall be organized in a specific format called a frame. The frame structure shall be based on International Standards Organization (ISO) Internal Standard (IS) 3309-1976 (E), Data Communications - High-Level Data Link Control Procedures - Frame Structure. The relay network frame structure information fields shall contain an integral number of 8-bit bytes in ASCII format.

30.9.1.2.3 Relay Data Link Controls. The data link shall permit any unit on the network to gain temporary control. The controlling unit shall be capable of broadcasting a message to all units on the network or sending a message to any selected subset of the terminals on the network. The controlling unit shall verify the reception of each control or data transmission frame by each receiving unit. The receiving unit shall indicate error-free reception by responding with an Acknowledge (ACK). The receiving unit shall indicate the reception of bad data by either no response or a Not Acknowledge (NAK) response. The reception of an invalid frame shall cause the receiving unit to discard the frame and not respond. The reception of a valid frame which fails the frame check sequence (Cyclic Redundancy Check) or contains invalid data shall be NAKed. A NAK response or no response shall cause the transmitting unit to retransmit the frame. If an ACK is not received after three transmission attempts, the transfer to that receiving device shall be aborted and an information message printed at the controlling unit (i.e., sender).

30.9.1.2.4 Relay Signal Speeds. The relay port shall operate at 9600 bps +/- 0.1 percent.

30.9.2 Clock INT/EXT/KG-30 Switch. Same as 3.9.2 and subpargraphs.

30.9.3 Detailed Requirements for Digital Interface Circuits. Same as 3.9.3 and subparagraphs 3.9.3.1 through 3.9.3.4.

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30.9.4 Relay Port Digital Interface Requirements. The relay port digital interface shall meet the requirements specified in paragraph 4 of EIA Standard RS-422, dated April 1975, with the following exceptions:

a. Paragraph 4.2.5, Multiple Receivers shall be "16" receivers.

b. Paragraph 4.2.6, Fail Safe operation shall not be applicable.

30.9.4.1 Relay Port Signals. The relay port signals shall be conditioned diphase.

30.9.5 Isolation of Chassis Ground, Signal, and Clock Lines. Same as 3.9.5.

30.9.6 System Interface. Same as 3.9.6 and subparagraphs. Relay Port Connector Pin Assignments are specified in Figure 15.

30.9.7 Power.

30.9.7.1 Power Consumption. The steady state power drain shall not exceed 135 watts.

30.9.7.2 Sources. Same as 3.9.7.2.

30.9.7.3 Overvoltage and Transients. Same as 3.9.7.3 and subparagraphs.

30.9.7.4 Isolation of Chassis Ground, Power Lines. Same as 3.9.7.4.

30.9.7.5 Battery Backup - Same as 3.9.7.5.

30.10 Keyboard Transmitter AN/UGC-74C(V) Requirements. Same as 3.10 and subparagraphs.

30.11 TEMPEST Requirements. Same as 3.11.

30.12 Acoustic Noise. Same as 3.12.

30.13 Electromagnetic Compatibility. Same as 3.13.

30.14 System Nuclear Survivability. Same as 3.14.

30.15 Environmental Service Conditions. The AN/UGC-74C(V) shall meet the requirements of 3.15 and subparagraphs with the exceptions and additions specified in the following paragraphs.

30.15.1 Temperature. The AN/UGC-74C(V) with the AMM detached shall meet the requirements of 3.15.1. The AN/UGC-74C(V) with the AMM attached shall have the following temperature limitations.

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30.15.1.1 Operating. The ambient temperature range shall be 32 F (0 C) to +131 F (+55 C) with the AMM attached.

b. Nonoperating. The nonoperating exposure range of the AN/UGC-74C(V) with the AMM attached or of the AMM itself shall be -40 F (-40 C) to +154 F (+68 C).

30.15.2 Humidity. Same as 3.15.2.

30.15.3 Altitude. Same as 3.15.3.

30.15.4 Immersion. Same as 3.15.4 with the addition of the AMM being capable of operation after being separately immersed.

30.15.5 Vibration. Same as 3.15.5.

30.15.5.1 Bounce, Loose Cargo. Same as 3.15.5. The AMM shall not be included as it is not contained in the transit case.

30.15.6 Shock.

30.15.6.1 Drop. Same as 3.15.6.1. In addition, the AMM itself shall withstand shocks encountered during normal handling. After normal installation procedures, the AMM must meet all requirements.

30.15.6.2 Bench Handling. Same as 3.15.6.2.

30.15.7 Ballistic Shock. Same as 3.15.7 except that electrical integrity of the AMM need not be maintained.

30.15.8 Dust. Same as 3.15.8.

30.15.9 Rain. Same as 3.15.9.

30.15.10 Salt Fog. Same as 3.15.10.

30.15.11 Orientation. Same as 3.15.11.

30.15.12 Fungus. Same as 3.15.12.

30.15.13 Leakage Air Seal. Same as 3.15.13.

30.16 Reliability. Same as 3.16.

30.17 Systems Safety Engineering. Same as 3.17 and subparagraphs.

30.18 Maintainability. Same as 3.18 and subparagraphs.

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APPENDIX

30.19 Weight. The weight of the AN/UGC-74C(V) including the AMM, cover and copy holder shall not exceed 107 pounds.

30.20 Workmanship. Same as 3.20.

40. QUALITY ASSURANCE PROVISIONS

40.1 Responsibility for Inspection. Same as 4.1.

40.1.1 Responsibility for Compliance. Same as 4.1.1

40.2 Classification of Inspection. Same as 4.2.

40.3 First Article. The AN/UGC-74C(V) First Article shall be as specified in 4.3 and subparagraphs with the exceptions and additions specified in the following subparagraphs.

40.3.1 First Article Units. Same as 4.3.1.

40.3.2 First Article Inspection. The First Article inspection shall consist of the inspections specified in the subsidiary documents covering the items listed in 40.4, preconditioning specified in 40.6, and the inspections specified for software verification, group A, group B, group C, and group D.

40.3.2.1 First Article Software Verification Inspection. Same as 4.3.2.1 except in accordance with Table XV inspections.

40.3.2.2 First Article Group A Inspection. Same as 4.3.2.2 except in accordance with Table XVI.

40.3.2.3 First Article Group B Inspection. Same as 4.3.2.3 except in accordance with Table XVII.

40.3.2.4 First Article Group C Inspection. First article units shall be subjected to group C inspection in accordance with Table XVIII. The selected units shall have passed group A and group B inspections.

40.3.2.5 First Article Group D Inspection. Same as 4.3.2.5.

40.4 Inspection Covered by Subsidiary Documents. Same as 4.4.

40.5 Quality Conformation Inspection of Equipment Before Packaging. Same as 4.5 except the inspections specified in 40.5.1 through 40.5.4 shall be performed.

40.5.1 Group A Inspection. Same as 4.5.1 except in accordance with Table XVI.

40.5.1.1 Order of Inspection Within Group A. Same as 4.5.1.1.

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40.5.2 Group B Inspection. Same as 4.5.2 except in accordance with Table XVII.

40.5.3 Group C Inspection. Same as 4.5.3 except in accordance with Table XIX.

40.5.3.1 Sampling for Group C Inspection. Same as 4.5.3.1 and subparagraphs.

40.5.4 Group D Inspection. Same as 4.5.4.

40.5.5 Reinspection of Conforming Group C and Group D Sample Units. Same as 4.5.5.

TABLE XV. SOFTWARE VERIFICATION INSPECTION. 1/

REQUIREMENT	REQ'T PARA.
Keyboard/Print Codes	3.8.5.1
Coded BAUDOT Printer Functions	3.8.5.3
Carriage Return	3.8.5.3.1
Line Feed	3.8.5.3.2
Letters	3.8.5.3.3
Figures	3.8.5.3.4
TAB	3.8.5.4.1
CR	3.8.5.4.2
LF	3.8.5.4.3
BS	3.8.5.4.4
Parity Error	3.8.5.4.5
Automatic Carriage Return	3.8.5.5
Automatic Line Feed	3.8.5.6
Print Pin Pressure Selection	3.8.5.7.1.2
Line Length	3.8.5.7.2
Compressed Text Mode	3.8.5.7.2.1
Multiple Character Fonts	3.8.5.7.3
Character Font ROM Capacity	3.8.5.7.3.1
Character Font Selection	3.8.5.7.3.2
Standard Character Font Symbol Set	3.8.5.7.3.3
OFF State	30.8.8.1
Cold Start State	30.8.8.2
Operation Validation/State Determination State	3.8.8.3
ICT State	3.8.8.4
RO State	3.8.8.5
KSR State	3.8.8.6
Battery Backup State	3.8.8.7
Self Test State	3.8.8.8.8
Factory Selftest State	3.8.8.9
Warm Restart State	3.8.8.10

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TABLE XV. SOFTWARE VERIFICATION INSPECTION. - Continued. 1/

REQUIREMENT	REQ'T PARA.
Message Processing	3.8.9
Message Storage	3.8.9.1
Memory Full	3.8.9.1.1
Operator Input Message Memory	3.8.9.1.1.1.1
Operator Input Aux Memory	3.8.9.1.1.1.2
Incoming Message (COMM)	3.8.9.1.1.2
Incoming Messages Relay Port	3.8.9.1.1.2.2
Message Receipt	30.8.9.2
Receive Message Process	30.8.9.2.1
Transmit Message Process	30.8.9.3
Relay Mesage Process	30.8.9.3.1
System Commands	3.8.9.4
PARAMETER Command	3.8.9.4.1
LINE Subcommand	3.8.9.4.1.1
SETAB Subcommand	3.8.9.4.1.2
CLRTAB Subcommand	3.8.9.4.1.3
DISPLAY Subcommand	3.8.9.4.1.4
CAPIT Subcommand	3.8.9.4.1.5
REMIND Subcommand	3.8.9.4.1.6
EOL Subcommand	3.8.9.4.1.7
LF Subcommand	3.8.9.4.1.8
SPACE Subcommand	3.8.9.4.1.9
PAPER Subcommand	3.8.9.4.1.10
RENVELOPE Subconmand	3.8.9.4.1.11
TENVELOPE Subcommand	3.8.9.4.1.12
EOMLF Subcommand	3.8.9.4.1.13
FONT Subcommand	3.8.9.4.1.14
COPIES Subcommand	3.8.9.4.1.15
DUPLEX Subcommand	3.8.9.4.1.16
HEADERS Subcommand	3.8.9.4.1.17
HELP Subcommand	3.8.9.4.1.18
PAGE Subcommand	3.8.9.4.1.19
SETUP Subcommand	3.8.9.4.1.20
AUTOS Subcommand	30.8.9.4.3
AUTOP Subcommand	30.8.9.4.4
EXIT Subcommand	3.8.9.4.1.21
GO Subcommand	3.8.9.4.1.22
EDIT Command	3.8.9.4.2
TOP Subcommand	3.8.9.4.2.1
BOTTOM Subconmand	3.8.9.4.2.2
UP Subcommand	3.8.9.4.2.3
DOWN Subcommand	3.8.9.4.2.4
BEFORE Subcommand	3.8.9.4.2.5
INSERT Subcommand	3.8.9.4.2.6

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TABLE XV. SOFTWARE VERIFICATION INSPECTION. - Continued. 1/

REQUIREMENT	REQ'T PARA.
DELETE Subcommand	3.8.9.4.2.7
FIND Subcommand	3.8.9.4.2.8
LIST Subcommand	3.8.9.4.2.9
REPLACE Subcommand	3.8.9.7.2.10
HEADER Subcommand	3.8.9.4.2.11
APPEND Subconmand	3.8.9.4.2.12
HELP Subcommand	3.8.9.4.2.13
EXIT Subcommand	3.8.9.4.2.14
PRINT Command	3.8.9.4.3
TRANSMIT Command	3.8.9.4.4
REMOVE Command	3.8.9.4.5
JUSTIFY Command	3.8.9.4.6
QUERY Command	3.8.9.4.7
STATUS Command	3.8.9.4.8
HELP Command	3.8.9.4.9
TTY Command	3.8.9.4.9
HELP Command	3.8.9.4.10
RESTART Command	3.8.9.4.11
CATALOG Command	30.8.9.4.5
GET Command	30.8.9.4.6
NEXT Command	30.8.9.4.7
SAVE Command	30.8.9.4.8
CLEAR Command	30.8.9.4.9
RENAME Command	30.8.9.4.10
CHANNEL Command	30.8.9.4.11
RELAY Command	30.8.9.4.12
CANCEL Command	30.8.9.4.13
LS Command	30.8.9.4.14
TST Command	30.8.9.4.15
NOFORM Command	30.8.9.4.16
Half Duplex Operation COMM	30.8.9.6.1
Half Duplex Operation Relay	30.8.9.6.2
Storage Capacity Msg Mem	3.8.10.2
Response Time	30.8.10.1
Storage Capacity Aux Mem	30.8.10.2
Error/Information Message List	30.8.11
Relay Network Control	30.9.1.2.1
Relay Communication Formats	30.9.1.2.2
Relay Data Link Controls	30.5.1.2.3
Battery Backup	3.9.7.5

1/ This document does not outline the sequence of tests necessary to demonstrate compliance with the requirements of the table.

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TABLE XVI. GROUP A INSPECTION - AN/UGC-74C(V).

INSPECTION	REQUIREMENT	REQ'T PARA.
Visual & Mechanical	Keyboard Arrangement	3.10.2 3/
	Workmanship	3.20 3/
	Air Seal	3.15.13 3/
Electro-mechanical	Visibility of Copy	3.8.2 4/
	Copy Illumination	3.8.3 4/
	Power Indicator	30.8.6.2 4/
	Self Test	3.8.7.1.4 4/
	External Power Switch	3.8.7.2.3 4/
	Cold Start State	3.8.8.2 4/
	Operation Validation/ State Determination State	30.8.8.3 4/
	KSR State	3.8.8.6 4/
	Self Test State	30.8.8.8 4/
	High Level Digital Signal Interface	3.9.3.3 1/ 4/
	Low Level Digital Interface	3.9.3.2 1/ 4/
Print Quality	Print Registration	3.8.10.4.1 2/ 4/
	Printed Character Quality	3.8.10.4.2 2/ 4/

1/ Functional loop back only.

2/ Note: Special Test Condition as follows: Print 36 consecutive lines with no character errors allowed. In the event of a print quality failure, adjustment of the printhead assembly, line feed assembly, and carriage assembly is permissible without reburn-in (see 6.4.8.).

3/ Use Sampling Plan 1 for this requirement (see TABLE VII).

4/ Use Sampling Plan 2 for this requirement (see TABLE VIII).

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APPENDIX

TABLE XVII. GROUP B INSPECTION - AN/UGC-74C(V).

REQUIREMENT	REQ'T PARA.	AQL
Visibility of Copy	30.8.2 3/	
ASCII/BAUDOT Control	3.8.4 3/	
Local Printer Function	3.8.5.2 3/	
Parity Error	3.8.5.4.5 3/	
Paper/Out	3.8.5.7.4 3/	
Audible Alarm Adjustment	3.8.6.1.1 3/	
Audible Alarm Actuation	30.8.6.1.2 3/	
Battery Backup Indicator	3.8.6.2.2 3/	
Paper Low Indicator	3.8.6.2.3 3/	
Line Indicator	3.8.6.2.4 3/	
Received Message Indicator	30.8.6.2.(b) 3/	
Transmit Indicator	3.8.6.2.6 3/	
Memory Full Warning Indicator	30.8.6.2.(a) 3/	
Parity Indicator	3.8.6.2.8 3/	
End of Line Indicator	3.8.6.2.9 3/	
Switches	30.8.7 3/	
PARITY Switch	3.8.7.1 3/	
STOP BITS Switch	3.8.7.1.2 3/	
STATE Switch	3.8.7.1.3 3/	
SELF TEST Switch	3.8.7.1.4 3/	
MODE Switch	3.8.7.1.5 3/	
FIGURES Switch	3.8.7.1.6 3/	
CLOCK (+/-) Switch	3.8.7.1.7 3/	
BAUD RATE Switch	3.8.7.1.8 3/	
REC MODE Switch	3.8.7.1.9 3/	
XMLT MODE Switch	3.8.7.1.10 3/	
SIGNALS Switch	3.8.7.1.11 3/	
CLOCK (INT/KG-30/EXT) Switch	3.8.7.1.12 3/	
ABORT Switch	30.8.7.2.1 3/	
Audio Alarm Reset	3.8.7.2.2 3/	
EXTERNAL POWER Switch	3.8.7.2.3 3/	
Illumination Control	3.8.7.2.4 3/	
TRANSFER Switch	3.8.7.2.5 3/	
PARITY RESET Switch	3.8.7.2.6 3/	
LINE FEED Switch	3.8.7.2.7 3/	
Audio Control	3.8.7.2.8 3/	
Operation Validation/State Determination State	30.8.8.3 3/	
Battery Backup State	30.8.8.7 2/	
Self Test State	30.8.8.8 2/	
Full Duplex Operation	3.8.9.5 3/	
Memory Capacity	30.8.10.2 3/	
Memory Retention	30.8.10.2.1 3/	
AMM Attachment and Removal	30.8.10.2.2	
Printing Speed	3.8.10.3 3/	

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TABLE XVII. GROUP B INSPECTION - AN/UGC-74C(V). - Continued.

REQUIREMENT	REQ'T PARA.	AQL
Print Quality	3.8.10.4 3/	
Print Legibility	3.8.10.5 3/	
Automatic Timeout	3.8.10.6 3/	
Relay Network Capacity 1/	30.8.10.7 3/	
ASCII Code Interface	3.9.1.1 3/	
Signal Speeds (ASCII)	3.9.1.1.1 3/	
External Clock Mode	3.9.1.1.2 3/	
BAUDOT Code Interface	3.9.1.2 3/	
Signal Speeds (BAUDOT)	3.9.1.2.1 3/	
Speed Accuracy	3.9.1.3 3/	
Relay Signal Speeds	30.9.1.2.4 3/	
Clock INT/EXT/KG-30 Switch	3.9.2 3/	
Internal Clock Mode - Transmit Clock	3.9.2.1 3/	
External Clock/Transmitted Data Relationship	3.9.2.2.1 3/	
External Clock/Received Data Relationship	3.9.2.2.2 3/	
Data Transmission	3.9.2.3 3/	
Detailed Requirements for Digital Interface Circuits	3.9.3 1/ 3/	
Signals	3.9.3.1 3/	
Conditioned Diphase	3.9.3.1.1 3/	
Low Level Digital Interface	3.9.3.2 3/	
Idle Transmit Condition	3.9.3.3.1 3/	
Transmitter Source Impedance	3.9.3.3.2 3/	
Receive Interface (130 Volt Operation)	3.9.3.3.3 3/	
Normal Input Keying (NIK) Interface	3.9.3.4 3/	
Relay Port Digital Interface	30.9.3.5 3/	
Relay Port Signals	30.9.3.5.1 3/	
Receive Equipment	3.9.4.1 3/	
Digital Distortion	3.9.4.1.1 3/	
Spurious Pulses	3.9.4.1.2 3/	
Transmit Equipment	3.9.4.2 3/	
Isolation of Chassis Ground, Signal, and Clock Lines	3.9.5 2/	
System Interface	30.9.6 3/	
Pin Assignment	3.9.6.1 3/	
Isolation of Chassis Ground, Power Lines	3.9.7.4 2/	
Two Key Rollover	3.10.3 3/	
Key Pressure	3.10.3.1 3/	
Lower Case Characters	3.10.4.1 3/	
Shift Key	3.10.4.2 3/	
Shift Lock Key	3.10.4.3 3/	
Machine Functions	3.10.4.4 3/	
Control Codes	3.10.4.5 3/	
Control Keys	3.10.4.6 3/	

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TABLE XVII. GROUP B INSPECTION - AN/UGC-74C(V). - Continued.

REQUIREMENT	REQ'T PARA.	AQL
Repeat Key	3.10.4.7 3/	
TAB Key	3.10.4.8 3/	
BS Key	3.10.4.9 3/	
Message Processing Keyboard	3.10.4.10 3/	
Safety	3.17 through 3.17.4 1/ 2/	

1/ First Article Test Only.

2/ Use Sampling Plan 2 for this requirement (see TABLE VIII).

3/ Use Sampling Plan 3 for this requirement (see TABLE X)

TABLE XVIII. FIRST ARTICLE GROUP C ENVIRONMENTAL INSPECTION.

	SAMPLE UNIT 1/ 3/								
INSPECTION	1,2,3	4	5	6	7	8	9,10	11,12	13	14,15
Immersion 4/	5,7									1,3
AC Overload	3									
Power Consumption	1									
Sources	2									
DC Overload & Transient	4									
Shock, Drop 4/	6									2
Temperature High		1					1			
Temperature Low		2					2			
Altitude		3								
Bench Handling		4								
Bounce (Loose Cargo) 4/		5					3			
Ballistic Shock		6					4			
Humidity 2/			1					1		
Vibration			2					2		
Rain			3							
Orientation			4					3		
Acoustic Noise			5					4		
Paper Usage			6					5		

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TABLE XVIII. FIRST ARTICLE GROUP C ENVIRONMENTAL INSPECTION. - Continued.

	SAMPLE UNIT 1/ 3/												
INSPECTION	1,2,3	4	5	6	7	8	9,10	11,12	13	14,15
Fungus 2/				1						
Salt Atmosphere 2/					1					
Maintainability Demonstration									1	
Systems Interface						1				
Dust 2/						2				

1/ The numbers in the unit column specify the order of inspections for the indicated unit.

2/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection.

3/ Sample units 14 and 15 are Auxiliary Memory Modules only.

4/ Auxiliary Memory Module is not attached to unit during this test.

TABLE XIX. GROUP C INSPECTION - AN/UGC-74C(V).

REQUIREMENT	REQ'T PARA.
Subgroup I	
Immersion	30.15.4
Subgroup II	
Unit 1	
Temperature High	30.15.1
Temperature Low	30.15.1
Altitude	3.15.3
Bounce, Loose Cargo	3.15.5.1
Bench Handling	3.15.6.2
Ballistic Shock	30.15.7
Unit 2	
Power Consumption	30.9.7.1
Sources	3.9.7.2
Overvoltage & Transients	3.9.7.3.2
AC Overload	3.9.7.3.1
Vibration	3.15.5

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TABLE XIX. GROUP C INSPECTION - AN/UGC-74C(V). - Continued.

REQUIREMENT	REQ'T PARA.
Immersion	30.15.4
Drop Shock	30.15.6.1
Immersion	30.15.4
Subgroup III	
Humidity	3.15.2
Rain	3.15.9
Orientation	3.15.11
Acoustic Noise	3.12
Paper Usage	3.8.1
Subgroup IV	
Unit 1	
Fungus	3.15.12
Unit 2	
Salt Atmosphere	3.15.10
Unit 3	
Dust	3.15.8
Subgroup V	
System Interface	30.9.6
Subgroup VI	
Software Verification	(See Table XII)
Subgroup VII	
Interchangeability	3.6

40.6 Preconditioning.

40.6.1 Bounce Preconditioning. Same as 4.6.1.

40.6.2 Burn-in Debugging. Same as 4.6.2 and subparagraphs.

40.7 Visual and Mechanical Inspection of Equipment. Same as 4.7.

40.8 Inspection for Dimensional Interchangeability.

40.9 Functional Requirements for Terminal, Communications AN/UGC-74C(V). The requirements identified within the subparagraphs of paragraph 30.8 of this specification shall be verified during First Article or the Quality Conformation Inspections.

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APPENDIX

40.10 AN/UGC-74C(V) Signal Characteristics and Electrical Requirements Test. The requirements identified within the subparagraphs of paragraph 30.9 of this specification shall be verified during First Article or the Quality Conformation Inspections. The test requirements shall be modified as follows:

40.10.1 Relay Port Digital Requirements Test. The equipment shall be tested for compliance with 30.9.3.2. The test procedure shall be in accordance with the applicable paragraphs of EIA Standard RS-422, dated April 1975 (exceptions in 30.9.3.2 shall be applicable).

40.11 Keyboard TT-745/UGC-74C(V) Requirements Tests. The requirements identified within the subparagraphs of paragraph 30.10 of this specification shall be verified during First Article or the Quality Conformation Inspections.

40.12 TEMPEST. Same as 4.12.

40.13 Electromagnetic Compatibility. Same as 4.13.

40.14 Nuclear Hardness Validation. Same as 4.14.

40.15 Environmental. Same as 4.15 with the exceptions and additions specified in the following subparagraphs.

40.15.1 Temperature Testing. Same as 4.15.1 and subparagraphs except as modified in the following subparagraphs.

40.15.1.1 High Temperature Testing. Same as 4.15.1.1 with the AMM not attached. A temperature stabilization and test of the AN/UGC-74C(V) with the AMM attached shall be performed at 55 C to verify AMM memory retention and operation per the approved test plan.

40.15.1.2 Low Temperature Testing. Same as 4.15.1.2 with the AMM not attached. After the Verification Test of the AN/UGC-74C(V) without AMM is performed at the low operating temperature -25 F (-32 C), the terminal shall be powered off and the AMM installed. A nonoperating temperature stabilization shall be performed at -40 F (-40 C) followed by a temperature stabilization at 32 F (0 C). The terminal shall then be operated to verify AMM memory retention and operation per the approved test plan.

40.15.2 Humidity Test. Same as 4.15.2.

40.15.3 Altitude Test. Same as 4.15.3.

40.15.4 Immersion Test. Same as 4.15.4 except the AMM shall separately be subjected to the test of Method 512.2, Procedure 1 of MIL-STD-810 (see 30.15.4).

40.15.5 Vibration Test. Same as 4.15.5.

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40.15.5.1 Bounce, Loose Cargo Test. Same as 4.15.5.1.

40.15.6 Shock Tests.

40.15.6.1 Drop Test. Same as 4.15.6.1 with the addition of subjecting the AMM separately to the test of Method 516.3, Procedure IV of MIL-STD-810 for equipment weighing under 100 pounds. Minor damage (see 6.4.1) is permitted (see 3.15.6.1).

40.15.6.2 Bench Handling Test. Same as 4.15.6.2.

40.15.7 Ballistic Shock Test. Same as 4.15.7 with the requirements specified in 30.15.7.

40.15.8 Dust Test. Same as 4.15.8.

40.15.9 Rain Test. Same as 4.15.9.

40.15.10 Salt Fog Test. Same as 4.15.10.

40.15.11 Orientation Test. Same as 4.15.11.

40.15.12 Fungus Test. Same as 4.15.12.

40.15.13 Leakage Air Seal Test. Same as 4.15.13.

40.16 Reliability Testing. Same as 4.16 and subparagraphs.

40.17 Acoustics Test. Same as 4.17.

40.18 Verification Test - AN/UGC-74C(V).

40.18.1 Purpose. The purpose of this test is to determine if any degradation of the equipnent occurred during or after any environmental test to which the equipment was subjected.

40.18.2 Equipment. Communications terminal under test.

40.18.3 Verification Test.

40.18.3.1 Pre-Test Verification.

a. Visually inspect equipment for damaged parts or assembly, corrosion and loose fasteners.

b. Turn power on (if necessary)

c. Perform Self Test per 30.8.8.8.

d. Verify correct operation per 30.8.8.8.

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40.18.3.2 Verification During Environmental Exposure.

a. Turn power on (if necessary).

b. Perform Factory Self Test per 30.8.8.9.

c. Verify correct operation per 30.8.8.9.

40.18.3.3 Post-Test Verification. Same as 40.18.3.1.

40.18.4 Acceptance. Same as 4.18.4 and subparagraphs.

40.19 Maintainability Demonstration. Same as 4.19.

40.20 Weight. Weigh the AN/UGC-74C(V) and verify conformance to 30.19.

40.21 Systems Safety Test. Same as 4.21.

50. PACKAGING

50.1 Packaging. Same as 5.1.

60. NOTES

60.1 Ordering Data. Same as 6.1.

60.2 Cables. Same as 6.2.

60.3 Verification Inspection. Same as 6.3.

60.4 Definitions. Same as 6.4.1 and subparagraphs.

60.5 Nomenclature. Same as 6.5.

60.6 Color. Same as 6.6.

60.7 Government-Furnished Equipment (GFE). Same as 6.7 and subparagraphs.

60.8 System Configurations. The AN/UGC-74C(V) terminal configurations are shown in Table XX.

60.9 Subject Term (Keyword) Listing. Same as 6.9.

60.10 Changes from Previous Issue. Same as 6.10.

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APPENDIX

TABLE XX. TERMINAL, COMMUNICATIONS AN/UGC-74C(V) CONFIGURATIONS.

NOMENCLATURE	TELEPRINTER TT-744B/ UGC-74C(V)	KEYBOARD TT-745A/ UGC-74C(V)	CASE, TELEPRINTER CY-7735/ UGC-74A(V)	COVER TELEPRINTER MX-10038/ UGC-74A(V)	AUXILIARY MEMORY MODULE MU-856/ UGC-74C(V)
TERMINAL, COMMUNICATION AN/UGC-74C(V)1	X		X	X	
TERMINAL, COMMUNICATION AN/UGC-74C(V)2	X		X		
TERMINAL, COMMUNICATION AN/UGC-74C(V)3	X	X	X	X	X
TERMINAL, COMMUNICATION AN/UGC-74C(V)4	X	X	X		X

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